Development, Analysis, and Validation of Parabolic Equation/Ray-Tracing Techniques in Railway Environments

by

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Graduate Department of Electrical and Computer Engineering
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Abstract

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The deployment of modern communication-based train control (CBTC) systems in railway networks requires a thorough understanding of the radio-wave propagation characteristics. Parabolic equation (PE) and ray-tracing (RT) are well-known techniques for generating propagation models. RT is versatile yet computationally expensive for long guiding structures whereas the PE method is efficient yet considers only forward propagation. Thus, for railway applications, it is of central interest to hybridize the two methods, to obtain a computationally efficient and versatile model.

In this thesis, a higher-accuracy PE solver is developed. Compared to traditional methods, it offers significantly better accuracy at no additional cost. Moreover, a systematic study of the numerical error and the relative performance of the PE methods is presented.

Finally, a hybrid modeling technique, combining the RT and PE methods, is presented. This hybrid model is validated against measured data of canonical tunnels and an actual deployment site of CBTC systems.

Dedication

To my parents

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Chapter 1

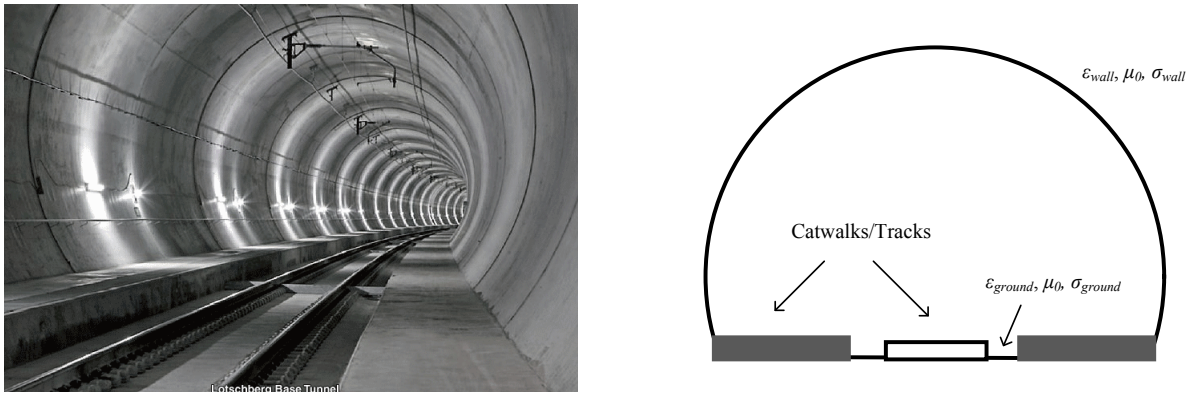
Introduction

Over the last few decades, wireless communication technologies have experienced rapid development. People, in their professional and personal lives, have begun to rely on and benefit greatly from the spreading of wireless communication systems. The thriving evolution is motivated by the desire to improve the performance of wireless communications steadily in terms of accessibility, speed, convenience, and reliability. Wireless communications in tunnels, including railways, urban roads, and pedestrian underpasses, is undoubtedly an indispensable part in the whole communication network architecture. The development of communication-based train control (CBTC) systems in tunnels has permanently connected railway signaling with advances in wireless communication technologies [1]. The scope of this thesis is to understand and characterize radio-wave propagation behaviors in railway environments, and provide an advanced propagation modeling tool for next generation railway signaling systems.

1.1 Railway Tunnel Propagation

The aim of modern communication-based train control (CBTC) systems for railway transportation is to provide reliable, wireless railway signaling and train navigation which can effectively cover the entire area of the railway network. Radio channel observation and characterization are indispensable for the development and deployment of modern CBTC systems in tunnels, and a thorough understanding of radio-wave propagation phenomena in tunnels is required to predict its performance accurately. This is of particular importance due to the safety-critical nature of railway signaling, and the rapid, reliable and cost-effective nature of system deployment. Over these years, a great deal of work has been carried out theoretically, numerically and experimentally to study the propagation characteristics in various tunnel scenarios [1–3]. On-site measurements, empirical and deterministic models for propagation prediction

are widely applied. Designers of next generation railway transportation signaling systems have benefited greatly from these models as it allows them to design systems with improved performance by configuring and placing the radio equipment optimally along railway tunnels. Fig. 1.1 shows the typical geometry for actual railway tunnels.



(a) (b)

Figure 1.1: Actual tunnel geometry. Source of (a): www.oil-electric.com.

Since railway transportation networks generally extend over tens to hundreds of kilometers, experimental characterization of radio-wave propagation, such as conducting a radio survey by channel sounding, is both exhaustively time-consuming and resource-intensive. Therefore, the development of numerical propagation prediction models that can be applied to wireless railway channels is as timely as ever. These models have to be computationally efficient in order to be applied to the electrically large problems that arise in tunnel propagation studies. Moreover, since they are aimed at guiding the deployment of radio equipment for train control systems whose function is safety-critical, their accuracy is of vital importance and cannot be overstated.

1.2 Wireless Channel Modeling

In communication systems, the channel refers to the pathway or medium transmitting information from one location to another. Wireless channel modeling is about representing the physical processes for transmitted signals and determine how a signal arrives at the end after interacting with a given channel. From the transmitter to the receiver, a signal can undergo a combination of phenomena, such as reflection, transmission, diffraction, refraction, and scattering. In principle, the electromagnetic fields impinging on the receiver antenna can be found by solving the related electromagnetic field equations. However,

in various environments, diverse and complicated propagation mechanisms need to be considered to accurately predict the signal propagation. In general, wireless propagation in channels depends on several important physical parameters such as the channel geometries, the electrical properties of the channel materials, and the location and orientation of the radio equipment.

Given an arbitrary environment, it is usually difficult to predict the propagation characteristics accurately and efficiently, and none of the existing models alone shall be well-suited to all scenarios. Therefore, various techniques have been developed to predict radio-wave propagation phenomena for different wireless communication systems. These channel models generally provide a trade-off between accuracy, complexity, and generality.

1.3 Modeling Techniques

Over the past half century, a number of modeling techniques for tunnel propagation have been used in the systems deployment and network planning process. Generally, these modeling techniques are categorized into two main streams: empirical and deterministic [4]. Historically, empirical channel modeling is the most widely recognized propagation prediction approach in tunnels. However, with the commercialization and rapid progress of modern computing technologies, more rigorous and efficient numerical channel modeling techniques are available in practice and deterministic modeling has exhibited great advantages over the empirical channel modeling.

- Empirical models, based on mathematical expressions which are determined via actual measurements obtained in the desired channel, are usually computationally efficient. The computational cost is independent of the electrical size and the complexity of the environments where measurements are carried out. In many cases, empirical models can provide very good insights into the physics and mechanisms of wave propagation in complicated indoor and urban environments. However, empirical models, restricted by the measurement accuracy and the conditions of certain environments, are not portable enough and sometimes offer rough or even unreliable estimations, especially in the cases where the materials and structures of the environments are not taken into account exactly. Therefore, more accurate and sophisticated models are needed and most of the efforts are concentrated on the deterministic channel modeling, where techniques evolve surprisingly fast.

- Deterministic models, derived rigorously from the laws of electromagnetic waves, are more portable and can yield more accurate results with exact theoretical laws applied, underlying assumptions

satisfied, and sufficient environmental features accounted for. For the deterministic models, usually adequate knowledge of the environments is required, including channel dimensions or geometries, wall roughness, and the electrical properties of boundary materials. Generally, the computational complexity increases as the electrical size grows and as specific details needs to be considered for the model. Waveguide theory, geometrical optics (GO), full-wave analysis, and parabolic equation (PE) are the mainly researched deterministic methods [5–9].

An overview of the most commonly applied modeling techniques for tunnel propagation is provided in the following diagram Fig. 1.2. The detailed comparison for the deterministic propagation prediction techniques is described in the following sections to offer a better view of their applicative scenarios and their advantages and drawbacks.

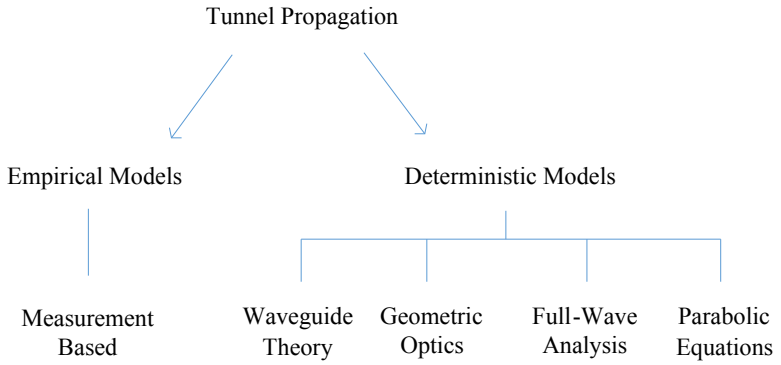


Figure 1.2: Diagram of the generally adopted modeling techniques for tunnel propagation.

1.3.1 Waveguide Theory

The propagation features of tunnels can be qualitatively determined and characterized via waveguide modal analysis [5, 10]. For these models, tunnels are initially approximated as canonical waveguide geometries, then field propagation and attenuation characteristics can be obtained by applying well-known analytical expressions in cylindrical and rectangular waveguides [4, 10, 11]. By employing lossy, bounded or unbounded dielectric walls [10], these waveguide models can be further extended to account for realistic tunnel materials (e.g. concrete, stone, etc.). Moreover, obstacles such as pedestrians and pillars can also be taken into consideration [12].

Since the field expressions are based on full-wave electromagnetic analysis, these models are very accurate and generally offer valuable physical insights of radio-wave propagation in tunnels. However,

even though they can illustrate characteristics of wave propagation in tunnels qualitatively, they are limited to simulating fields in canonical rectangular or circular waveguides, as shown in Fig. 1.3. Since the eigenfunctions for actual tunnels with complex geometries are difficult to be determined, waveguide mode analysis is hardly applicable to actual scenarios, and even if approximations are made, it is difficult to guarantee satisfying accuracy [9].

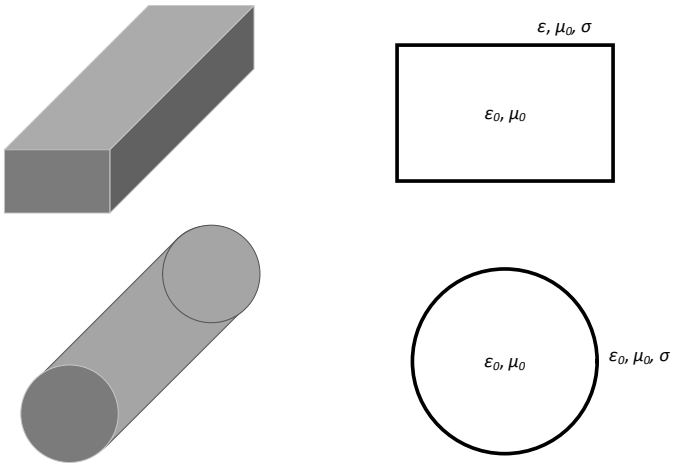


Figure 1.3: Waveguide theory based models.

1.3.2 Geometric Optics

Methods based on geometric optics, for example ray-tracing, have emerged as efficient simulation tools for propagation modeling [13]. These methods approximate the fields as reflecting or refracting rays along the propagation path. Typically, ray paths undergo multiple reflections or refractions before reaching the receiver. To calculate the total field, the contributions of all idealized rays that can reach the receiver are summed up.

Even though ray-tracing methods are not as accurate as full-wave solvers, they offer useful insights to study radio-wave propagation in arbitrary geometries and provide a good trade-off between accuracy, computational efforts, and generality. Therefore, geometric optics methods are generally adopted for propagation modeling [14]. One of the drawbacks is that ray-tracing becomes becomes quite computationally expensive for guiding structures such as tunnels at long distances, where an increasing number of reflections is needed for convergence. Moreover, the computational cost grows significantly as the number of the objects present along the channel increases and ray-tracing method fails for caustic regions [15].

1.3.3 Full-Wave Analysis

Full-wave models based on the solution of Maxwell's equations on a discrete grid, for example the finite-difference time-domain (FDTD) and the finite element method (FEM), can provide highly accurate results [16]. They can fully take into account all effects such as radiation, reflection, and diffraction and can solve wave propagation problems in complex geometries. Moreover, since the medium constitutive relations are automatically incorporated into the solution of Maxwell's equations, these methods are well-suited to study wave interactions in complex media.

However, to solve the continuous electromagnetic fields over electrically large areas, large amounts of memory are required to keep track of the fields at all locations and exhaustively many calculations are needed to update the fields at successive instants. Full three-dimensional FDTD or FEM simulations for large scenarios such as tunnels are not practical or realistic, even with the use of a supercomputer or cluster. Therefore, usage of these full-wave models has been generally limited to the modeling of two-dimensional or smaller scenarios such as indoor channels within buildings [17, 18].

1.3.4 Parabolic Equations

The parabolic equation method, an approximation of the Helmholtz wave equation, has proved to be a computationally efficient and accurate approach to study radio-wave propagation characteristics in straight and curved tunnels with arbitrary cross-section geometries [9, 15, 19, 20]. It is based on the paraxial approximation, which assumes that the wave propagation occurs mainly along or close to the axis of the tunnel. This enables the reduction of Helmholtz equation, from an elliptical equation into a parabolic equation with respect to the transverse components of the fields. Consequently, the complexity of numerical implementation is substantially reduced. Complex tunnel cross-section characteristics can be incorporated in parabolic equation solvers and full three-dimensional analysis can be accomplished accurately and efficiently [2, 21]. Moreover, coupled with tensor impedance boundary conditions [22], the parabolic equation method can capture wave depolarization effects that play an important role in path loss calculations, especially in curved tunnels [21].

The benefit of the parabolic equation method is that since it is based on the wave equation, it provides advantages similar to those of full-wave solvers without requiring as much execution time or memory. Furthermore, the slow varying nature of low order modes enables the use of implicit finite difference methods (FDMs) and marching techniques where large discretizations along the axis of propagation are allowed. However, this method cannot model high-order modes and corresponding rapid fluctuations accurately [23]. Generally, the parabolic equation method provides a compromise between the versa-

tile, yet computationally intensive ray and full-wave methods, and the efficient, yet inherently limited, waveguide mode analysis method.

1.4 Motivations and Objectives

Nowadays, people begin to rely on and benefit from the convenience brought by wireless communication technologies. A considerable number of wireless devices have been installed to enable extensive coverage of wireless communication systems. Wireless communications in railway tunnels scenarios is undoubtedly an indispensable part within the whole communication network architecture. The main objective of this thesis is to develop an advanced propagation prediction technique to estimate radio-wave propagation characteristics in complex railway environments.

Existing modeling techniques have to make a trade-off between accuracy, complexity and generality, particularly when considering electrically large problems, such as tunnels and railway systems. On one hand, ray-tracing has been successfully applied to studying wave propagation in complex environments, yet, due to the computational complexity, it is not suitable for modeling long tunnels. The PE methods, on the other hand, are far more computationally efficient and effective for these relatively uniform structures. However, the PE methods consider only forward propagation. Therefore, a hybrid model, which combines the strengths of these two methods and avoids their inherent limitations, should be developed to solve wave propagation for the whole railway system with high efficiency and accuracy.

1.5 Contributions

Two three-dimensional parabolic equation solvers are developed from the ground up. The first is the scalar parabolic equation solver, which can simulate wave propagation in rectangular tunnel with perfectly conducting walls. The second is the vector parabolic equation solver, where the PE method is coupled with impedance boundary conditions to be able to analyze wave propagation in actual tunnels with lossy walls. They have proved to be effective and computationally efficient modeling tools for those long, relatively uniform guideway tunnels. The results are validated against known analytical solutions as well as measured data in actual existing tunnels.

Moreover, an improved parabolic equation method, which has substantial advantages over the traditional techniques that have been employed in tunnel scenarios, is presented. This method provides significantly better accuracy at no additional computational cost. Furthermore, a systematic study of the numerical error and the relative performance of the PE methods, which provides guidelines for the choice

of the parameters and the trade-offs involved in solving parabolic equations, is described. Numerical investigations and assessments are accompanied by a case study of wave propagation in tunnels.

Additionally, the applicability of an existing ray-tracing solver in solving wave propagation of complicated environments is demonstrated. Finally, a hybrid model, combining the versatility of ray-tracing and the computational efficiency and accuracy of the PE method, is developed. This hybrid model offers a prospect for simulating wave propagation in the whole railway network, including stations, tunnels, and open-air sections.

1.6 Thesis Outline

The thesis is organized as follows:

Chapter 2 provides the fundamentals of the ray-tracing method. To demonstrate its applicability for modeling complicated environments, ray-tracing is then applied to solving wave prorogation in the railway stations and validated with the measured data. Moreover, limitations and weaknesses of ray-tracing are discussed and illustrated by modeling wave propagation in the long straight Massif Central tunnel at south-central France.

Chapter 3 introduces the parabolic approximation theory, where the derivations for the scalar and vector parabolic equations are presented. Subsequently, finite-difference schemes, such as the Crank-Nicolson (CN) and Alternating-Direction-Implicit (ADI) methods, are described and compared. Numerical investigations and validations are conducted for both the scalar and vector PE methods.

Chapter 4 presents a high-accuracy PE method, which has substantial advantages over the traditional techniques. The formulation of this method is presented, and verifications are accompanied by the case study of a rectangular tunnel with perfectly conducting walls and an actual tunnel with lossy walls.

Chapter 5 provides a comparative study of the commonly adopted numerical schemes for parabolic equations. Their computational efficiencies and accuracies are compared in detail.

Chapter 6 introduces a hybrid channel modeling technique, which combines the ray-tracing and PE methods. The proposed model is validated with the measured data, collected in canonical tunnels and an actual deployment site of communication-based train control systems.

Chapter 7 concludes this thesis and presents an outline of future research activities .

Chapter 2

Ray-Tracing Method

2.1 Introduction

Ray-tracing methods, based on geometrical optics, first became well-known for their applications in modeling computer graphics [13]. When applied to modeling radio-wave propagation problems across wireless channels, on the basis of a high frequency approximation, ray-tracing methods are adopted to study the reflection and refraction process of electromagnetic waves on the wall surfaces. A transmitting antenna emits a set of rays, representing various points on the wavefront, and at a given frequency, phase change and path loss are associated with these rays as they travel away from the transmitting antenna. Before reaching the receiver, these rays can undergo multiple reflections or refractions, where interactions with a media boundary are calculated according to the laws of Fresnel that govern geometric optics. Moreover, the edge effects are considered by combination with the geometrical theory of diffraction. For the calculation of the total field at the receiver, the contribution of all rays that can reach the receiver after being radiated from the transmitter need to be taken into account.

The ray-tracing method has proved to be a successful approach for a variety of applications such as characterization of indoor or urban propagation channels [24,25]. The main advantage of the ray-tracing method is that it can be applied to model the wave propagation in quite complicated environments with reasonable accuracy. The main drawback is that it has high demand of computational resources for long guideway structures, such as railways or urban road tunnels. Therefore, ray-tracing methods are generally applied to wave propagation modeling of indoor scenarios. One possible solution for ray-tracing to play a role within a wider range of applications in tunnels is to hybridize it with other channel modeling techniques that are suitable and efficient for guiding-way structures, such as the parabolic

equation method which will be discussed in the following chapters.

In this chapter, the fundamentals of the ray-tracing method are briefly introduced. Two commonly applied ray-tracing techniques are described and compared. Then, ray-tracing is applied to modeling wave propagation in railway stations to assess its feasibility for complicated environments. The simulated results are validated with the measured data at actual deployment sites of communication-based train control systems. Finally, the limitations and weaknesses of ray-tracing are discussed and illustrated by modeling wave propagation in the long straight Massif Central tunnel at south-central France.

2.2 Fundamentals of the Ray-Tracing Method

Generally, based on how the ray paths are taken into account, there are two commonly applied ray-tracing techniques: shoot, bounce and reflect (SBR) based ray-tracing method, and the image theory (IT) based ray-tracing method [13].

2.2.1 Shoot, Bounce and Reflect based Ray-Tracing

SBR-based ray-tracing methods, essentially launch rays in all directions from the source in order to illuminate the surrounding environments sufficiently, and keep track of all the rays to obtain the rays which can ultimately reach the receiver. The amplitude of each ray decays as it propagates in the environment, and the ray is terminated when its amplitude drops below certain threshold. Moreover, through interactions with the objects present in the environment, reflected, transmitted or diffracted rays can be spawned, which will undergo the same process until their amplitudes are too weak to make any contributions to the total fields. For each interaction with the media boundary, corresponding coefficients based on Fresnel's laws are calculated.

For numerical implementation, due to the discrete nature of the method, only a finite number of rays can be generated at the source. Therefore, if the receiver is assumed to be a point, the probability for it to receive any rays is very low. To better determine whether or not a given ray will reach the receiver, a reception sphere around the receiver point is defined to sum up all the incoming rays that intercept the given area or volume. The reception sphere must be electrically small, typically smaller than the wavelength by at least a factor of 10. However, if the reception sphere is too small and the angular resolution of the source is not fine enough, potential rays which would reach the receiver are not detected. Hence, the angular resolution where the rays are launched from the source needs to be adjusted to the radius of the reception sphere, accordingly.

Once the geometrical tracing of the rays is completed and all ray paths that reach the given reception

sphere identified, the total received field can be calculated by superposing the contributions from each received ray. Fig. 2.1 illustrates how the rays are transmitted and received with this SBR-based ray-tracing method, where Tx and Rx are respectively the transmitter and the receiver.

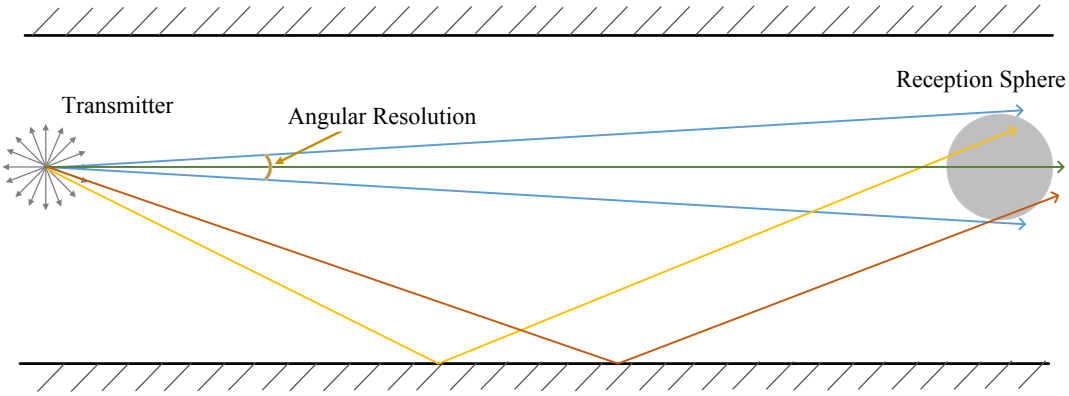


Figure 2.1: The SBR-based ray-tracing diagram, where two walls are considered and limited number of rays are plotted.

Conceptually, this SBR-based ray-tracing method is easier to understand and implement than other ray-tracing techniques. Its simplicity and effectiveness make it a logical initial choice while investigating different ray-tracing techniques. As the computational complexity of the SBR method depends on the number of objects present in the environment and the number of radiated rays launched from the source, hence, in complex or large environments, this inherent nature results in exponential computational complexity. Therefore, further research has focused on improved techniques such as the image theory (IT) based ray-tracing to overcome these problems.

2.2.2 Image Theory based Ray-Tracing

IT-based ray-tracing methods, instead of launching rays in all directions, are guided by image sources which are generated by reflecting the original source through all the planar facets present in the environment. As can be seen from Fig. 2.2, the reflected ray is represented as a ray radiated directly from the virtual source located symmetrically to the transmitter with respect to the plane that contains the facet. These virtual image sources relate to the ray paths that have undergone corresponding number of reflections as the order of this image source. In environments such as tunnels which have parallel facets, the number of virtual image sources can be infinite. Hence, in order to keep the complexity of the

calculation within a reasonable level, the number of image sources or reflection levels has to be limited. Generally, up to a specified number of reflections when the obtained fields converge at the receiver, it guarantees all the reflected paths are accounted for.

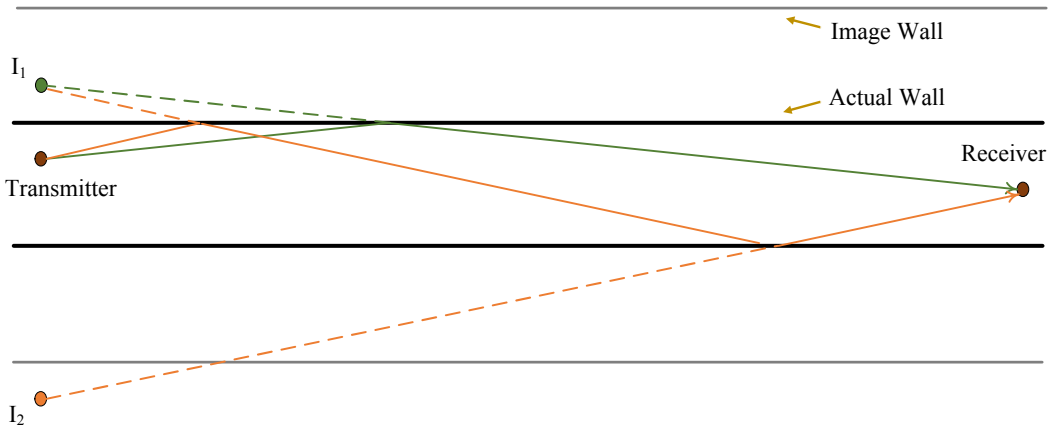


Figure 2.2: Image theory based ray-tracing diagram, where I_1 and I_2 are the image sources.

For the numerical implementation, this IT-based ray-tracing method calculates the ray-path starting from the receiver and incrementally builds the path geometry all the way back to the transmitter. Based on the positions of the transmitter and the receiver, this IT-based method provides a systematic way for the computation of the ray-path geometry and focuses only on the rays which will actually propagate to the receiver point. Therefore, no reception sphere needs to be defined at the receiver and no time is spent in computing ray paths that will not contribute to the total fields.

Compared to the SBR-based ray-tracing method, on one hand, since the construction of the ray-path geometry concerns only the rays that will reach the receiver point and does not require the launching of rays in all directions at the transmitter, this IT-based ray-tracing method is computationally more efficient and yields relatively more accurate results. On the other hand, this method does not suffer from many of those issues related to the reception sphere used in the SBR method, such as missing or double counting the number of rays at the receiver. Admittedly, this IT-based ray-tracing method has its own limitations as well. Since this method is based on image theory, it is restricted to environments that can be fully described using only planar facets. However, this problem can be solved, at a reasonable sacrifice of computational cost and accuracy, by approximating the curved surfaces with a series of planar facets.

The SBR-based ray-tracing method is an easy yet inefficient approach. However, with regard to

wave propagation modeling in railway tunnel environments, it rapidly becomes impractical even for simple canonical problems, as the distance between the transmitter and the receiver becomes very large. Moreover, it cannot guarantee that all the existing ray paths are properly accounted for and therefore the results generally lack of desired accuracy. On the contrary, the IT-based ray-tracing method is far more efficient than the SBR method. For these reasons, in the following discussion, the IT-based ray-tracing method is adopted for simulation and comparison.

2.3 Numerical Investigations at Railway Stations

The ray-tracing method has been widely applied to the characterization of indoor or urban propagation channels and has proved to be a powerful and effective technique [13, 24, 25]. In this section, ray-tracing is applied to model wave propagation in a more complicated environment, the railway station. Two sections, the Corona-Grandin section and the Bay-Corona section as shown in Figs. 2.3 and 2.6, from Edmonton railway network are chosen for simulation and verification. These sections are actual deployment sites of communication-based train control systems and both ends are connected with long straight tunnels. The simulated results are verified with the measured data, provided by Thales Canada.

2.3.1 Corona-Grandin Section

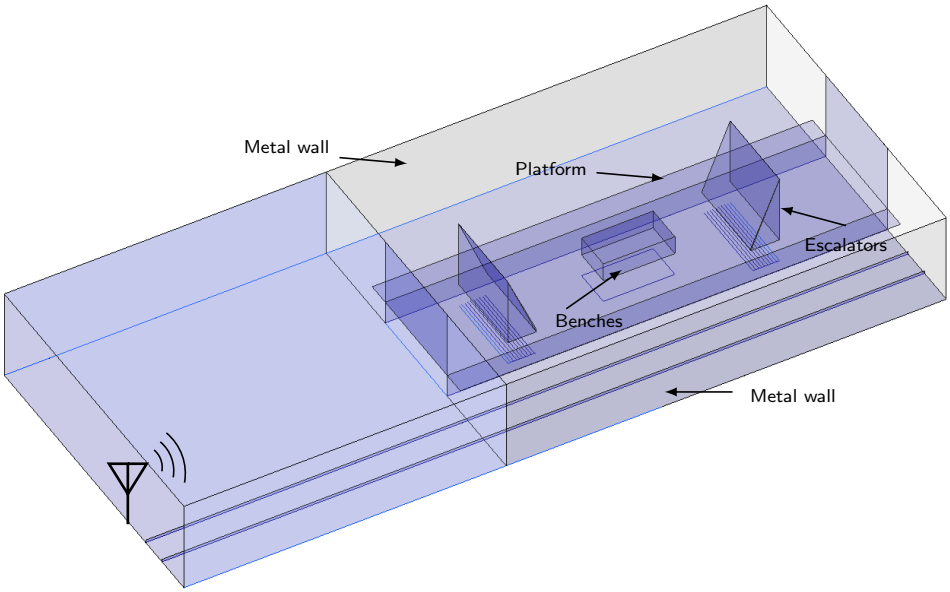
The Corona-Grandin section is mainly composed of of two segments, the two-way rectangular tunnel and the station platform, as illustrated in Fig. 2.3, where (a) is the on-site image and (b) is the modeling structure diagram of this section. For the station platform area, the environment structure is quite complicated, where obstructions such as escalators, elevators, stairs, catwalks, pillars, and rail tracks exist in this scenario. The walls present in this scenario mainly consist of large blocks of smooth stones or concrete and certain areas such as the bilateral walls of the station and partial of the ceilings, are composed of metal.

The transmitting antenna, operating at the frequency of 2.41 GHz, is placed at the beginning of the rectangular tunnel. The measurement data are obtained by fixing the receiving equipment on a trolley which moves along the track to receive the signal, as shown in Fig. 2.4. The received power is calculated by taking the maximum value received by the two antennas on the trolley.

The simulated results are validated against the measured data, as illustrated in Fig. 2.5. As can be seen from the graphs, they have a good agreement, where the received powers are of the same level, and the profile of the ray-tracing results captures the main maxima and minima of those of the measured data.



(a) On-site image.



(b) The modeling structure diagram.

Figure 2.3: Corona-Grandin section, where the double-way rectangular tunnel and the station platform are respectively 90 m and 123 m in length. The cross-section dimensions are of 17.86 m × 7.14 m and the height of the platform is 0.9 m. The transmitting antenna is placed 3.1 m in height and horizontally 4.9 m to the right wall. Source of (a): Thales Canada.

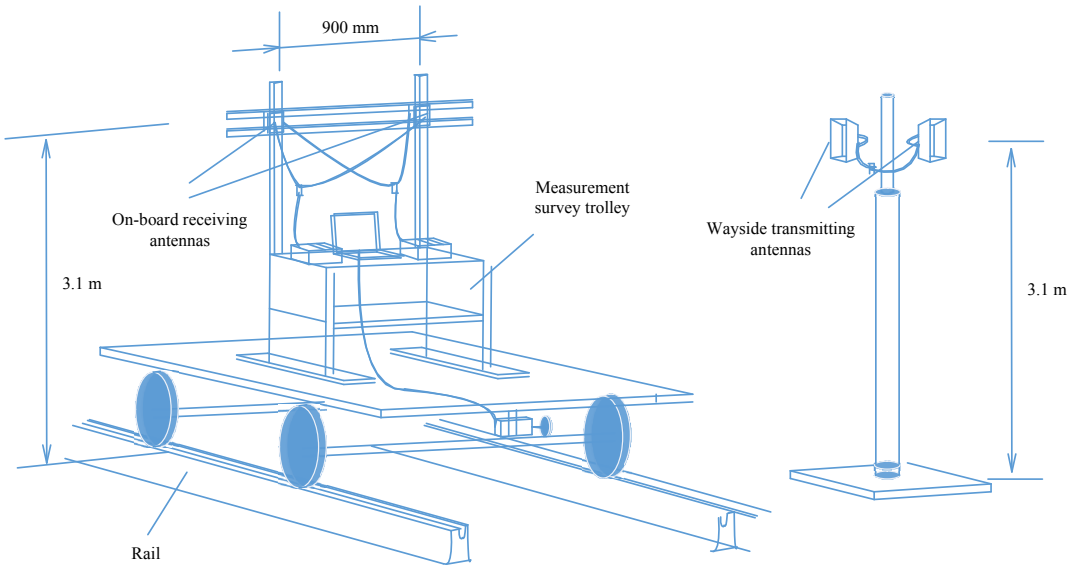


Figure 2.4: Layout for the transmitting antenna and the receiving antenna adopted for the measurement. The rail track is 2.81 m in width and 0.1 m in height, and the center axis of the rail track to the right wall is 3.48 m. Two receiving antennas are fixed on the trolley with 0.9 m separation and 3.1 m to the top of the rail track. Source: Thales Canada.

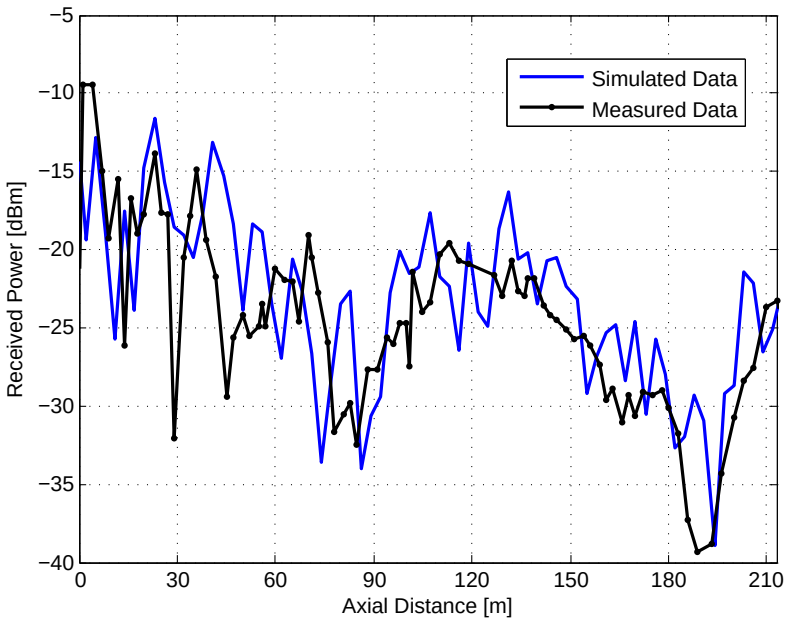


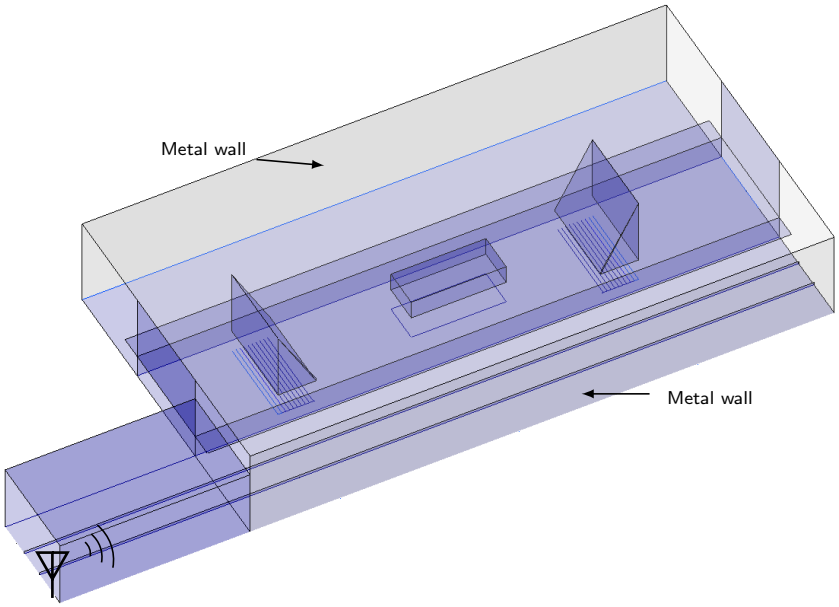
Figure 2.5: Received power for the Corona to Grandin section.

2.3.2 Bay-Corona Section

The Bay-Corona section also consists of of two parts, yet instead of a wide double-way rectangular tunnel, it is initiated with a narrow one-way rectangular tunnel, as illustrated in Fig. 2.6.



(a) On-site image.



(b) The modeling structure diagram.

Figure 2.6: Bay-Corona section, where the narrow one-way rectangular tunnel and the station platform are respectively 40 m and 123 m in length. The equivalent cross-section dimension for the one-way rectangular tunnel is 5.4 m × 5.2 m. The dimensions of the station platform are the same as the Corona-Grandin section. The transmitting antenna is placed 3.1 m in height and horizontally 0.5 m to the right wall. Source of (a): Thales Canada.

For this section, the transmitting antenna is also placed at the beginning of the rectangular tunnel yet operates at the frequency of 2.45 GHz. Moreover, instead of the center of the tunnel, the transmitting antenna is placed horizontally 0.5 m from the right wall. The simulated data is obtained through the same precess as the Corona-Grandin section. As can be seen in Fig. 2.7, the simulated results and the measured data match very well.

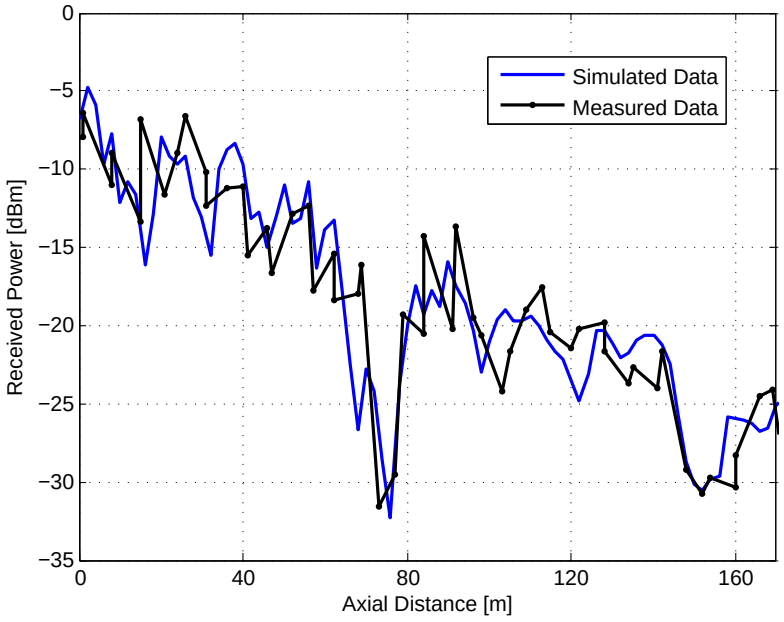


Figure 2.7: Received power for the Corona to Grandin section

2.4 Limitations and Weaknesses

From the examples presented in the previous section, ray-tracing has proved to be able to approximate wave propagation in very complicated railway environments with reasonably accurate results. However, the ray-tracing model also has its inherent limitations and weaknesses. It typically suffers from an exponential computational complexity for guiding structures such as tunnels at long distances, where an increasing number of reflections is required for convergence. To better illustrate its limitations and weaknesses, in this section, it is applied to simulate wave propagation in the long straight Massif Central tunnel in south-central France.

The Massif Central tunnel, previously studied by Dudley *et al.* [10], is a straight, 3.5 km long tunnel consisting of large blocks of smooth stones or concrete. In [10], the equivalent cross-section dimensions

of 7.8 m × 5.3 m are adopted, and the electrical parameters are chosen as $\varepsilon_r = 5$ and $\sigma_0 = 0.01\,\mathrm{S/m}$. According to Dudley *et al.* [10], the near zone of the tunnel is characterized by rapid fluctuations caused by the interaction of many modes, and the fields in the far zone consist primarily of the first few lower-order modes.

The simulated distance is 2.5 km. To compare the received power, a receiver point is defined at height $y = 2\,\mathrm{m}$ and at one-quarter of the width of the tunnel horizontally. The received power at both 900 MHz and 2100 MHz is compared to the measured data, presented in [10]. This comparison is shown in Figs. 2.8 and 2.9.

As can be seen from Fig. 2.8, the results did not converge even with a reflection level of 25, with which the simulation takes about 21 hours performed on a workstation. At a higher frequency of 2100 MHz, the ray-tracing results converge faster as a function of the maximum allowed reflections. However, there is still much difference between the two received power levels at distances beyond 2 km, as illustrated in Fig. 2.9. Therefore, it can be seen that the exponentially increasing execution time is an important limitation of the ray-tracing method.

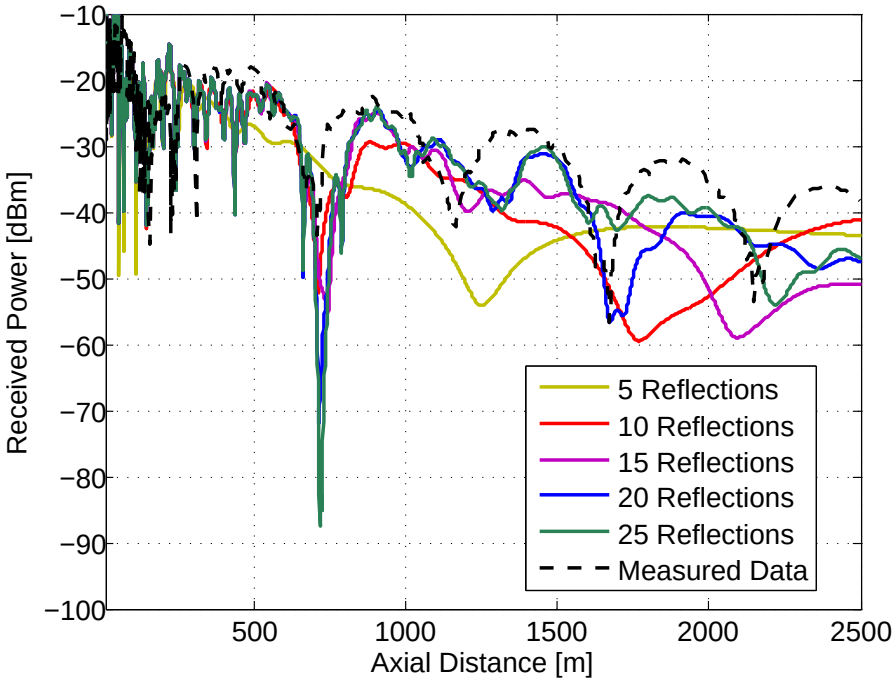


Figure 2.8: Received power at 900 MHz

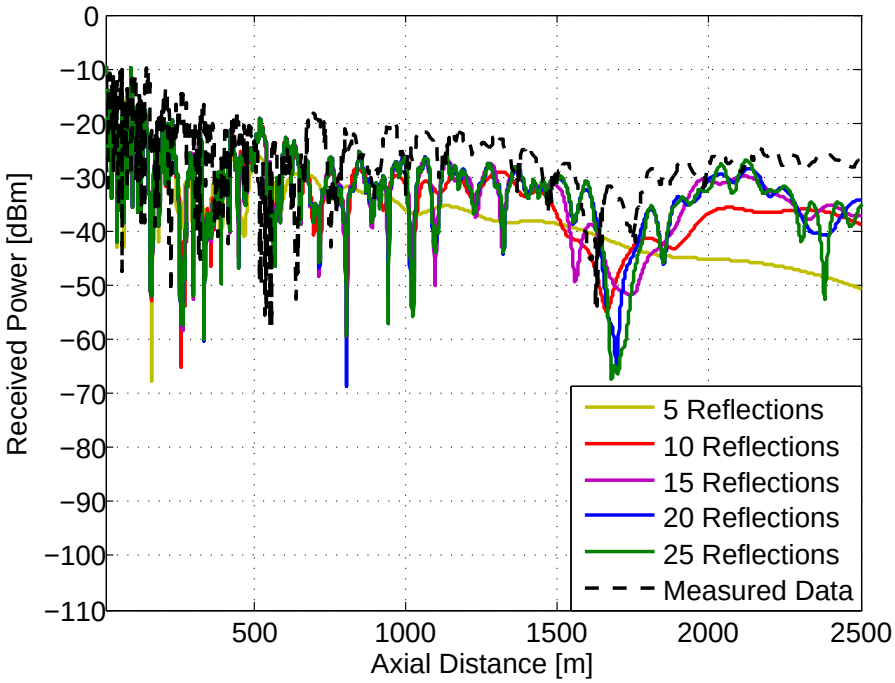


Figure 2.9: Received power at 2100 MHz

2.5 Conclusion

This chapter introduced the fundamentals of the ray-tracing method, where the SBR-based ray-tracing and the IT-based ray-tracing method were briefly described and compared. The ray-tracing method proved to successfully model wave prorogation in quite complicated environments. However, as for the long tunnels, an increasing number of reflections are required for convergence and ray-tracing is restricted by the resulting exponential computational complexity.

Chapter 3

Fundamentals of the Parabolic Equation Method

3.1 Introduction

As discussed in the previous chapter, ray-tracing can become a very inefficient tool for wave propagation modeling in electrically large guideway structures, such as railway tunnels. Therefore, a computationally more efficient technique, the parabolic equation method, is studied in this chapter.

The parabolic equation (PE) method was first introduced to study radio-wave propagation by Leontovich and Fock in 1946 [26, 27]. The standard parabolic equation is an approximation of the Helmholtz equation that assumes fields having slow variations along the propagation direction. This assumption leads to a more efficient numerical implementation by allowing the reduction from the Helmholtz wave equation, an elliptical equation, into a parabolic equation with respect to the transverse components of the fields. Extensive research has been focused on the development of parabolic equation methods. With the emergence of advanced mathematical techniques and thorough physical analysis, the parabolic equation method became an efficient and universal simulation approach. It has been widely applied to engineering issues such as underwater and atmosphere propagation [28, 29].

In the work done by Kuttler and Dockery [30, 31], splitting algorithms were applied to parabolic equations to study electromagnetic propagation in the troposphere. Moreover, iterative methods were combined with PE techniques by Zelly and Constantinou to solve atmospheric propagation problems [32]. These improvements enabled the widespread use of PE methods in solving radio-wave propagation problems involving tropospheric refraction and terrain diffraction. Subsequently, the introduction of

impedance boundary conditions propelled the application of PE methods for radio-wave propagation in urban environments, such as railway or road tunnels. Complex tunnel cross-section characteristics can be incorporated in these solvers [2, 21]. Further development then aimed at refined PE modifications to account for irregular [15, 21–23], depolarization effects [22], rough surfaces [23], and transparent boundaries [33, 34].

In this chapter, first the theory behind the parabolic approximation method is introduced and derivations for the scalar and vector parabolic equations are presented. The scalar parabolic equation is the standard formulation of the PE method, where transverse fields are solved independently. This is the case study for tunnels with perfectly conducting walls. However, since railway tunnels are typically composed of lossy materials such as concrete or stone, the vector parabolic equation (VPE) method along with the impedance boundary condition (IBC) is introduced. Subsequently, finite-difference schemes, such as the Crank-Nicolson (CN) and Alternating-Direction-Implicit (ADI) methods, are described and presented. Their formulation, order of accuracy, and stability issues are discussed as well. Finally, numerical investigations and validations are conducted for both the scalar and vector PE methods.

3.2 Parabolic Approximation Theory

The parabolic equation method is an approximation of the Helmholtz wave equation that is applied to simulate fields which mainly travel along or close to the axis of propagation [9]. The derivation of the parabolic equation starts with the Helmholtz wave equation for free-space

$$\left(\frac{\partial^2}{\partial x^2}+\frac{\partial^2}{\partial y^2}+\frac{\partial^2}{\partial z^2}+k_0^2\right)\phi(x,y,z)=0 \tag{3.1}$$

where ϕ is the scalar potential (electric or magnetic) and k_0 is the free-space wave number. All field components can be calculated through the scalar potential ϕ [35, 36]. Assuming propagation is predominantly along the z-axis, the components of transverse magnetic fields can be obtained from the magnetic scalar potential ϕ^m, where the relationships are:

$$E_x=\frac{1}{jw\mu_0\varepsilon_0}\frac{\partial^2\phi^m}{\partial x\partial z} \tag{3.2a}$$

$$E_y=\frac{1}{jw\mu_0\varepsilon_0}\frac{\partial^2\phi^m}{\partial y\partial z} \tag{3.2b}$$

$$E_z = \frac{1}{jw\mu_0\varepsilon_0}\left(\frac{\partial^2\phi^m}{\partial z^2} + k_0^2\right) \tag{3.2c}$$

$$H_x = \frac{1}{\mu_0}\frac{\partial\phi^m}{\partial y} \tag{3.2d}$$

$$H_y = -\frac{1}{\mu_0}\frac{\partial\phi^m}{\partial x} \tag{3.2e}$$

$$H_z = 0 \tag{3.2f}$$

where ε_0 and μ_0 are the free-space permittivity and permeability, respectively. Similarly, the components of transverse electric fields can be obtained from the electric scalar potential ϕ^e and the corresponding equations are

$$E_x = -\frac{1}{\varepsilon_0}\frac{\partial\phi^e}{\partial y} \tag{3.3a}$$

$$E_y = \frac{1}{\varepsilon_0}\frac{\partial\phi^e}{\partial x} \tag{3.3b}$$

$$E_z = 0 \tag{3.3c}$$

$$H_x = \frac{1}{jw\mu_0\varepsilon_0}\frac{\partial^2\phi^e}{\partial x\partial z} \tag{3.3d}$$

$$H_y = \frac{1}{jw\mu_0\varepsilon_0}\frac{\partial^2\phi^e}{\partial y\partial z} \tag{3.3e}$$

$$H_z = \frac{1}{jw\mu_0\varepsilon_0}\left(\frac{\partial^2\phi^e}{\partial z^2} + k_0^2\right) \tag{3.3f}$$

3.2.1 Scalar Parabolic Equation

Assuming that the z-axis is the propagation direction, solutions to the Helmholtz wave equation Eqn. (3.1) can be cast in the form [2]

$$\phi(x, y, z) = u(x, y, z)\,\mathrm{e}^{-jk_0 z} \tag{3.4}$$

where the fast varying phase term in the propagation direction is factored out, and $u(x, y, z)$ represents the reduced plane wave solution which varies slowly along the propagation direction.

Assuming [2]

$$\left|\frac{\partial^2 u}{\partial z^2}\right| \ll k_0\left|\frac{\partial u}{\partial z}\right| \tag{3.5}$$

and substituting Eqn. (3.4) into Eqn. (3.1), the standard three-dimensional parabolic equation is obtained:

$$\frac{\partial u}{\partial z} = \frac{1}{2jk_0}\left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2}\right)u \tag{3.6}$$

Compared to the Helmholtz wave equation (an elliptical equation), the parabolic equation method reduces the second-order derivative with respect to the direction of propagation (here, the z-direction) into a first-order derivative, which greatly reduces the computational cost of solving large-scale propagation problems and allows for the application of numerical finite-difference methods for parabolic equations, including unconditionally stable solvers. However, the parabolic equation method is not suitable for all kinds of wave propagation problems. The assumption Eqn. (3.5) is only valid for waves traveling paraxially with respect to the propagation direction. Moreover, the parabolic equation method cannot take the backward traveling waves into account [9]. According to Zelly and Constantinou [32], the range of angles over which the parabolic equation method is valid can be assessed by substituting the solution of the reduced plane wave $u(x, y, z)$ into Eqn. (3.6) and evaluating the produced error expression. A solution of the scalar potential ϕ that satisfies the Helmholtz wave equation Eqn. (3.1) can be of the form

$$\phi(x, y, z) = C\,\mathrm{e}^{-j(k_x x + k_y y + k_z z)} \tag{3.7}$$

where C is a constant amplitude. According to Eqn. (3.4), the reduced plane wave solution $u(x, y, z)$ can be expressed as

$$u(x, y, z) = C\,\mathrm{e}^{-j[k_x x + k_y y + (k_z - k_0) z]} \tag{3.8a}$$

$$= C\,\mathrm{e}^{-jk_0[\cos\varphi \sin\theta\, x + \sin\varphi \sin\theta\, y + (\cos\theta - 1) z]} \tag{3.8b}$$

where $k_x = k_0 \cos\varphi \sin\theta$, $k_y = k_0 \sin\varphi \sin\theta$, $k_z = k_0 \cos\theta$, and φ is the azimuth angle and θ is the angle of propagation measured from the z-axis. By substituting Eqn. (3.8b) into Eqn. (3.6), the relative difference between two sides is obtained as

$$\text{error} = 1 - \cos\theta - \frac{\sin^2\theta}{2} \tag{3.9}$$

The desired level of accuracy will determine the range of angles over which the parabolic equation method is valid. Generally, the angle range of $\pm 15^\circ$ to the propagation direction is adopted [32]. Moreover, a more comprehensive error analysis is presented in Chapter 5.

3.2.2 Vector Parabolic Equation

The scalar PE method is typically applied to tunnels with perfectly conducting walls, where the transverse magnetic and electric fields propagate independently. However, the railway tunnels are generally

composed of lossy materials and coupling between the *TE* and *TM* modes exists. Hence, field components cannot be solved independently anymore with the use of the scalar PE. Consequently, the vector parabolic equation (VPE) along with the impedance boundary condition (IBC) are developed to address the polarization effects and loss issues for railway tunnels.

The vector PE coupled with the IBC has proved to successfully model radio-wave propagation in tunnels and urban environments [44, 45]. Here, the standard VPE formulation outlined by Popov *et al.* [45] is adopted:

$$\frac{\partial \boldsymbol{W}}{\partial z} = \frac{1}{2jk_0}\left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2}\right)\boldsymbol{W} \tag{3.10}$$

where $\boldsymbol{W}$ is the complex wave amplitude of the reduced plane wave propagating along the z-axis, as shown in the following expression:

$$\boldsymbol{E}_{\text{trans}} = \begin{bmatrix} E_x \\ E_y \end{bmatrix} = \boldsymbol{W}\mathrm{e}^{-jk_0 z} \tag{3.11}$$

In this VPE model, the Leontovich impedance boundary condition is used for the calculation of fields coupling on the tunnel walls and the characterization of the wall electrical parameters [47]. It is defined as

$$\hat{\boldsymbol{n}} \times \boldsymbol{E} = -Z\hat{\boldsymbol{n}} \times (\hat{\boldsymbol{n}} \times \boldsymbol{H}) \tag{3.12}$$

where Z is the relative surface impedance and $\hat{\boldsymbol{n}}$ is the outward normal on the tunnel wall. For a wall with relative permittivity ε_r and conductivity σ_0 (in S/m), Z is approximated as [46]

$$Z = \frac{\sqrt{\varepsilon_{rc} - 1}}{\varepsilon_{rc}} \tag{3.13}$$

where $\varepsilon_{rc} = \varepsilon_r - j\sigma_r$ and $\sigma_r = \sigma_0/w\varepsilon_0$, are, respectively, the complex permittivity and relative conductivity. At the boundary, the relation is given by

$$\boldsymbol{W}|_{\text{boundary}} = \frac{j}{k_0}\mathbf{T_0}\,\mathbf{G_0}\,\mathbf{T_0}\,\left.\frac{\partial \boldsymbol{W}}{\partial n}\right|_{\text{boundary}} \tag{3.14}$$

where $\mathbf{T_0}$ and $\mathbf{G_0}$ are matrices defined by

$$\mathbf{T_0} = \begin{bmatrix} n_x & n_y \\ n_y & -n_x \end{bmatrix} \tag{3.15}$$

$$\mathbf{G_0} = \begin{bmatrix} 1/Z & 0 \\ 0 & Z \end{bmatrix} \tag{3.16}$$

where n_x and n_y are x- and y-direction components of the unit normal vectors on the boundary.

3.3 Finite-Difference Schemes

For the numerical implementation of parabolic equations, typically implicit finite-difference approaches such as the Crank-Nicolson (CN) and the Alternating-Direction-Implicit (ADI) method are adopted [2]. In this section, the formulation and the stability characteristics for these two methods are discussed.

3.3.1 Crank-Nicolson Method

The traditional finite-difference technique that is widely adopted in previous work is the Crank-Nicolson (CN) method [2]. The popularity of the CN method comes from the fact that it is unconditionally stable and second-order accurate in x-, y- and z-directions [37].

Formulation

The derivatives in Eqn. (3.6) are approximated using the Taylor's theorem [39] and a centered-difference scheme is applied to calculate the partial derivatives, for example in x-direction

$$\frac{\partial^2 u_i}{\partial x^2} \approx \frac{u_{i+1} - 2u_i + u_{i-1}}{\Delta x^2} + \mathcal{O}\big(\Delta x^2\big) \tag{3.17}$$

where $\mathcal{O}(\cdot)$ is the order of accuracy. By substituting these centered-difference approximations into Eqn. (3.6), the standard CN scheme is obtained:

$$\begin{aligned} \frac{u_{i,j}^{k+1} - u_{i,j}^{k}}{\Delta z} = \frac{1}{2jk_0}\left(\frac{u_{i+1,j}^{k+1} - 2u_{i,j}^{k+1} + u_{i-1,j}^{k+1}}{2\Delta x^2} + \frac{u_{i+1,j}^{k} - 2u_{i,j}^{k} + u_{i-1,j}^{k}}{2\Delta x^2}\right) \\ + \frac{1}{2jk_0}\left(\frac{u_{i,j+1}^{k+1} - 2u_{i,j}^{k+1} + u_{i,j-1}^{k+1}}{2\Delta y^2} + \frac{u_{i,j+1}^{k} - 2u_{i,j}^{k} + u_{i,j-1}^{k}}{2\Delta y^2}\right) \\ + \mathcal{O}\big(\Delta z^3\big) + \mathcal{O}\big(\Delta z \Delta x^2\big) + \mathcal{O}\big(\Delta z \Delta y^2\big) \end{aligned} \tag{3.18}$$

which can be simplified and rewritten as

$$\left(1-\frac{r_x}{4jk_0}\delta_x-\frac{r_y}{4jk_0}\delta_y\right)u_{i,j}^{k+1}=\left(1+\frac{r_x}{4jk_0}\delta_x+\frac{r_y}{4jk_0}\delta_y\right)u_{i,j}^{k} + \mathcal{O}(\Delta z^3)+\mathcal{O}(\Delta z\Delta x^2)+\mathcal{O}(\Delta z\Delta y^2) \quad (3.19)$$

where $r_x=\Delta z/\Delta x^2$ and $r_y=\Delta z/\Delta y^2$ are the mesh ratios, and δ_x and δ_x are given by

$$\delta_x u_{i,j}^k = u_{i+1,j}^k - 2u_{i,j}^k + u_{i-1,j}^k \quad (3.20)$$

$$\delta_y u_{i,j}^k = u_{i,j+1}^k - 2u_{i,j}^k + u_{i,j-1}^k \quad (3.21)$$

The CN scheme is shown in Fig. 3.1, where the dots represent the nodes that are invoked in the finite-differences of Eqn. (3.19) during one step. The updating process includes finite differences that are taken along both directions on the transverse plane, where a large matrix is formed from the updating equation for the desired positions and the updating matrix for the CN method is not in tridiagonal form. Hence, the CN method can be quite computationally intensive for problems with fine meshes or when a large number of propagation steps is needed.

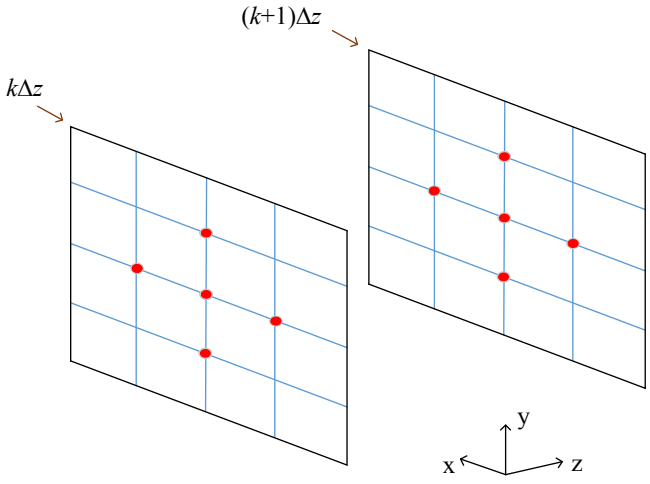


Figure 3.1: The three-dimensional grid for the CN method, where the red dots represent the nodes that are invoked in the finite-differences of Eqn. (3.19) at the k-th step.

Stability

The stability of finite-difference schemes is closely associated with the numerical error. A finite-difference scheme is stable if the error generated at each iteration step of the calculation does not lead to the total error increasing as the computation is continued. So if the finite-difference scheme is stable, the error is bounded. On the contrary, if the error grows, the scheme is unstable. Here, stability analysis is performed via the von Neumann method, a powerful technique to check the stability of finite-difference schemes [37, 38].

The von Neumann method is based on the decomposition of the solution into Fourier series. The solution at the $z = k\Delta z$ plane can be expressed as

$$u^k = g^k \, \mathrm{e}^{jk_x x} \, \mathrm{e}^{jk_y y} \tag{3.22}$$

where g is the growth factor. If $|g| \leq 1$, then the error is bounded and the finite-difference scheme is stable. Substituting Eqn. (3.22) into the finite-difference equation Eqn. (3.19), the expression of the growth factor for the CN method can be obtained as

$$|g| = \left| \frac{1 + \frac{r}{jk_0} \left[\sin^2 (k_x x) + \sin^2 (k_y y) \right]}{1 - \frac{r}{jk_0} \left[\sin^2 (k_x x) + \sin^2 (k_y y) \right]} \right| = 1 \tag{3.23}$$

where the growth factor is always bounded for any mesh ratio. Therefore, the CN method is unconditionally stable.

3.3.2 Alternating-Direction-Implicit Method

Since the update matrix for the CN method is not tridiagonal, the computational cost of this method becomes significantly high in electrically large problems, motivating the investigation on more efficient alternative techniques. Recently, a computationally more efficient approach, the Alternating-Direction-Implicit (ADI) method, has been successfully introduced to the PE-based modeling of tunnel propagation [40, 41, 43].

The ADI method is a simple modification of the CN method and is developed to address the problem of computational complexity [15]. It is also unconditionally stable and second-order accurate in x-, y- and z-directions. However, it reduces the computational cost substantially by splitting the one-step CN equation into two steps, where the fields along x- and y-directions are solved consecutively in the first and second step. Compared to the CN method, each step of the ADI method generates much smaller tridiagonal matrices which can be inverted more efficiently.

Formulation

The ADI formulation for the parabolic equation can be obtained directly by adding $r_x r_y/(4jk_0)^2 \delta_x \delta_y$ to Eqn. (3.19), resulting in

$$\begin{aligned}\left[1-\frac{r_x}{4jk_0}\delta_x-\frac{r_y}{4jk_0}\delta_y+\frac{r_x r_y}{(4jk_0)^2}\delta_x\delta_y\right]u_{i,j}^{k+1} \\ =\left[1+\frac{r_x}{4jk_0}\delta_x+\frac{r_y}{4jk_0}\delta_y+\frac{r_x r_y}{(4jk_0)^2}\delta_x\delta_y\right]u_{i,j}^{k}+\frac{r_x r_y}{(4jk_0)^2}\delta_x\delta_y\left(u_{i,j}^{k+1}-u_{i,j}^{k}\right) \\ +\mathcal{O}(\Delta z^3)+\mathcal{O}(\Delta z\Delta x^2)+\mathcal{O}(\Delta z\Delta y^2)\ \underbrace{+\mathcal{O}(\Delta z^3)}_{\text{new error term}}\end{aligned} \tag{3.24}$$

The ADI method introduces a new error term $\mathcal{O}(\Delta z^3)$, which is of the same order as the other three error terms. Eqn. (3.24) can be simplified and rewritten as

$$\begin{aligned}\left(1-\frac{r_x}{4jk_0}\delta_x\right)\left(1-\frac{r_y}{4jk_0}\delta_y\right)u_{i,j}^{k+1}=\left(1+\frac{r_x}{4jk_0}\delta_x\right)\left(1+\frac{r_y}{4jk_0}\delta_y\right)u_{i,j}^{k} \\ +\mathcal{O}(\Delta z^3)+\mathcal{O}(\Delta z\Delta x^2)+\mathcal{O}(\Delta z\Delta y^2)\ \underbrace{+\mathcal{O}(\Delta z^3)}_{\text{new error term}}\end{aligned} \tag{3.25}$$

Eqn. (3.25) can be split into two steps to obtain the fields for each direction separately. The commonly used splitting technique was first proposed by Peaceman and Rachford [41], which is also known as the Peaceman-Rachford (PR)-ADI method

$$\left(1-\frac{r_x}{4jk_0}\delta_x\right)\tilde{u}_{i,j}^{k+\frac{1}{2}}=\left(1+\frac{r_y}{4jk_0}\delta_y\right)u_{i,j}^{k} \tag{3.26a}$$

$$\left(1-\frac{r_y}{4jk_0}\delta_y\right)u_{i,j}^{k+1}=\left(1+\frac{r_x}{4jk_0}\delta_x\right)\tilde{u}_{i,j}^{k+\frac{1}{2}} \tag{3.26b}$$

where the index $k+1/2$ corresponds to an intermediate plane introduced between the $z=k\Delta z$ and $z=(k+1)\Delta z$ planes.

Note that Eqn. (3.26a) is implicit in the x-direction and explicit in the y-direction while Eqn. (3.26b) is implicit in the y-direction and explicit in the x-direction, as illustrated in Fig. 3.2. The element values on the intermediate plane are updated line by line along the x-direction. After all the element values are generated for the intermediate plane, then the element values on the $z=(k+1)\Delta z$ plane are updated line by line along the y-direction.

An alternative way to split Eqn. (3.25) is the Locally-One-Dimensional (LOD)-ADI method [42].

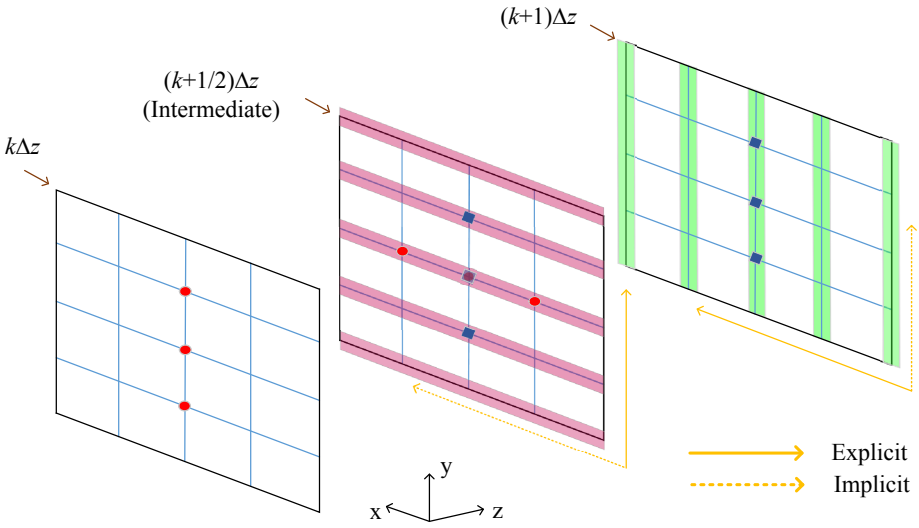


Figure 3.2: The three-dimensional grid for the ADI method. The red circular dots represent the elements used to update Eqn. (3.26a) at a time. The blue square dots represent the nodes used to update Eqn. (3.26b) at a time.

The LOD method update equations are similar to the PR-ADI method. However, for this "locally-one-dimensional" scheme, each sub-step includes spatial finite differences with respect to either x or y only. The update equations are:

$$\left(1-\frac{r_y}{4jk_0}\delta_y\right)\tilde{u}_{i,j}^{k+\frac{1}{2}}=\left(1+\frac{r_y}{4jk_0}\delta_y\right)u_{i,j}^{k} \tag{3.27a}$$

$$\left(1-\frac{r_x}{4jk_0}\delta_x\right)u_{i,j}^{k+1}=\left(1+\frac{r_x}{4jk_0}\delta_x\right)\tilde{u}_{i,j}^{k+\frac{1}{2}} \tag{3.27b}$$

Evidently, Eqn. (3.27a) includes only finite-differences with respect to y, whereas Eqn. (3.27b) includes only finite-differences with respect to x. This "locally-one-dimensional" nature of the method is depicted in Fig. 3.3.

Note that the sequence to update the nodes is in the same direction for each equation. Generally, for equations with time variable and three-direction spatial variables involved, the LOD scheme splits the whole equation into three sub-equations, where only one-direction variable is updated at each step, and it saves significant computational cost. However, Eqn. (3.6) cannot be split into three sub-equations independently with only spatial variable in one-direction involved. It is similar to a "pseudo" LOD scheme and the computational effort is similar as the PR-ADI method.

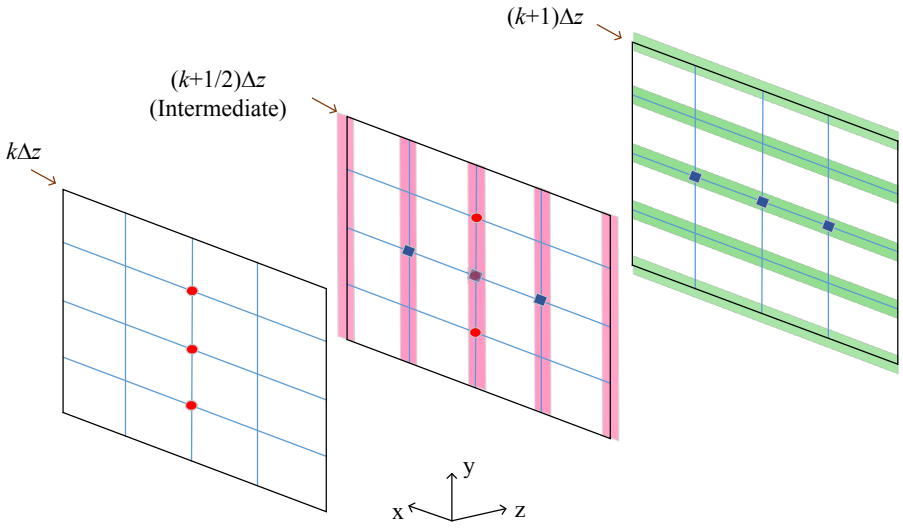


Figure 3.3: The three-dimensional grid for the LOD method. Similar to the PR-ADI method, the red circular dots represent the nodes used to update Eqn. (3.27a) at a time and the blue square dots represent the nodes used to update Eqn. (3.27b) at a time.

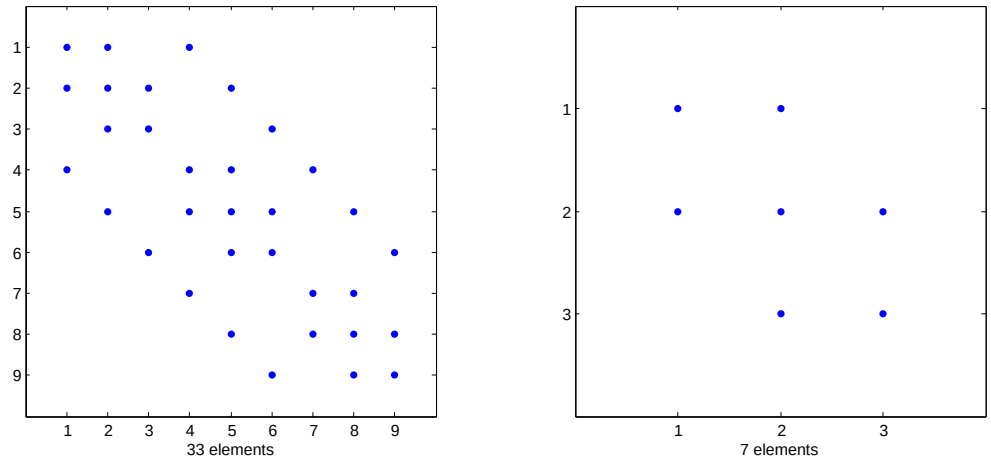
Stability

The PR-ADI and LOD are also unconditionally stable [37], which can easily be validated using the von Neumann method. The process is similar to the CN method and by substituting Eqn. (3.22) into the finite-difference equation Eqn. (3.25), the expression of the growth factor can be obtained, where the growth factor is bounded for any mesh ratio.

$$|g| = \left| \frac{\left(1 + \frac{r}{jk_0} \sin^2 \frac{k_x x}{2}\right)\left(1 + \frac{r}{jk_0} \sin^2 \frac{k_y y}{2}\right)}{\left(1 - \frac{r}{jk_0} \sin^2 \frac{k_x x}{2}\right)\left(1 - \frac{r}{jk_0} \sin^2 \frac{k_y y}{2}\right)} \right| = 1 \tag{3.28}$$

3.3.3 Computational Efficiency Comparison

The ADI method not only remains unconditionally stable and second-order accurate as the CN method, but also reduces the computational cost substantially by splitting the single-step CN equation into two steps, where the fields along x- and y-directions are solved consecutively in the first and second step. For a simple comparison of the computational efficiency between the CN method and the PR-ADI (LOD) method, when $N_x = N_y = 5$ the matrix element configurations of the two methods are compared in Fig. 3.4, where N_x and N_y are, respectively, the number of sampling points along x and y-direction.



(a) CN (b) ADI

Figure 3.4: Typical elements configuration for CN and ADI methods.

As can be seen from the graphs, even for a very coarse mesh, the matrix for the CN method is still significantly larger that the matrix of the ADI method: there are 33 elements in the 9×9 matrix for the CN method and 7 elements in the 3×3 matrix for the ADI method. Moreover, the updating matrix for the CN method is not in tridiagonal form. Therefore, the ADI method is computationally more efficient than the CN method for problems requiring fine meshes or when a large number of propagation steps is needed.

3.4 Numerical Investigations

In this section, the scalar parabolic equation method is applied to the case of a rectangular tunnel with perfectly conducting walls, where simple Dirichlet and Neumann boundary conditions are enforced. Subsequently, an actual tunnel case with lossy walls is considered. Since the CN and ADI methods are of the same order of accuracy, yet the former is far less efficient than the latter, here only the ADI method is simulated and validated against measured data. The comparative study of different numerical PE schemes applied to tunnel propagation modeling will be discussed in detail in Chapter 5.

3.4.1 Scalar Parabolic Equation Methods

The accuracy of the scalar ADI-PE method is assessed by comparing numerical simulations of rectangular tunnels with known analytical solutions. The tunnel cross-section dimensions are $40\lambda \times 40\lambda$ at the maximum operating frequency of 3 GHz, as illustrated in Fig. 3.5. The simulated distance is 100 m.

The transverse plane discretizations are chosen as $\Delta x = \Delta y = 0.4\lambda$, and for the propagation direction, $\Delta z = 5\lambda$ is adopted. A unit-strength Gaussian source with standard deviation (in the space domain) of 3.5λ is placed at the center of the transverse plane as an initial condition. With these parameters, the propagating fields in the tunnel will mainly consist of lower-order waveguide modes. Hence, the basic assumption for the applicability of the parabolic equation is satisfied.

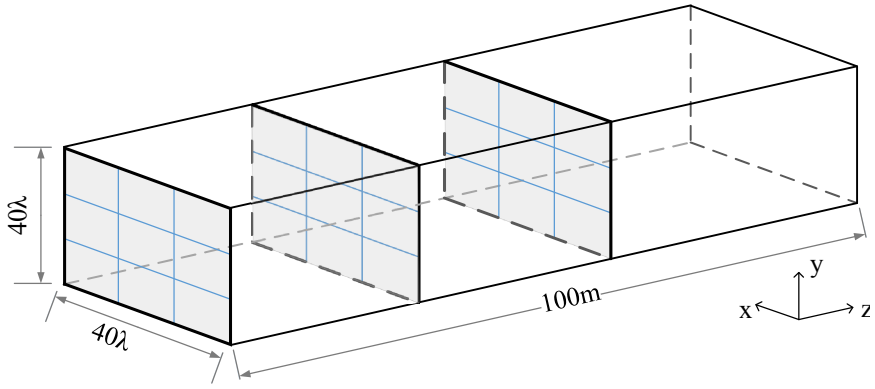


Figure 3.5: Geometry of the three-dimensional rectangular tunnel.

By enforcing either the Dirichlet or Neumann boundary conditions, the magnetic or electric scalar potential is simulated. Their analytical expressions are available and the results for this problem are compared to those obtained from the ADI method, via the relative error in the Euclidean norm

$$E_{\text{rms}} = \frac{\sqrt{\frac{1}{N}\sum_i \sum_j \left| u_{i,j}^{k_f,\text{num}} - u_{i,j}^{k_f,\text{exact}} \right|^2}}{\sqrt{\frac{1}{N}\sum_i \sum_j \left| u_{i,j}^{k_f,\text{exact}} \right|^2}} \tag{3.29}$$

where u^{num} denotes numerically and u^{exact} denotes analytically computed fields, respectively; N is the number of the discretization points on the cross-section at $z = k_f \Delta z$ and k_f is the sequence for the plane along z-direction.

Dirichlet Boundary Conditions

The propagating field will be of the TM type when the Dirichlet boundary conditions are enforced on the walls, and if the Neumann Boudanry Conditions are enforced, the propagating field will be of the TE type. The analytical solutions of the scalar potential for a rectangular waveguide with Dirichlet

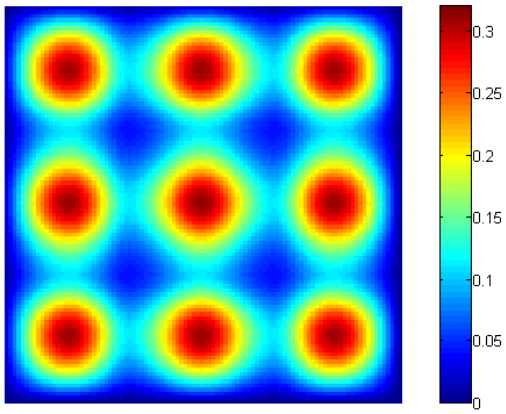
boundary conditions are given by

$$\phi^{\mathrm{m}} = \sum_{m=1}^{\infty}\sum_{l=1}^{\infty} B_{m,l}\,\sin\frac{m\pi x}{a}\sin\frac{l\pi y}{b}\,\mathrm{e}^{-jk_z z} \tag{3.30}$$

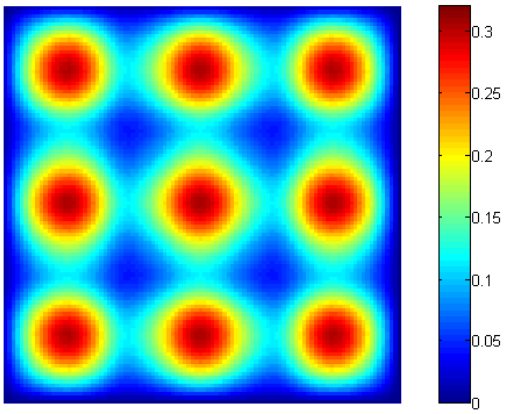
$$B_{m,l} = \frac{8\pi\sigma^2}{a\,b}\sin\frac{m\pi}{2}\sin\frac{l\pi}{2}\,\mathrm{e}^{-\frac{\left(a^2m^2+b^2l^2\right)\pi^2\sigma^2}{2a^2b^2}} \tag{3.31}$$

where a and b are, respectively, the width and height of a rectangular tunnel, and m and l specifies the mode indices and represent field variations along the x and y-directions, respectively.

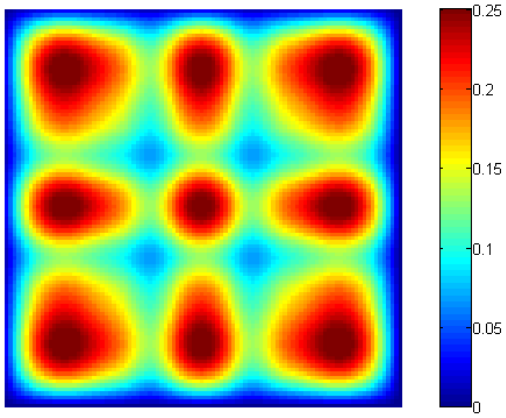
The analytical and numerical transverse-field patterns are compared in Fig. 3.6. As can be seen, the field patterns obtained using the ADI method agree well with the analytical results.



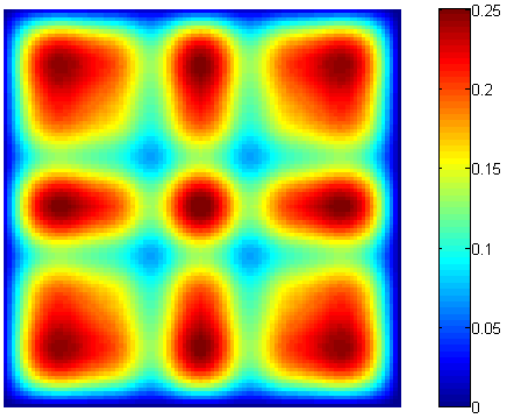
(a) 50 m: analytical solution



(b) 50 m: ADI method (rms error 2.47%)



(c) 100 m: analytical solution



(d) 100 m: ADI method (rms error 4.92%)

Figure 3.6: Plane wave field patterns at 50 m and 100 m from the initial plane in a rectangular tunnel with Dirichlet boundary conditions. The relative error is calculated through the Euclidean norm Eqn. (3.29).

Neumann Boundary Conditions

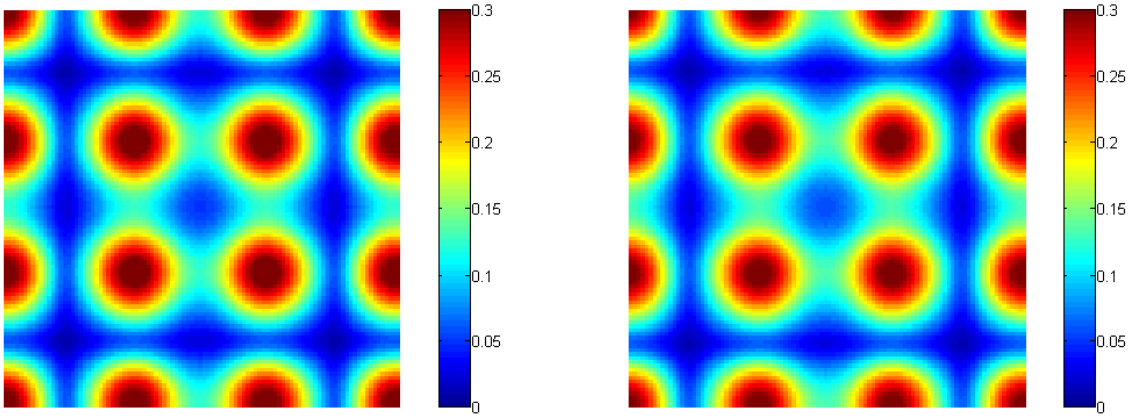
The analytical solutions of the scalar potential for the rectangular waveguide with Neumann boundary conditions are given by

$$\phi^{\mathrm{e}} = \sum_{m=0}^{\infty} \sum_{l=0}^{\infty} c_{m,l}\, B_{m,l}\, \cos\frac{m\pi x}{a} \cos\frac{l\pi y}{b}\, \mathrm{e}^{-jk_z z} \tag{3.32}$$

$$B_{m,l} = \frac{8\pi\sigma^2}{a\,b} \cos\frac{m\pi}{2} \cos\frac{l\pi}{2}\, \mathrm{e}^{-\frac{\left(a^2m^2+b^2l^2\right)\pi^2\sigma^2}{2a^2b^2}} \tag{3.33}$$

where $c_{\,0,\,0} = 0.25, c_{m\neq 0,\,0} = c_{\,0,\,l\neq 0} = 0.5$, and $c_{m\neq 0,\,l\neq 0} = 1$.

Fig. 3.7 shows the analytical and numerical transverse-field patterns in the rectangular tunnel with Neumann boundary conditions, where again the simulated and analytical patterns are in good agreement.



(a) 50 m: analytical solution

(b) 50 m: ADI method (rms error 3.13%)

(c) 100 m: analytical solution

(d) 100 m: ADI method (rms error 7.09%)

Figure 3.7: Plane wave field patterns at 50 m and 100 m from the initial plane in a rectangular tunnel with Neumann boundary conditions. The relative error is calculated through the Euclidean norm Eqn. (3.29).

3.4.2 Vector Parabolic Equation Methods

In actual tunnels, the walls are typically composed of certain lossy materials, which are approximated using the impedance boundary conditions. Tunnels with impedance walls can be very difficult to model and closed form solutions are not available [48]. Wait [49] has shown and Mahmoud [50] has verified that for a rectangular tunnel surrounded by lossy medium, an exact analytical solution is not possible. However, useful approximate solutions do exist and have been elaborated by Laakmann *et al.* [51] and Mahmoud [52]. In this section, the accuracy of the vector ADI-PE method is validated by comparing numerical simulations with approximate analytical solutions as well as the measured data of the Massif Central tunnel at south-central France.

Approximate Analytical Solutions

The impedance boundary conditions are applied to model lossy tunnels by treating the walls as imperfect metal with relative surface impedance of a good conductor. If the magnitude of this relative surface impedance is much less than unity, then perturbation techniques can be adopted for the derivation of the approximate fields [48]. Within such structures, instead of pure TE or TM modes, hybrid modes are supported for propagation, which can be divided into the x-polarized hybrid modes, $EH^x_{m,l}$, and y-polarized hybrid modes, $EH^y_{m,l}$ [51].

Considering the case where all the four walls of the tunnel are characterized by a surface impedance, field component $E(x, y, z)$ can be expressed as a modal sum as [10]

$$E(x, y, z) = \sum_{m=1}^{\infty} \sum_{l=1}^{\infty} A_{m,l}\, e_{m,l}(x, y)\, \mathrm{e}^{-\gamma_{m,l}\, z} \tag{3.34}$$

where $A_{m,l}$ is the complex amplitude, $e_{m,l}(x, y)$ is the modal eigenfunction, and $\gamma_{m,l}$ is the complex propagation constant, given by

$$\gamma_{m,l} = \alpha_{m,l} + j\beta_{m,l} \tag{3.35}$$

where $\alpha_{m,l}$ and $\beta_{m,l}$ are the attenuation and propagation constants, respectively.

For the case of a vertical (y-direction) dipole as the source, y-polarized hybrid modes are excited and based on the assumptions

$$\left(\frac{m\pi}{a}\right)^2 + \left(\frac{l\pi}{b}\right)^2 \ll k_0^2 \tag{3.36a}$$

$$\left|\frac{\varepsilon_{rc}}{\varepsilon_0}-1\right|^{\frac{1}{2}} \gg \frac{m\lambda}{2a} \tag{3.36b}$$

$$\frac{\varepsilon_0}{\varepsilon_{rc}}\left|\frac{\varepsilon_{rc}}{\varepsilon_0}-1\right|^{\frac{1}{2}} \gg \frac{l\lambda}{2b} \tag{3.36c}$$

the approximate fields are given by [51]

$$\begin{aligned} e_{m,l}(x,y) = C_0 &\left\{\cos\left(\frac{m\pi x}{a}+\varphi_x\right) - \sin\left[\frac{2j}{k_0 a\sqrt{\varepsilon_{rc}}}\left(\frac{m\pi x}{a}\right)\right]\sin\left(\frac{m\pi x}{a}+\varphi_x\right)\right\} \\ &\left\{\sin\left(\frac{l\pi y}{b}+\varphi_y\right) - \sin\left[\frac{2j}{k_0 b\sqrt{\varepsilon_{rc}}}\left(\frac{l\pi y}{b}\right)\right]\sin\left(\frac{l\pi y}{b}+\varphi_y\right)\right\} \end{aligned} \tag{3.37}$$

where C_0 is a normalization constant, and φ_x and φ_y are

$$\varphi_x = \begin{cases} \pi/2, & \text{if } m \text{ is even} \\ 0, & \text{if } m \text{ is odd} \end{cases}$$

$$\varphi_y = \begin{cases} 0, & \text{if } l \text{ is even} \\ \pi/2, & \text{if } l \text{ is odd} \end{cases}$$

The approximate analytical expressions for the attenuation and propagation constants can be obtained and are given by [51]

$$\alpha_{m,l} = \frac{2}{a}\left(\frac{m\lambda}{2a}\right)^2 \mathrm{Re}\left(\frac{1}{\sqrt{\varepsilon_{rc}-1}}\right) + \frac{2}{b}\left(\frac{l\,\lambda}{2b}\right)^2 \mathrm{Re}\left(\frac{\varepsilon_{rc}}{\sqrt{\varepsilon_{rc}-1}}\right) \tag{3.38}$$

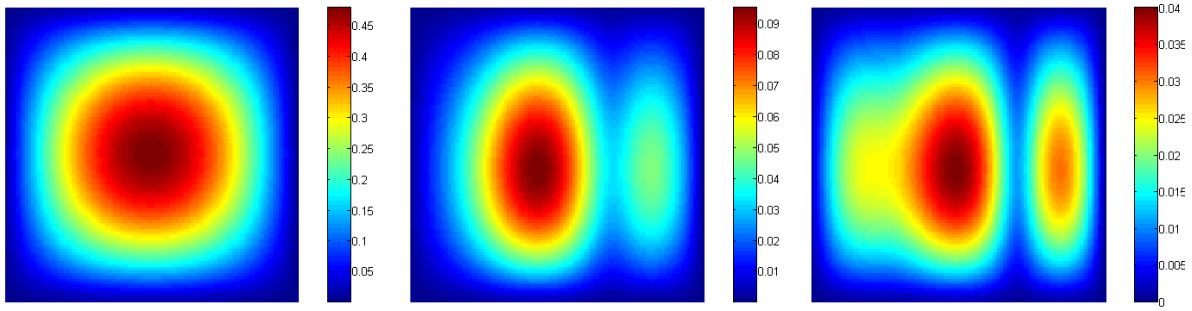
$$\beta_{m,l} = \frac{2\pi}{\lambda}\left[1 - \frac{1}{2}\left(\frac{m\lambda}{2a}\right)^2 - \frac{1}{2}\left(\frac{l\,\lambda}{2b}\right)^2\right] \tag{3.39}$$

and similar expressions for the x-polarized hybrid modes and the corresponding fields can be found in [52]

Here, the accuracy of the vector ADI-PE method is validated by modeling wave propagation in a rectangular tunnel with impedance boundary conditions, and the numerical results are compared to the approximate analytical solutions. To guarantee that the approximate analytical solution provides a valid and accurate reference, the tunnel cross-section dimensions are $40\lambda \times 40\lambda$ at the maximum operating frequency of 2.4 GHz and the electrical parameters are chosen as $\varepsilon_r = 12$ and $\sigma_0 = 0.01\,\mathrm{S/m}$. With these parameters, the assumptions for these analytical expressions are satisfied.

For the transverse plane discretization, $\Delta x = \Delta y = 0.4\lambda$, and for the propagation direction, $\Delta z = 8\lambda$. The simulated distance is 3 km and the y-polarized hybrid fields are adopted as the initial source, where a superposition of different number of modes (1, 16, and 40 modes with equal weight) are included as

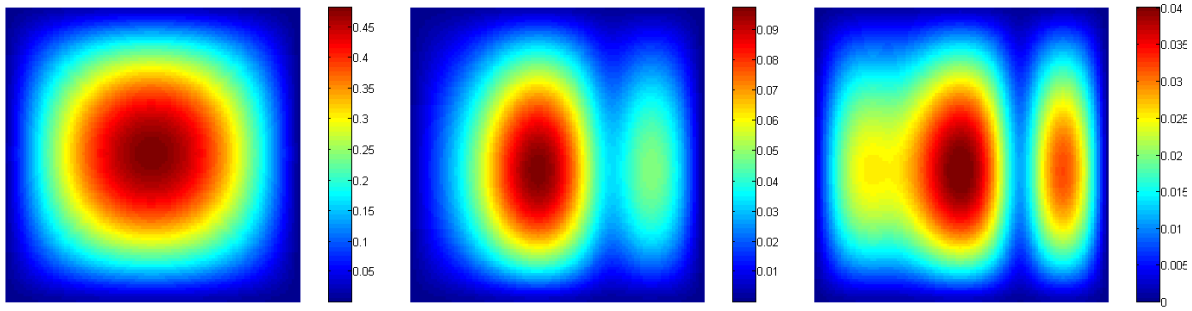
the initial condition. Fig. 3.8 and Fig. 3.9 show the analytical and numerical transverse-field patterns and the received power.



(a) 1 mode: analytical

(c) 16 modes: analytical

(e) 40 modes: analytical



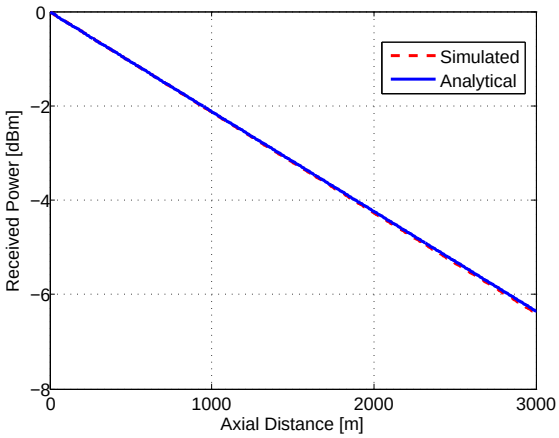
(b) 1 mode: ADI method (rms error 3.07%)

(d) 16 modes: ADI method (rms error 6.13%)

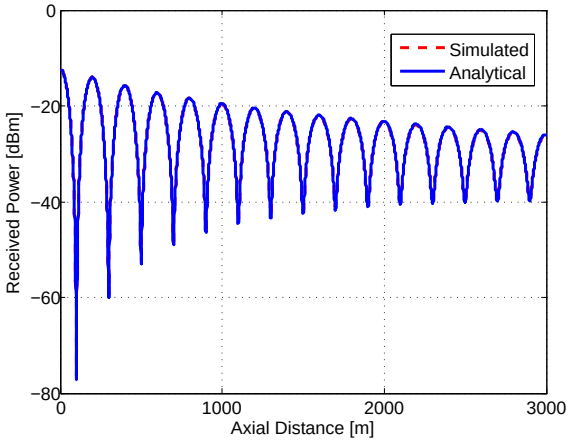
(f) 40 modes: ADI method (rms error 10.44%)

Figure 3.8: Plane wave field patterns at 3 km from the initial plane in a rectangular tunnel with lossy walls. The initial plane respectively includes a superposition of 1, 16, and 40 number of modes with equal weight as the source, and the relative error is calculated through the Euclidean norm Eqn. (3.29).

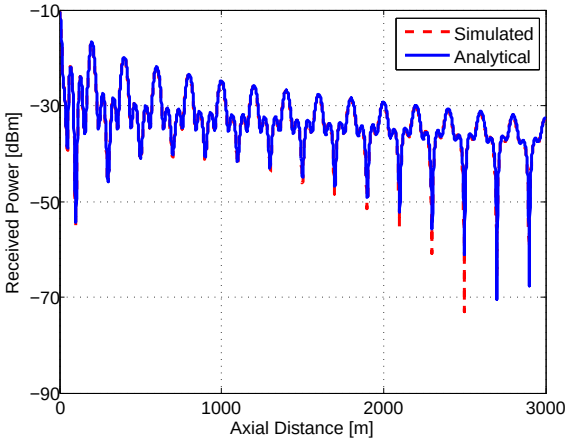
Evidently, the numerical results can capture the field pattern and the received power profiles are in good agreement with the analytical ones, even when lots of higher-order modes are excited. This is because higher-order modes are more severely attenuated over large distances due to multiple reflections, while lower-order modes, which propagate at small angles relative to the axis, make the most significant contribution to the field over large distances. It can also be seen from Eqn. (3.38) that attenuation constants of higher-order modes are much larger than those of lower-order modes. Therefore, the VPE method, which is based on paraxial propagation, is an efficient and accurate method for wave propagation modeling in long, relatively uniform tunnels with lossy walls.



(a) 1 mode



(b) 16 modes



(c) 40 modes

Figure 3.9: Received power at 2400 MHz and the initial plane respectively includes a superposition of 1, 16, and 40 number of modes with equal weight as the source.

Measured Data of an Actual Tunnel

To further validate the accuracy of the vector ADI-PE method, it is applied to simulate wave propagation in the Massif Central tunnel, as studied in the previous chapter. For the transverse plane discretization, $\Delta x = \Delta y = 0.3\lambda$ is adopted, and for the propagation direction, $\Delta z = \lambda$ is adopted. The simulated results for the received power at 900 MHz and 2100 MHz are compared to both the measured data and the ray-tracing simulation with a level of 25 reflections.

As illustrated in Figs. 3.10 and 3.11, the near zone of the tunnel is characterized by rapid fluctuations caused by the interaction of many modes, and as higher-order modes attenuate more severely, the fields in the far zone consist primarily of the first few lower-order modes. Moreover, it can been seen that the

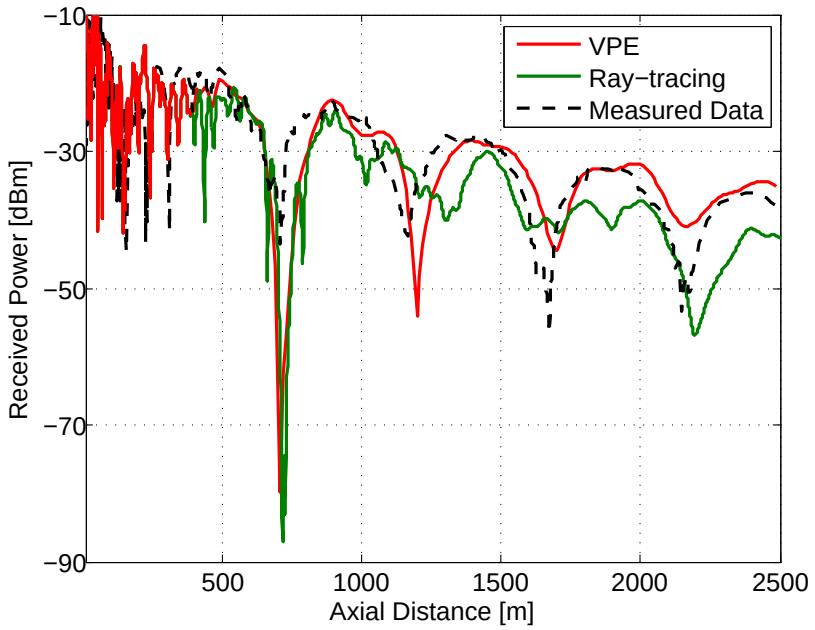


Figure 3.10: Received power at 900 MHz over 2.5 km. The equivalent cross-section dimensions of 7.8 m × 5.3 m are adopted, and the electrical parameters are chosen as $\varepsilon_r = 5$ and $\sigma_0 = 0.01\,\mathrm{S/m}$. Ray-tracing is applied to generate initial source fields for the VPE and the source plane is set at the 400 m.

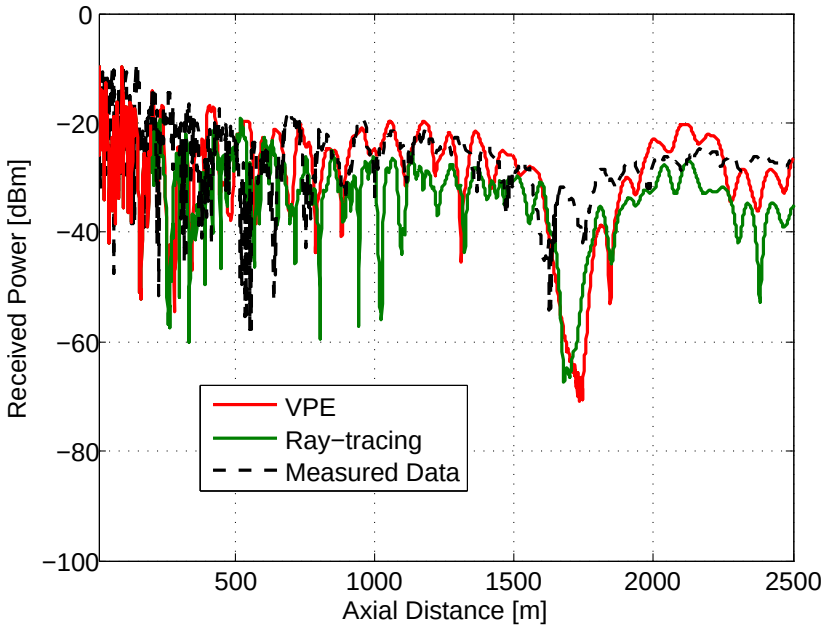


Figure 3.11: Received power at 2100 MHz over 2.5 km. The equivalent cross-section dimensions of 7.8 m × 5.3 m are adopted, and the electrical parameters are chosen as $\varepsilon_r = 5$ and $\sigma_0 = 0.01\,\mathrm{S/m}$. Ray-tracing is applied to generate initial source fields for the VPE and the source plane is set at the 200 m.

simulated results of VPE are in better agreement with the measured data than ray-tracing. Moreover, compared to ray-tracing, with which the simulation took more than 21 hours running on a workstation for the 900 MHz case, the vector ADI-PE took less than 18 seconds, corresponding to a speed up of at least a factor of 4000. On the whole, the VPE method is a more efficient and accurate approach for the wave propagation modeling of long tunnels.

3.5 Conclusion

This chapter introduced the fundamentals of the parabolic approximation theory. The formulations for both the scalar and vector parabolic equations were presented and the impedance boundary condition which is coupled with the latter for realistic tunnel applications was described. Moreover, the commonly adopted finite-difference schemes, CN and ADI, were discussed and compared. At last, the parabolic methods were validated by the case study of a rectangular tunnel with perfectly conducting walls and an actual tunnel with lossy walls. The parabolic equation method has proved to be an effective and computationally efficient modeling approach for these long, relatively uniform guiding structures.

Chapter 4

A Higher-Accuracy ADI Scheme for the Parabolic Equation Method

4.1 Introduction

The traditional and widely adopted finite-difference method for parabolic equations is the CN scheme [37, 45]. Recently, the ADI method, which is computationally more efficient, has been successfully applied to the parabolic equation based modeling of tunnel propagation problems [15, 23]. Just as the ADI method is derived via a relatively simple modification of the CN scheme, ADI itself can be modified to significantly improve its accuracy, via a scheme that was originally introduced by Mitchell and Fairweather [53, 54]. To the best of our knowledge, this Mitchell-Fairweather (MF)-ADI scheme has not previously been applied to solving radio-wave propagation problems in wireless channels. This chapter is dedicated to demonstrating the advantages of this MF-ADI method over the classical formulation of the PR-ADI method.

In this chapter, first the formulation and stability of this high-accuracy MF-ADI method are discussed. Then, numerical verifications for its advantages over the standard PR-ADI method are accompanied by the case study of a rectangular tunnel with perfectly conducting walls and an actual tunnel geometry.

4.2 The MF-ADI Method

The MF-ADI method, first introduced by Mitchell and Fairweather in 1964 [53], is second-order accurate in the propagation direction but fourth-order accurate in the transverse directions. Compared to the standard PR-ADI method, this scheme provides significantly better accuracy at no additional memory or execution time requirements.

4.2.1 Formulation

Instead of the second-order approximation given in Eqn. (3.17), the MF-ADI employs the following fourth-order approximation for the second-order derivatives $\partial^2 u/\partial^2 \xi$, $\xi = \{x, y\}$, with respect to the transverse plane coordinates [53]

$$\frac{1}{\Delta\xi^2}\left(1+\frac{1}{12}\delta_\xi\right)^{-1}\delta_\xi u = \frac{\partial u^2}{\partial \xi^2} + \mathcal{O}\left(\Delta\xi^4\right) \tag{4.1}$$

Multiplying both sides of Eqn. (3.25) by the term

$$\left(1+\frac{1}{12}\delta_x\right)\left(1+\frac{1}{12}\delta_y\right) \tag{4.2}$$

leads to

$$\begin{aligned}\left(1+\frac{1}{12}\delta_x\right)\left(1+\frac{1}{12}\delta_y\right)\left(1-\frac{r_x}{4jk_0}\delta_x\right)\left(1-\frac{r_y}{4jk_0}\delta_y\right)u_{i,j}^{k+1} \\ = \left(1+\frac{1}{12}\delta_x\right)\left(1+\frac{1}{12}\delta_y\right)\left(1+\frac{r_x}{4jk_0}\delta_x\right)\left(1+\frac{r_y}{4jk_0}\delta_y\right)u_{i,j}^{k}\end{aligned} \tag{4.3}$$

which can be rewritten as

$$\begin{aligned}\left[1+\frac{1}{12}\delta_x-\frac{\Delta z}{4jk_0}\left(1+\frac{1}{12}\delta_x\right)\frac{\partial^2}{\partial x^2}\right]\left[1+\frac{1}{12}\delta_y-\frac{\Delta z}{4jk_0}\left(1+\frac{1}{12}\delta_y\right)\frac{\partial^2}{\partial y^2}\right]u_{i,j}^{k+1} \\ = \left[1+\frac{1}{12}\delta_x+\frac{\Delta z}{4jk_0}\left(1+\frac{1}{12}\delta_x\right)\frac{\partial^2}{\partial x^2}\right]\left[1+\frac{1}{12}\delta_y+\frac{\Delta z}{4jk_0}\left(1+\frac{1}{12}\delta_y\right)\frac{\partial^2}{\partial y^2}\right]u_{i,j}^{k}\end{aligned} \tag{4.4}$$

Subsequently, by replacing the term

$$\left(1+\frac{1}{12}\delta_\xi\right)\frac{\partial^2}{\partial \xi^2} \tag{4.5}$$

by $\delta_\xi/\Delta\xi^2 + \mathcal{O}(\Delta\xi^4)$, $\xi = \{x, y\}$, the following scheme is obtained

$$\left[1 - \left(\frac{r_x}{4jk_0} - \frac{1}{12}\right)\delta_x\right]\left[1 - \left(\frac{r_y}{4jk_0} - \frac{1}{12}\right)\delta_y\right] u_{i,j}^{k+1} = \left[1 + \left(\frac{r_y}{4jk_0} + \frac{1}{12}\right)\delta_y\right]\left[1 + \left(\frac{r_x}{4jk_0} + \frac{1}{12}\right)\delta_x\right] u_{i,j}^{k} + \mathcal{O}(\Delta z^3) + \mathcal{O}(\Delta z \Delta x^4) + \mathcal{O}(\Delta z \Delta y^4) \quad (4.6)$$

As can be seen from Eqn. (4.6), it is second-order accurate in z-direction and fourth-order accurate in the transverse x and y-directions. This formulation can also be split into two steps

$$\left[1 - \left(\frac{r_x}{4jk_0} - \frac{1}{12}\right)\delta_x\right] \tilde{u}_{i,j}^{k+\frac{1}{2}} = \left[1 + \left(\frac{r_y}{4jk_0} + \frac{1}{12}\right)\delta_y\right] u_{i,j}^{k} \quad (4.7a)$$

$$\left[1 - \left(\frac{r_y}{4jk_0} - \frac{1}{12}\right)\delta_y\right] u_{i,j}^{k+1} = \left[1 + \left(\frac{r_x}{4jk_0} + \frac{1}{12}\right)\delta_x\right] \tilde{u}_{i,j}^{k+\frac{1}{2}} \quad (4.7b)$$

Similar to the PR-ADI method, the resulting matrices for the MF-ADI method are tridiagonal and the overall computational cost remains the same. Hence, the improvement in this higher-order of accuracy is achieved without any overhead on operation count.

4.2.2 Stability

The MF-ADI method is also unconditionally stable [37], which can be easily validated using the von Neumann method. The process is similar to that of the PR-ADI method and by substituting Eqn. (3.22) into the finite-difference equation Eqn. (4.6), the expression of the growth factor can be obtained:

$$|g| = \left| \frac{\left[1 + 4\left(\frac{r}{4jk_0} + \frac{1}{12}\right)\sin^2\frac{k_x x}{2}\right]\left[1 + 4\left(\frac{r}{4jk_0} + \frac{1}{12}\right)\sin^2\frac{k_y y}{2}\right]}{\left[1 - 4\left(\frac{r}{4jk_0} - \frac{1}{12}\right)\sin^2\frac{k_x x}{2}\right]\left[1 - 4\left(\frac{r}{4jk_0} - \frac{1}{12}\right)\sin^2\frac{k_y y}{2}\right]} \right| \quad (4.8a)$$

$$= \left| \frac{\left[\left(1 + \frac{1}{3}\sin^2\frac{k_x x}{2}\right) + \frac{r}{jk_0}\sin^2\frac{k_x x}{2}\right]\left[\left(1 + \frac{1}{3}\sin^2\frac{k_y y}{2}\right) + \frac{r}{jk_0}\sin^2\frac{k_y y}{2}\right]}{\left[\left(1 + \frac{1}{3}\sin^2\frac{k_x x}{2}\right) - \frac{r}{jk_0}\sin^2\frac{k_x x}{2}\right]\left[\left(1 + \frac{1}{3}\sin^2\frac{k_y y}{2}\right) - \frac{r}{jk_0}\sin^2\frac{k_y y}{2}\right]} \right| \quad (4.8b)$$

$$= 1 \quad (4.8c)$$

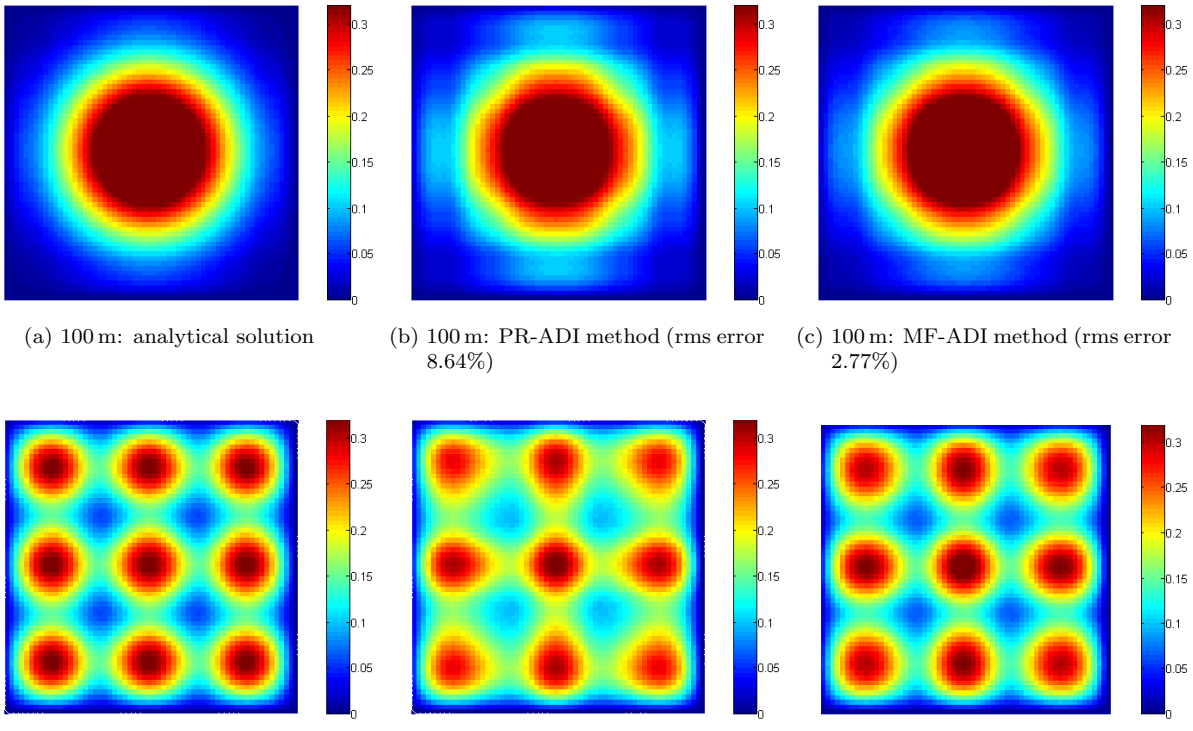
where the growth factor is always bounded for any mesh ratio. Therefore, the MF-ADI method is unconditionally stable.

4.3 Comparisons

To verify the advantages of this MF-ADI method over the classical PR-ADI method for wave propagation problems in tunnel scenarios, these two methods are applied to and compared for the case of a rectangular tunnel with perfectly conducting walls (Dirichlet boundary conditions); subsequently, an actual tunnel case, with lossy walls, is considered.

4.3.1 Rectangular Tunnel with Perfectly Conducting Walls

The tunnel cross-section dimensions are $30\lambda \times 30\lambda$ at the maximum operating frequency of 2.4 GHz. The transverse-plane discretizations are chosen as $\Delta x = \Delta y = 0.5\lambda$, and in the propagation direction, $\Delta z = 2\lambda$. A unit-strength Gaussian source with standard deviation (in the space-domain) of 3λ is placed at the center of the transverse plane as an initial condition. Dirichlet boundary conditions are enforced on the rectangular tunnel walls. Analytical results for this problem are available and compared to those obtained from the PR-ADI and MF-ADI methods; the transverse-field patterns are shown in Fig. 4.1.



(a) 100 m: analytical solution (b) 100 m: PR-ADI method (rms error 8.64%) (c) 100 m: MF-ADI method (rms error 2.77%)

(d) 300 m: analytical solution (e) 300 m: PR-ADI method (rms error 23.02%) (f) 300 m: MF-ADI method (rms error 8.03%)

Figure 4.1: Plane-wave field patterns at 100 m and 300 m from the initial plane in a rectangular tunnel with Dirichlet boundary conditions.

Evidently, the MF-ADI method has a much better error performance than the PR-ADI method. To quantify this improvement, the error norm Eqn. (3.29) and CPU time for the two methods is plotted as a function of the discretization rate on the transverse plane (with $\Delta x = \Delta y$, $r_x = r_y = 10$) in Fig. 4.2. Since the rms error and CPU time vary with the propagation distance, they are both calculated for a tunnel length of one wavelength. The graphs in Fig. 4.2 confirm that MF-ADI outperforms PR-ADI and presents an asymptotic error behavior that is consistent with fourth-order accuracy.

Practically, as illustrated in Fig. 4.2, the MF-ADI method allows for a larger Δx, Δy to be chosen than the PR-ADI method. With the same error, this turns the error performance advantage into a significant computation time advantage for MF-ADI. Furthermore, Fig. 4.3 indicates that with a fixed computation time, MF-ADI method outperforms PR-ADI in terms of accuracy. Moreover, the figure shows that the errors of both methods converge to a floor, set by the error of the approximation of the wave equation Eqn. (3.1) with the parabolic equation Eqn. (3.6).

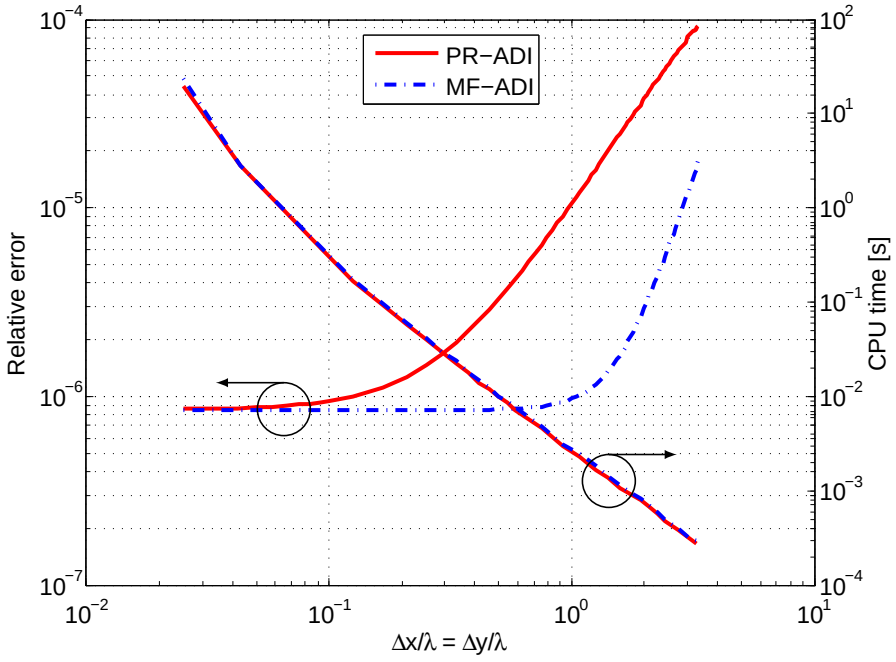


Figure 4.2: Performance comparison of accuracy and CPU time vs. Δx for the PR-ADI and the MF-ADI methods in a rectangular tunnel with perfectly conducting walls.

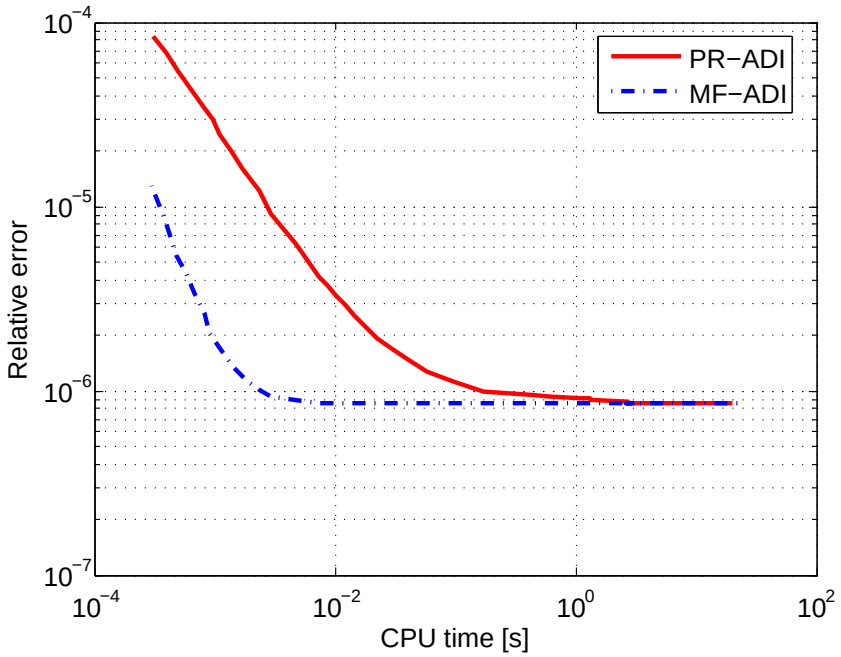


Figure 4.3: Performance comparison of accuracy vs. CPU time for the PR-ADI and the MF-ADI methods in a rectangular tunnel with perfectly conducting walls.

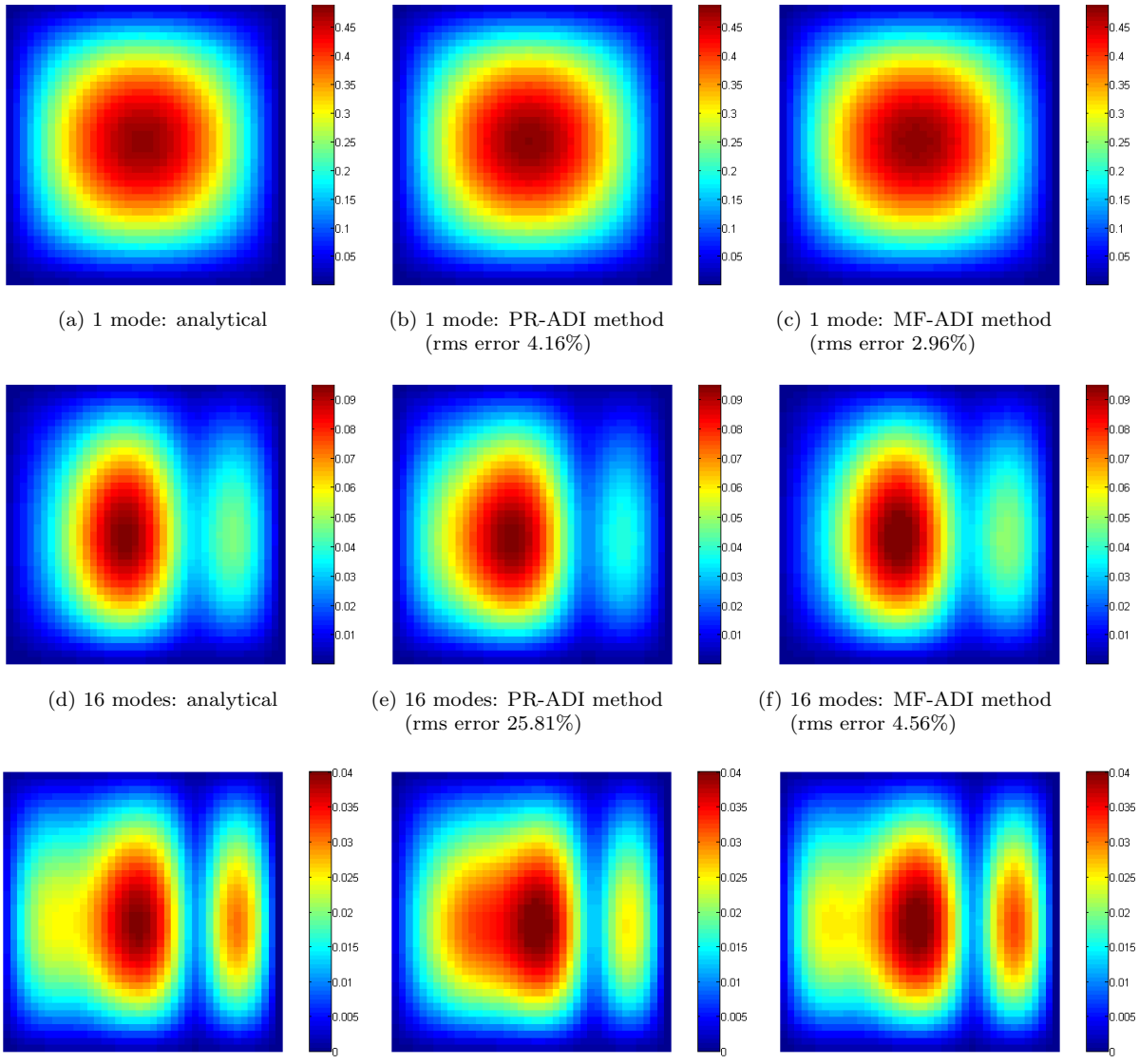
4.3.2 Actual Tunnels with Lossy Walls

To verify the advantages of this MF-ADI method over the PR-ADI method, wave propagation in an actual tunnel with lossy walls is considered for simulation. First, the simulated results of the PR-ADI and MF-ADI methods are compared to the approximate analytical solutions for this problem. Then, their performance comparison is verified with the measured data obtained from the Massif Central tunnel.

Analytical Solutions

The cross-section dimensions for this lossy tunnel are $40\lambda \times 40\lambda$. The maximum operating frequency is 2.4 GHz and the electrical parameters are chosen as $\varepsilon_r = 12$ and $\sigma_0 = 0.01\,\mathrm{S/m}$. To better show the outperformance of MF-ADI over PR-ADI, the spatial discretizations are selected as $\Delta x = \Delta y = \lambda$, and $\Delta z = 8\lambda$. The simulated distance is 3 km and the y-polarized hybrid fields are adopted as initial source, where a superpostion of a series number of modes, 1, 16, and 40, with equal weight, are included as the initial condition. The transverse-field patterns obtained from the two methods are compared to the analytical ones, as illustrated in Fig. 4.4.

Evidently, the error performance of the MF-ADI method is significantly better than that of the PR-ADI method. Moreover, even with such a coarse mesh discretization, the field-pattern results obtained



(a) 1 mode: analytical

(b) 1 mode: PR-ADI method (rms error 4.16%)

(c) 1 mode: MF-ADI method (rms error 2.96%)

(d) 16 modes: analytical

(e) 16 modes: PR-ADI method (rms error 25.81%)

(f) 16 modes: MF-ADI method (rms error 4.56%)

(g) 40 modes: analytical

(h) 40 modes: PR-ADI method (rms error 40.37%)

(i) 40 modes: MF-ADI method (rms error 6.03%)

Figure 4.4: Plane wave field patterns at 3 km from the initial plane in a rectangular tunnel with lossy walls. The simulated results from the MF-ADI and PR-ADI methods are compared to the analytical solution. The initial plane respectively includes a superposition of 1, 16, and 40 number of modes with equal weight as the source.

from MF-ADI still are in good agreement with the analytical ones, which shows the significantly better computational efficiency of the MF-ADI method.

Measurement

The two ADI methods are applied to the modeling of radio-wave propagation at 900 MHz in the Massif Central tunnel of south-central France, which has previously been used during validation of the ADI

method in Chapter 2. Here, the cell sizes are chosen as $\Delta x = \Delta y = 1.5\lambda$, and $\Delta z = 4\lambda$. The simulated distance is 2.5 km and the results are compared to measured data, presented in [10]. This comparison is depicted in Fig. 4.5.

The received power obtained using the MF-ADI method is in significantly better agreement with the measured data than the one obtained using the PR-ADI method. Therefore, this case study confirms the superior performance of MF-ADI, compared to its standard ADI counterpart.

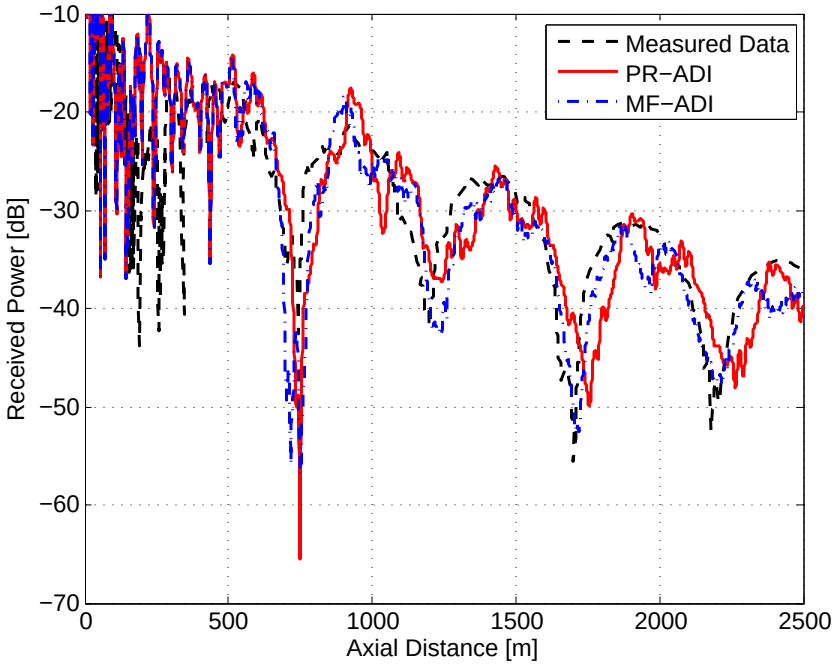


Figure 4.5: Performance comparison of the PR-ADI and the MF-ADI methods in the Massif Central tunnel with lossy walls. Ray-tracing is applied to generate initial source fields for the VPE and the source plane is set at the 500 m.

4.4 Conclusion

This chapter presented the formulation and application of the MF-ADI method to the solution of parabolic equations in tunnel propagation scenarios. The formulation itself and the numerical results indicated that this version of ADI method has substantial advantages over the traditional techniques that have been employed in such simulations over the past few years. Compared to the standard PR-ADI method, this scheme proves to offer significantly better accuracy at no additional memory or execution time requirements. Based on the existing PR-ADI solvers, only a straightforward modification is needed to implement this MF-ADI scheme and realize the performance gains that were noted in this chapter.

Chapter 5

Comparative Study of PE Methods for Tunnel Propagation Modeling

5.1 Introduction

Parabolic equation methods have been widely applied to the modeling of wireless propagation in railway or road tunnel environments, striking a good balance between accuracy and efficiency [21–23]. With the multitude of numerical methods for PE problems, a question that naturally arises is what is their relative performance and how can their parameters be pre-determined in order to control the numerical errors in the solution, coming both from numerical dispersion and from the PE approximation of the wave equation. Surprisingly, the existing literature contains little more than mere rules of thumb with regards to this question and there are no concrete guidelines for the choice of the parameters of these methods and the trade-offs involved.

Since PE is a paraxial approximation to the wave equation, therefore, it carries an approximation error, which is different from the numerical dispersion error that is commonly studied when finite-difference solvers for parabolic equations are considered. This chapter provides a comprehensive analysis of the two sources of error that arise when PE methods are employed for the modeling of radio-wave propagation scenarios: the well-known numerical dispersion error stemming from the finite-difference solvers for PE and the approximation error stemming from the use of PE for the solution of wave propagation problems that are subject to Maxwell's equations.

Moreover, a systematic study of the numerical error and the relative performance of the aforementioned four methods, CN, PR-ADI, LOD and MF-ADI, are presented. Specific guidelines on how to

choose their discretization parameters are formulated, leading to robust criteria in realistic propagation scenarios, as shown via numerical examples. These guidelines are successfully tested in specific tunnel cases, including Massif Central tunnel for which measured path-loss data are available.

5.2 Error Analysis and Comparative Study

In this section, the numerical errors involved in the application of the aforementioned four numerical schemes to wave propagation problems are analyzed and compared. This analysis is divided in two parts. First, an error analysis which includes both the standard dispersion error of parabolic solvers and the error introduced by the PE approximation of the wave equation is presented. The results of this analysis can indeed guide the choice of the cell sizes Δx, Δy and Δz for each scheme, as well as illustrate their relative performance. For completeness, the standard numerical dispersion analysis for parabolic equation solvers (i.e. one that simply focuses on the numerical dispersion error, disregarding the approximation error) is presented as well.

5.2.1 Complete Error Analysis

As discussed in Chapter 3, the exact solution of the scalar potential ϕ that satisfies the Helmholtz wave equation is Eqn. (3.7). By substituting the exact plane-wave solution Eqn. (3.8a), in a discrete-space form, into the four numerical schemes for the parabolic equation, the numerical wavenumber $\tilde{k}_0$ can be extracted. Note that the components of the numerical wavevector at a direction defined by the angles (φ, θ) of the spherical coordinate system (where θ is the angle with respect to the direction of propagation and φ is the azimuth angle) are given by:

$$\tilde{k}_x = \tilde{k}_0 \cos\varphi \sin\theta \tag{5.1a}$$

$$\tilde{k}_y = \tilde{k}_0 \sin\varphi \sin\theta \tag{5.1b}$$

$$\tilde{k}_z = \tilde{k}_0 \cos\theta \tag{5.1c}$$

The exact analytical k_0 is given by the consistency condition

$$k_0^2 = k_x^2 + k_y^2 + k_z^2 \tag{5.2}$$

and the analytical phase shift (in radians) per wavelength is $k_0\lambda = 2\pi$.

Then, the deviation of the numerical wavenumber $\tilde{k}_0$ from the actual wavenumber k_0 can be determined as a function of the cell size. In the following, this calculation is performed for each of the four numerical methods under consideration.

The CN Method

Substituting Eqn. (3.8a) into Eqn. (3.19), the following numerical dispersion relation for the CN method is obtained:

$$\tan\frac{(k_0-k_z)\Delta z}{2}=\frac{\Delta z}{k_0\Delta x^2}\sin^2\frac{k_x\Delta x}{2}+\frac{\Delta z}{k_0\Delta y^2}\sin^2\frac{k_y\Delta y}{2} \tag{5.3}$$

Letting $\Delta x=\Delta y$, $\alpha=\Delta z/\left(k_0\Delta x^2\right)$, and $N_\lambda=\lambda/\Delta x=2\pi/\left(k_0\Delta x\right)$ (where N_λ is a parameter controlling the number of nodes on the cross-section, as well as along the direction of propagation, per wavelength) and substituting Eqn. (5.1) into Eqn. (5.3), then the following equation is derived, with respect to the normalized numerical wavenumber $\tilde{k}_0\lambda$:

$$\tan\frac{\tilde{k}_0\lambda(1-\cos\theta)\alpha\pi}{N_\lambda^2}=\alpha\sin^2\frac{\tilde{k}_0\lambda\sin\varphi\sin\theta}{2N_\lambda}+\alpha\sin^2\frac{\tilde{k}_0\lambda\cos\varphi\sin\theta}{2N_\lambda} \tag{5.4}$$

This nonlinear equation is easily solved via the Newton-Raphson method. Since analytically $k_0\lambda=2\pi$, the absolute value of the relative error in the numerical wavenumber is:

$$\mathcal{E}=|\tilde{k}_0-k_0|/k_0 \tag{5.5}$$

Obviously, $\mathcal{E}$ produces a phase and magnitude error of the propagating wave along the direction of propagation. This metric is used as an indicator of the relative accuracy of the four methods under comparison.

The PR-ADI Method

Repeating the process of the previous section for the PR-ADI method, Eqn. (3.8a) is entered into Eqn. (3.26a) and Eqn. (3.26b). Then, the following numerical dispersion relations are obtained for the PR-ADI sub-steps:

$$\mathrm{e}^{-j\left(k_z-k_0\right)\Delta z_1}=\frac{1+j\frac{\Delta z}{k_0\Delta y^2}\sin^2\frac{k_y\Delta y}{2}}{1-j\frac{\Delta z}{k_0\Delta x^2}\sin^2\frac{k_x\Delta x}{2}} \tag{5.6a}$$

$$\mathrm{e}^{-j\left(k_z-k_0\right)\Delta z_2}=\frac{1+j\frac{\Delta z}{k_0\Delta x^2}\sin^2\frac{k_x\Delta x}{2}}{1-j\frac{\Delta z}{k_0\Delta y^2}\sin^2\frac{k_y\Delta y}{2}} \tag{5.6b}$$

where $\Delta z_1 = \Delta z_2 = \Delta z/2$. Further through similar steps, the normalized wavenumbers for the two sub-steps $k_{0,1}\lambda$ and $k_{0,2}\lambda$ are derived through the following expressions:

$$\mathrm{e}^{\frac{j\tilde{k}_{0,1}\lambda\left(1-\cos\theta\right)\alpha\pi}{N_\lambda^2}} = \frac{1+j\alpha\sin^2\frac{\tilde{k}_0\lambda\sin\varphi\sin\theta}{2N_\lambda}}{1-j\alpha\sin^2\frac{\tilde{k}_0\lambda\cos\varphi\sin\theta}{2N_\lambda}} \tag{5.7a}$$

$$\mathrm{e}^{\frac{j\tilde{k}_{0,2}\lambda\left(1-\cos\theta\right)\alpha\pi}{N_\lambda^2}} = \frac{1+j\alpha\sin^2\frac{\tilde{k}_0\lambda\cos\varphi\sin\theta}{2N_\lambda}}{1-j\alpha\sin^2\frac{\tilde{k}_0\lambda\sin\varphi\sin\theta}{2N_\lambda}} \tag{5.7b}$$

The LOD Method

Substituting Eqn. (3.8a) into Eqn. (3.27a) and Eqn. (3.27b), the following numerical dispersion relation for the LOD method is obtained:

$$\mathrm{e}^{-j\left(k_z-k_0\right)\Delta z_1} = \frac{1+j\frac{\Delta z}{k_0\Delta y^2}\sin^2\frac{k_y\Delta y}{2}}{1-j\frac{\Delta z}{k_0\Delta y^2}\sin^2\frac{k_y\Delta y}{2}} \tag{5.8a}$$

$$\mathrm{e}^{-j\left(k_z-k_0\right)\Delta z_2} = \frac{1+j\frac{\Delta z}{k_0\Delta x^2}\sin^2\frac{k_x\Delta x}{2}}{1-j\frac{\Delta z}{k_0\Delta x^2}\sin^2\frac{k_x\Delta x}{2}} \tag{5.8b}$$

yielding the following equations for the normalized wavenumbers of the two sub-steps (following the same notation as in the PR-ADI case):

$$\mathrm{e}^{\frac{j\tilde{k}_{0,1}\lambda\left(1-\cos\theta\right)\alpha\pi}{N_\lambda^2}} = \frac{1+j\alpha\sin^2\frac{\tilde{k}_0\lambda\sin\varphi\sin\theta}{2N_\lambda}}{1-j\alpha\sin^2\frac{\tilde{k}_0\lambda\sin\varphi\sin\theta}{2N_\lambda}} \tag{5.9a}$$

$$\mathrm{e}^{\frac{j\tilde{k}_{0,2}\lambda\left(1-\cos\theta\right)\alpha\pi}{N_\lambda^2}} = \frac{1+j\alpha\sin^2\frac{\tilde{k}_0\lambda\cos\varphi\sin\theta}{2N_\lambda}}{1-j\alpha\sin^2\frac{\tilde{k}_0\lambda\cos\varphi\sin\theta}{2N_\lambda}} \tag{5.9b}$$

The MF-ADI Method

Substituting Eqn. (3.8a) into Eqn. (4.7a) and Eqn. (4.7b), the following numerical dispersion relation is derived for the MF-ADI method:

$$\mathrm{e}^{-j\left(k_z-k_0\right)\Delta z_1} = \frac{1+\left(j\frac{\Delta z}{k_0\Delta y^2}-\frac{1}{3}\right)\sin^2\frac{k_y\Delta y}{2}}{1-\left(j\frac{\Delta z}{k_0\Delta x^2}+\frac{1}{3}\right)\sin^2\frac{k_x\Delta x}{2}} \tag{5.10a}$$

$$\mathrm{e}^{-j\left(k_z-k_0\right)\Delta z_2} = \frac{1+\left(j\frac{\Delta z}{k_0\Delta x^2}-\frac{1}{3}\right)\sin^2\frac{k_x\Delta x}{2}}{1-\left(j\frac{\Delta z}{k_0\Delta y^2}+\frac{1}{3}\right)\sin^2\frac{k_y\Delta y}{2}} \tag{5.10b}$$

Then, the normalized wavenumbers of the two sub-steps are:

$$\mathrm{e}^{\frac{j\tilde{k}_{0,1}\lambda\left(1-\cos\theta\right)\alpha\pi}{N_\lambda^2}} = \frac{1+\left(j\alpha-\frac{1}{3}\right)\sin^2\frac{\tilde{k}_0\lambda\sin\varphi\sin\theta}{2N_\lambda}}{1-\left(j\alpha+\frac{1}{3}\right)\sin^2\frac{\tilde{k}_0\lambda\cos\varphi\sin\theta}{2N_\lambda}} \tag{5.11a}$$

$$\mathrm{e}^{\frac{j\tilde{k}_{0,2}\lambda\left(1-\cos\theta\right)\alpha\pi}{N_\lambda^2}} = \frac{1+\left(j\alpha-\frac{1}{3}\right)\sin^2\frac{\tilde{k}_0\lambda\cos\varphi\sin\theta}{2N_\lambda}}{1-\left(j\alpha+\frac{1}{3}\right)\sin^2\frac{\tilde{k}_0\lambda\sin\varphi\sin\theta}{2N_\lambda}} \tag{5.11b}$$

Since for the PR-ADI, LOD, and MF-ADI schemes, the equations are split into two steps, the numerical phase per cell is the sum of the two steps,

$$\tilde{k}_0\Delta z = \left(\tilde{k}_{0,1}+\tilde{k}_{0,2}\right)\Delta z/2 \tag{5.12}$$

Numerical Results

Fig. 5.1 illustrates the phase error Eqn. (5.5) for the four numerical schemes as a function of the number of points per wavelength N_λ and the zenith angle θ, where $\alpha = 4$ and $\varphi = 45°$. It can be observed that

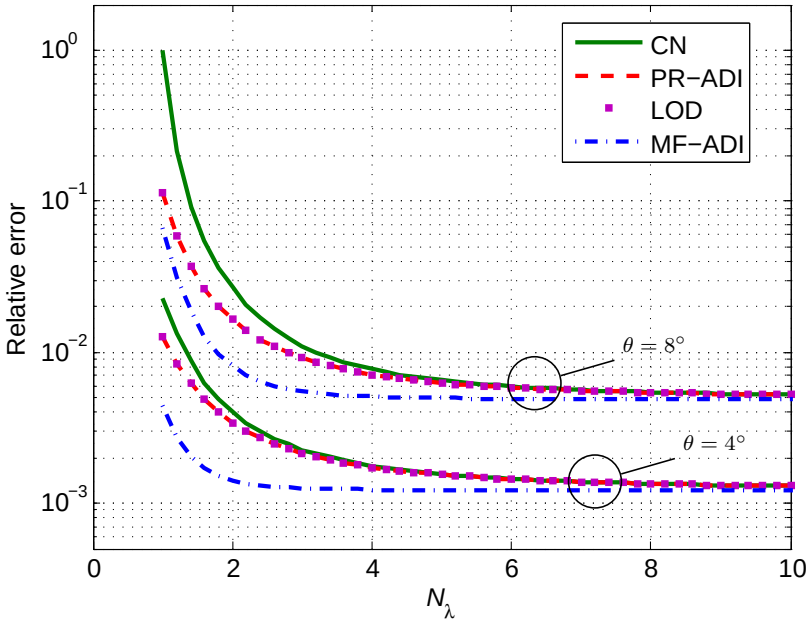


Figure 5.1: Comparison of dispersion phase error per λ vs. discretization between different numerical schemes.

the PR-ADI and LOD schemes have the same performance. The MF-ADI scheme has the lowest phase error, and the CN scheme has the largest phase error. Moreover, the phase error reduces as N_λ increases.

Asymptotically, all four methods converge to the same error floor, which represents the approximation error of the parabolic equation. This is an insightful result that cannot be extracted via conventional dispersion analysis of parabolic equation solvers and indicates the value of the proposed approach. As expected, this floor is reduced as the paraxial propagation assumption of the parabolic equation is better met. Note, for example, that this error floor is much lower (and close to zero) for $\theta = 4°$ than for $\theta = 8°$.

To better illustrate how the approximation error evolves as a function of the angle θ for a fixed finite-difference scheme and discretization rate, Fig. 5.2 shows the phase error with respect to θ, with $N_\lambda = 5$, $\alpha = 4$, $\varphi = 45°$. A general "rule of thumb", which was previously stated in [2], was that the wavevector should be confined within 15° from the z-axis for the PE approximation error to be acceptable. Notably, these results offer a more comprehensive, quantitative description of this error.

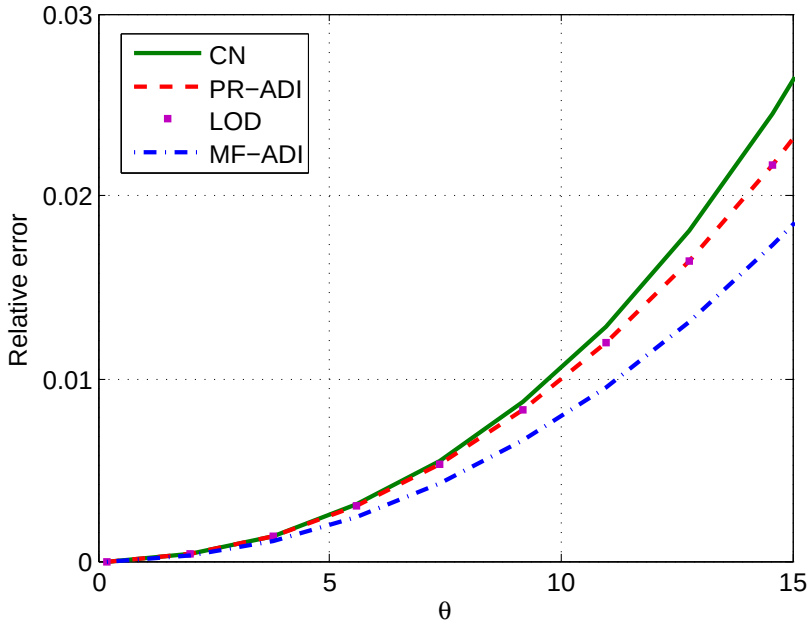


Figure 5.2: Comparison of dispersion phase error per λ vs. the angle θ (in deg) between different numerical schemes.

Fig. 5.3 illustrates the phase error as a function of α, where $N_\lambda = 5$, $\theta = 8°$, and $\varphi = 45°$. It can be seen that with increasing α, advantages for PR-ADI, LOD, and MF-ADI schemes over CN scheme become more obvious, which can be further proved by the discretization error analysis in Section 5.2.2.

Moreover, Fig. 5.4 depicts the phase error as a function of the azimuth angle φ (in deg), with $N_\lambda = 5$, $\alpha = 4$, and $\theta = 8°$. Minimum error is observed for wave propagation along the diagonal ($\varphi \approx 45°, 135°$). This effect is due to the numerical dispersion error of finite differences rather than the PE approximation and is encountered in other finite-difference methods as well, such as the Finite-Difference Time-Domain (FDTD) technique. Assume, for example, that a target error of 10^{-3} is set for a case when $\theta \leq 4°$. Then,

$N_\lambda \geq 5$ would allow the CN, PR-ADI and LOD methods to meet the target, whereas $N_\lambda > 2$ would suffice for the MF-ADI. Further calibration of the error characteristics of these methods is possible by tuning α.

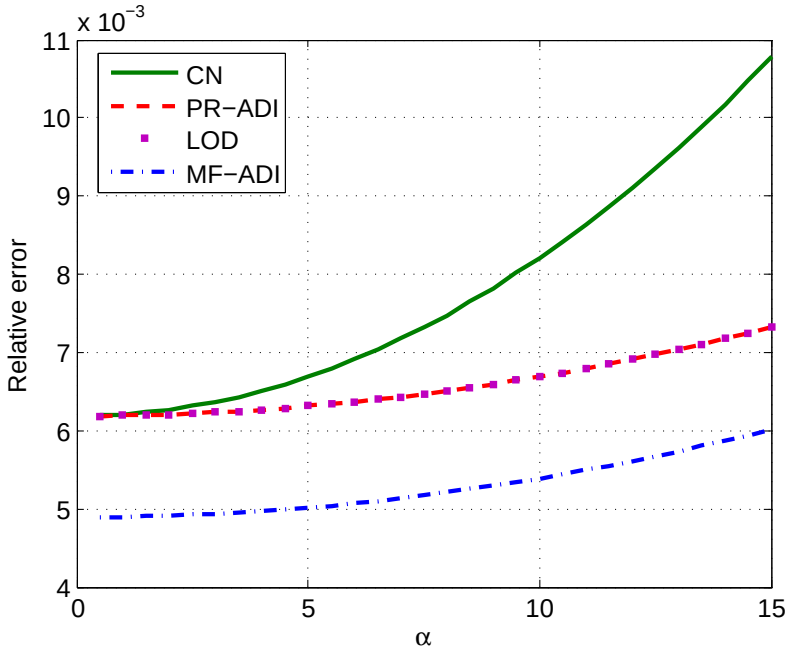


Figure 5.3: Comparison of dispersion phase error per λ vs. α between different numerical schemes.

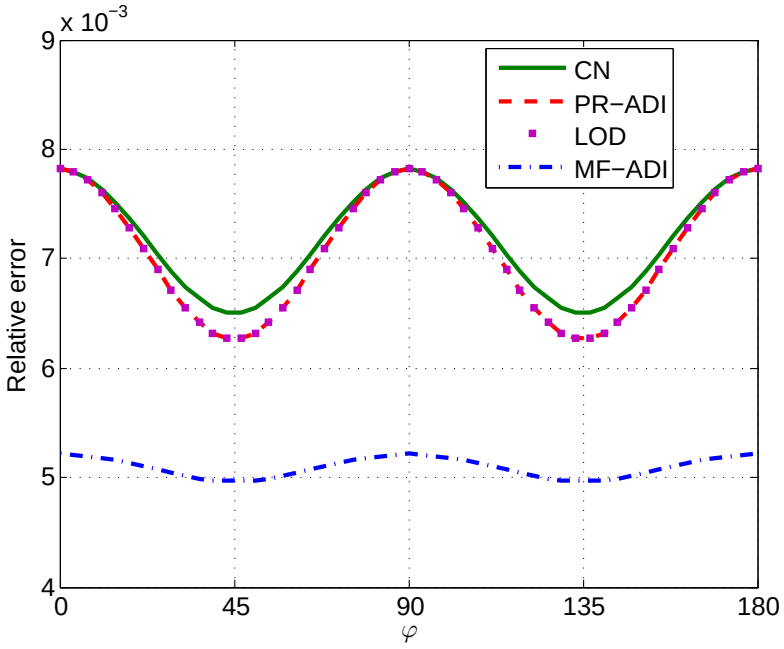


Figure 5.4: Comparison of dispersion phase error per λ vs. the azimuth angle φ (in deg) between different numerical schemes.

5.2.2 Numerical Dispersion Error Analysis

For completeness, we also present the standard numerical dispersion analysis of the four PE solvers under comparison, which isolates their truncation error from the PE approximation error. Through the method of separation of variables, the reduced plane wave solution

$$u(x, y, z) = X(x)Y(y)Z(z) \tag{5.13}$$

can be decomposed into

$$2jk_0\frac{Z'}{Z} = \frac{X''}{X} + \frac{Y''}{Y} = -k^2 \tag{5.14}$$

and by substituting it into Eqn. (3.6), the general form of the solution to the parabolic equation is obtained:

$$u(x, y, z) = C\,\mathrm{e}^{\pm jk_x x}\mathrm{e}^{\pm jk_y y}\mathrm{e}^{jk_z z} \tag{5.15}$$

where $k_x^2 + k_y^2 = k^2$, $k_z = k^2/(2k_0)$, and C is a constant coefficient. The theoretical dispersion relation for the parabolic equation Eqn. (3.6) is

$$2k_0k_z = k_x^2 + k_y^2 = k^2 \tag{5.16}$$

The truncation errors of the four schemes, defined as in [42], can then be obtained. The CN scheme is second-order accurate in x-, y- and z-directions and the leading terms of the truncation error are

$$T_1 = -\frac{1}{12}D\Delta x^2 u_{xxxx} - \frac{1}{12}D\Delta y^2 u_{yyyy} + \frac{1}{24}\Delta z^2 u_{zzz} - \frac{1}{8}D\Delta z^2 u_{xxzz} - \frac{1}{8}D\Delta z^2 u_{yyzz} \tag{5.17}$$

where $D = 1/(2jk_0)$ is a constant coefficient.

The PR-ADI and LOD schemes for the parabolic equation are obtained by adding $(r_x r_y/(4jk_0)^2)\delta_x\delta_y$ to Eqn. (3.19), which introduces a new error term $\mathcal{O}\left(\Delta z^3\right)$. But the new error term is of the same order as the other three error terms. Thus, the PR-ADI and LOD schemes are also second-order accurate in x-, y- and z-directions. Moreover, since the discretization equation for the PR-ADI and LOD schemes are the same before splitting, the truncation errors for them will also be the same. The leading terms of the truncation error are

$$T_2 = -\frac{1}{12}D\Delta x^2 u_{xxxx} - \frac{1}{12}D\Delta y^2 u_{yyyy} + \frac{1}{24}\Delta z^2 u_{zzz} - \frac{1}{8}D\Delta z^2 u_{xxzz} - \frac{1}{8}D\Delta z^2 u_{yyzz} \underbrace{+\frac{1}{4}D^2\Delta z^2 u_{xxyyz}}_{\text{introduced error term}} \quad (5.18)$$

Substituting Eqn. (5.15) into Eqn. (5.18), the truncation error for the PR-ADI and LOD schemes becomes

$$T_2 = -\frac{1}{8}\left(\frac{1}{3}jk_z^3 + Dk_x^2k_z^2 + Dk_y^2k_z^2\right)\Delta z^2 u - \frac{1}{12}Dk_x^4\,\Delta x^2 u - \frac{1}{12}Dk_y^4\,\Delta y^2 u \underbrace{+\frac{1}{4}D^2 jk_z\,k_x^2\,k_y^2\,\Delta z^2 u}_{\text{introduced error term}} \quad (5.19)$$

Using $k_x = k\cos\varphi$ and $k_y = k\sin\varphi$, Eqn. (5.19) can be simplified to

$$T_2 = \frac{j}{32}\left(\frac{1}{3} \underbrace{-\cos^2\varphi\sin^2\varphi}_{\text{introduced error term}}\right)\frac{k^6}{k_0^3}\,\Delta z^2 u + \frac{j}{24}\frac{k_x^4}{k_0}\,\Delta x^2 u + \frac{j}{24}\frac{k_y^4}{k_0}\,\Delta y^2 u \quad (5.20)$$

In Eqn. (5.20), the new error term introduced by the PR-ADI and LOD schemes ranges from $-1/4$ to 0, which reduces the first error term. So the dispersion errors of the PR-ADI and LOD schemes are smaller compared to the CN scheme. Therefore, the PR-ADI and LOD schemes have better performances than the CN scheme.

For the MF-ADI scheme, after simplification, the leading terms of the truncation error are

$$T_3 = \frac{j}{32}\left(\frac{1}{3} - \cos^2\varphi\sin^2\varphi\right)\frac{k^6}{k_0^3}\,\Delta z^2 u + \frac{j}{288}\left(\frac{k_x^2k_y^2k^2}{k_0}\,\Delta x^2\Delta y^2 + \frac{k_x^4k^2}{k_0}\,\Delta x^4 + \frac{k_y^4k^2}{k_0}\,\Delta y^4\right)u \quad (5.21)$$

Comparing Eqn. (5.21) with Eqn. (5.20), the first truncation error term for z-direction is the same, but the latter truncation error terms for x- and y-directions are of higher-order. The MF-ADI scheme is second-order accurate in z-direction and fourth-order accurate in x- and y-directions. However, these standard results are only partially helpful for PE error analysis in propagation modeling. The following section demonstrates that the analysis of Section 5.2.1 effectively captures the dependence of the total error (numerical dispersion and approximation) that arises in finite-difference schemes for the PE, when these are applied to propagation problems.

5.3 Assessments

5.3.1 Rectangular Tunnel with Perfectly Conducting Walls

The purpose served by this section is two-fold. First, the proposed error analysis of Section 5.2.1 is corroborated by numerical and analytical results, derived from modal analysis of a rectangular waveguide with perfect electric conductor walls. Second, the same structure is used to compare the performance of the four methods under study, for Dirichlet boundary conditions applied on the walls of the waveguide.

Numerical Validation of Error Analysis

The following numerical test has been designed to validate the analysis of Section 5.2.1. A square waveguide is simulated at 2.4 GHz. This structure supports guided modes. An (m, n) mode has the propagation constant:

$$k_z^{(m,n)} = 2\pi\sqrt{\varepsilon_0\mu_0}\sqrt{f^2 - f_{c,(m,n)}^2}$$

where

$$f_{c,(m,n)} = \frac{1}{2\sqrt{\varepsilon_0\mu_0}}\sqrt{\left(\frac{m}{a}\right)^2 + \left(\frac{n}{b}\right)^2}$$

is the cut-off frequency of the mode and a and b are the width and height of the guide, respectively.

The cross-section of the waveguide is $15\lambda \times 15\lambda$ at the simulated frequency of 2.4 GHz. All simulations use a grid with $\Delta x = \Delta y = 0.3\lambda$, and $\Delta z = \lambda$. Sampling the electric field $\mathbf{E}(x, y, z)$ along the z-direction and taking the Fourier transform with respect to z, the one-dimensional Fourier transform $\mathcal{F}_z\{\mathbf{E}(x, y, k_z)\}$ is derived. The spectrum of $\mathcal{F}_z\{\mathbf{E}(x, y, k_z)\}$ consists of peaks that correspond to the propagation constants of the guided modes. Hence, we can determine the cut-off frequencies $f_{c,(m,n)}$ in three ways: first, analytically, second, via the dispersion analysis of Section 5.2.1 by setting $\tilde{k}_x = 2\pi m/a$, $\tilde{k}_y = 2\pi n/b$, $\tilde{k}_z = 0$ and finding the frequency, and third, from the peaks of the spectrum of $\mathcal{F}_z\{\mathbf{E}(x, y, k_z)\}$. These three sets of results are collected in Table 5.1. Notably, theory, error analysis and simulation results are in good agreement with each other for all the modes included in Table 5.1.

Comparative Performance Evaluation

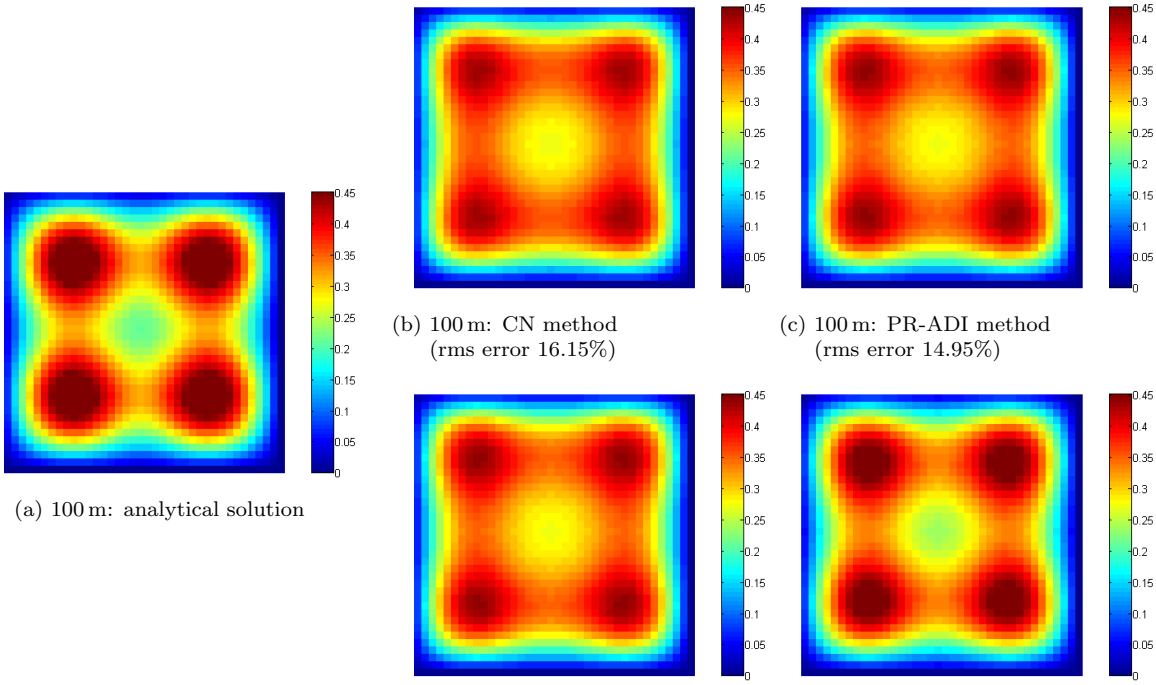
The combined error and computation time analysis of the four schemes under study are proceeded. Since the rectangular waveguide permits an analytical solution, the Euclidean error norm Eqn. (3.29) can be adopted, invoking the analytical field solutions.

Fig. 5.5 and 5.6 illustrate the analytical and numerical transverse-field patterns respectively at $z = 100\,\text{m}$ and $z = 200\,\text{m}$ (excitation plane is $z = 0$). The tunnel cross-section's dimensions are still $15\lambda \times 15\lambda$

Table 5.1: Comparison between resonant frequencies (in 10^6 Hz) from theory, the error analysis of Section 5.2.1 and simulation.

Mode	**Theory**	**CN**		**PR-ADI (LOD)**		**MF-ADI**	
(m,n)	f_{mn}	f_{mn} Section 5.2.1	f_{mn} simulation	f_{mn} Section 5.2.1	f_{mn} simulation	f_{mn} Section 5.2.1	f_{mn} simulation
$(1,1)$	1.1314	1.1309	1.1309	1.1309	1.1310	1.1311	1.1312
$(2,1)$	1.7888	1.7866	1.7865	1.7866	1.7868	1.7876	1.7874
$(2,2)$	2.2627	2.2587	2.2588	2.2587	2.2589	2.2602	2.2602
$(3,1)$	2.5298	2.5228	2.5227	2.5228	2.5228	2.5262	2.5260
$(3,2)$	2.8844	2.8755	2.8754	2.8756	2.8756	2.8791	2.8791
$(3,3)$	3.3941	3.3801	3.3800	3.3805	3.3805	3.3855	3.3854

at the maximum operating frequency of 2.4 GHz, where $\Delta x = \Delta y = 0.4\lambda$, $\alpha = 4$. A unit strength Gaussian source with 2.5λ standard deviation is placed at the center of the $z = 0$ plane as the initial condition. With these parameters, the field solution primarily consists of low-order waveguide modes.



(a) 100 m: analytical solution

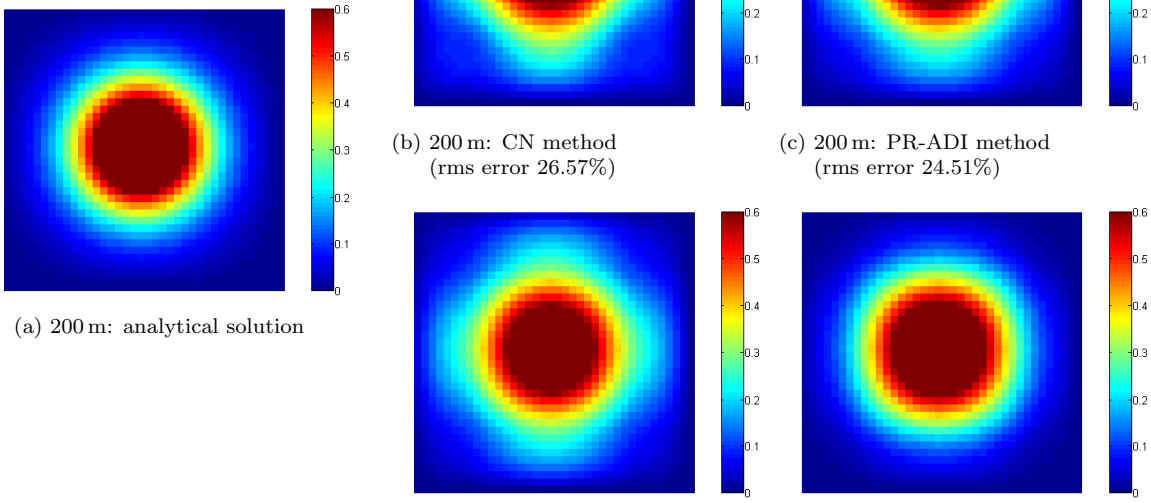
(b) 100 m: CN method (rms error 16.15%)

(c) 100 m: PR-ADI method (rms error 14.95%)

(d) 100 m: LOD method (rms error 14.95%)

(e) 100 m: MF-ADI method (rms error 8.27%)

Figure 5.5: Magnitude of $u(x, y, z = 100m)$ (i.e. at 100 m from the source plane) in a rectangular waveguide with Dirichlet boundary conditions.



(a) 200 m: analytical solution

(b) 200 m: CN method (rms error 26.57%)

(c) 200 m: PR-ADI method (rms error 24.51%)

(d) 200 m: LOD method (rms error 24.51%)

(e) 200 m: MF-ADI method (rms error 14.93%)

Figure 5.6: Magnitude of $u(x, y, z = 200m)$ (i.e. at 200 m from the source plane) in a rectangular waveguide with Dirichlet boundary conditions.

As can been seen, the MF-ADI method has better agreement with the analytical results and the rms error is much smaller than the other three methods. The PR-ADI and LOD methods have the same rms error, and the CN method has the largest rms error.

Then, the field solutions obtained at $z = 100$ m by the four numerical schemes are compared with the analytical solution, obtained with the Helmholtz equation and the parabolic equation. In the former case, the error includes both approximation and dispersion error, while in the latter case it includes only the dispersion error. Hence, the results of this analysis illustrate how these errors vary as the discretization parameters change in the four schemes.

The error norm Eqn. (3.29) is plotted as a function of $\Delta x/\lambda$ (with $\alpha = 4$) in Fig. 5.7 and as a function of α (with $\Delta x = 0.3\lambda$) in Fig. 5.8. The thick lines represent the errors of these schemes with respect to the analytical solution of the wave equation. The thin lines represent errors with respect to the analytical solution of the parabolic equation.

As can be seen from Figs. 5.7 and 5.8, the MF-ADI method has the best performance; the PR-ADI and LOD methods have the same performance, and the CN method has the worst performance among

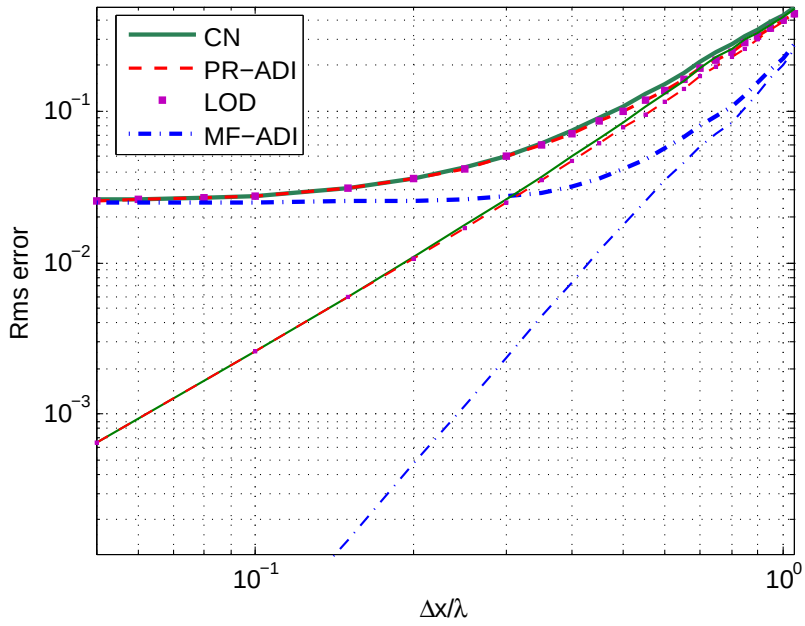


Figure 5.7: Performance comparison of rms error Eqn. (3.29) vs. varying transverse discretization $\Delta x/\lambda$ for the four numerical methods solving wave propagation. Thick lines: the rms error for these schemes applied to approximate the Helmholtz wave equation. Thin lines: the rms error for these schemes applied to solve the parabolic equation.

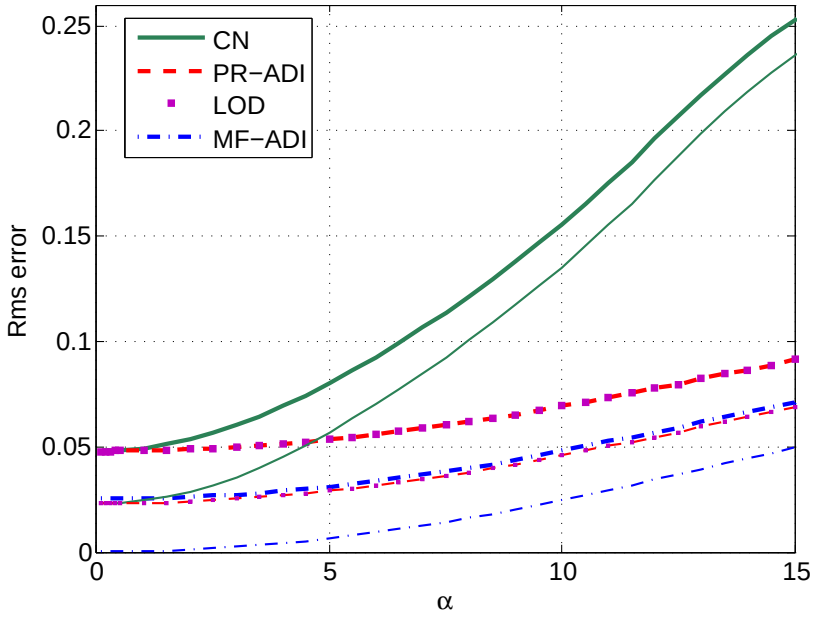


Figure 5.8: Performance comparison of rms error vs. varying α for the four numerical methods solving wave propagation. Thick lines: the rms error for these schemes applied to approximate the Helmholtz wave equation. Thin lines: the rms error for these schemes applied to solve the parabolic equation.

the four methods. Moreover, it can be observed from Fig. 5.7 that the slope of the error curves for the MF-ADI method is about twice the slope of the other three methods. This agrees with the fact that the MF-ADI method is fourth-order accurate and the others are second-order accurate with respect to the transverse directions. It can also be seen that the MF-ADI method converges much faster than the others. This allows for a larger Δx to be chosen for the MF-ADI method for the same error target.

Additionally, it can be observed from Fig. 5.7 that when Δx decreases, unlike the thin lines, the rms errors for the thick lines converge to constant values instead of going to zero. This error floor, shared by all methods, is set by the approximation error of the parabolic equation. In general, all results are in agreement with the basic conclusions of our error analysis, providing an additional confirmation of its usefulness and validity.

Finally, Fig. 5.9 illustrates the relation between accuracy and CPU time (in sec) for the four methods. It can be seen that for a given computation time, the MF-ADI method is always more accurate than the other three methods. The figure also shows how the computational cost of the CN method grows as the number of cells increases.

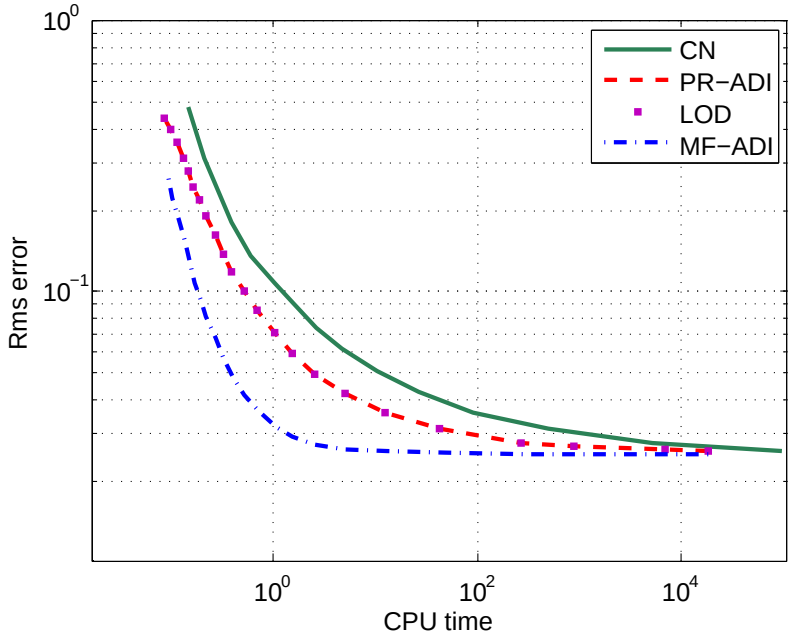


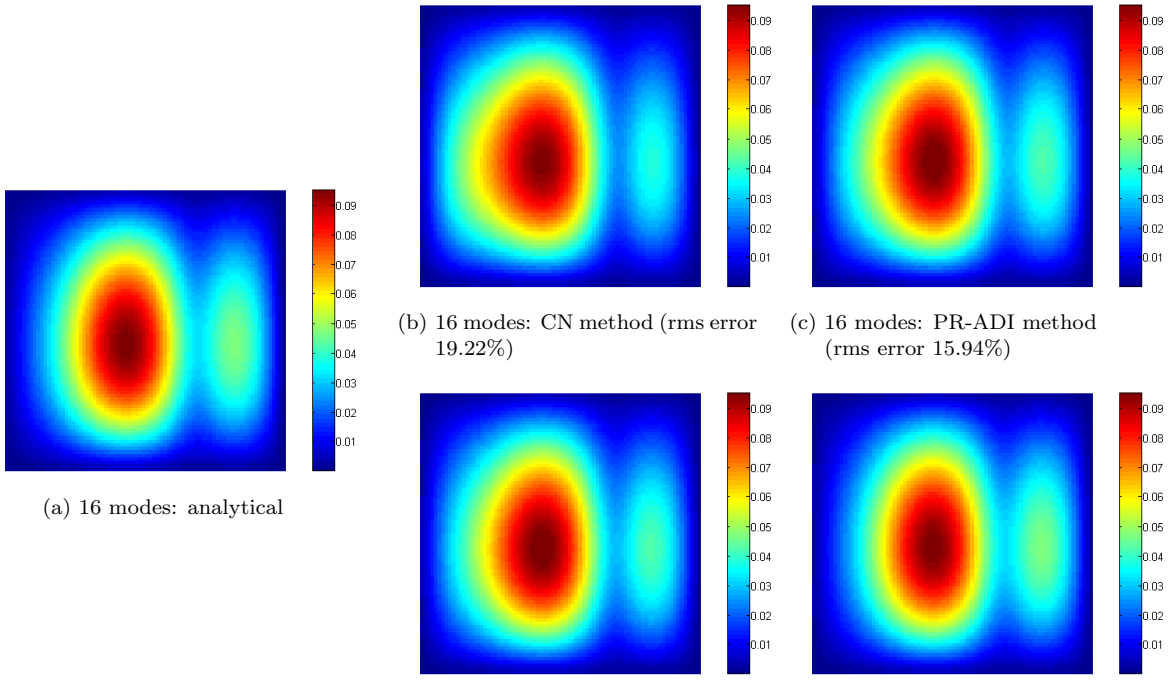
Figure 5.9: Performance comparison of accuracy vs. CPU time (in sec) for the four numerical methods solving wave propagation.

5.3.2 Actual Tunnels with Lossy Walls

To further assess the performance of these numerical methods with regard to accuracy, wave propagation in an actual tunnel with lossy walls is considered for simulation.

Analytical Solutions

The cross-section dimensions, operating frequency, and electrical parameters are the same as adopted in Section 4.3.2. To better assess the performance difference of the four methods, the spatial discretizations are chosen as $\Delta x = \Delta y = 0.6\lambda$, and $\alpha = 10$. The y-polarized hybrid fields are adopted as the initial source, where 16 and 40 modes with equal weight, are respectively included as the initial condition. The transverse-field patterns obtained from the two methods at 3 km are compared to the analytical ones, as illustrated in Fig. 5.10 and Fig. 5.11.



(a) 16 modes: analytical

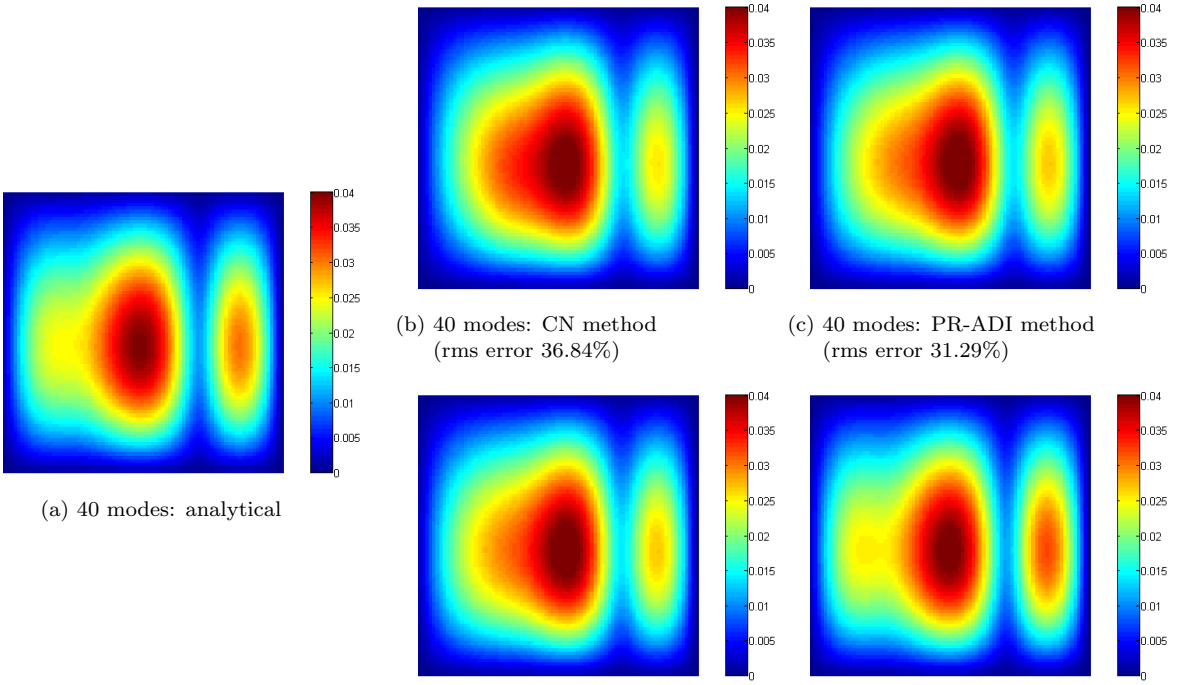
(b) 16 modes: CN method (rms error 19.22%)

(c) 16 modes: PR-ADI method (rms error 15.94%)

(d) 16 modes: LOD method (rms error 15.94%)

(e) 16 modes: MF-ADI method (rms error 5.26%)

Figure 5.10: Plane wave field patterns at 3 km from the initial plane in a rectangular tunnel with lossy walls. The simulated results from the four methods are compared to the analytical solution. The initial plane respectively includes 16 modes with equal weight as the source, and the relative error is calculated through the Euclidean norm Eqn. (3.29).



(a) 40 modes: analytical

(b) 40 modes: CN method (rms error 36.84%)

(c) 40 modes: PR-ADI method (rms error 31.29%)

(d) 40 modes: LOD method (rms error 31.29%)

(e) 40 modes: MF-ADI method (rms error 4.32%)

Figure 5.11: Plane wave field patterns at 3 km from the initial plane in a rectangular tunnel with lossy walls. The simulated results from the four methods are compared to the analytical solution. The initial plane respectively includes 40 number of modes with equal weight as the source, and the relative error is calculated through the Euclidean norm Eqn. (3.29).

Measurement

The simulated results are compared to the measured data at 900 MHz in the Massif Central tunnel, as depicted in Fig. 5.12. The straight lines represent the corresponding attenuation, which is calculated through the slope of the least square fit line using data at distances in the far zone (0.5-2.5 km).

Table 5.2 presents the attenuation obtained from the measured and simulated data for the Massif Central tunnel. This case study further confirms the superior performance of the MF-ADI method, compared to the other three methods, and the CN method has the worst performance among them.

Table 5.2: The measured and simulated attenuation for the Massif Central tunnel.

Attenuation	**Measurement**	**CN**	**PR-ADI (or LOD)**	**MF-ADI**
dB/km	8.5	9.2	9.1	8.7

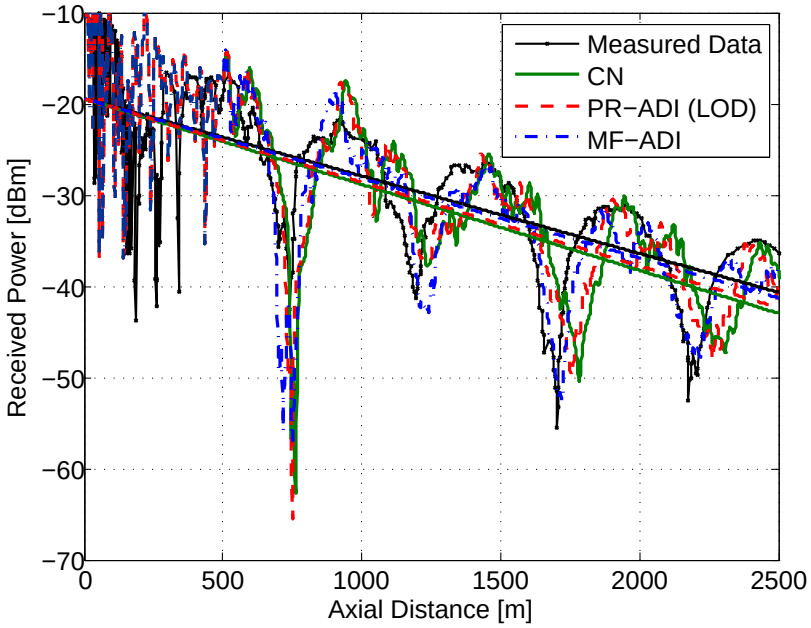


Figure 5.12: Performance comparison of the CN, PR-ADI, LOD, and MF-ADI methods in the Massif Central tunnel with lossy walls. For transverse discretizations, Δx and Δy are chosen as 1.5λ, and $\Delta z = 4\lambda$. Ray-tracing is applied to generate initial source fields for the VPE and the source plane is set at the 500 m.

5.4 Conclusion

Parabolic equation solvers have been developed and successfully applied to wireless propagation modeling for many years. However, the rich literature on this topic was still missing a comprehensive error analysis of established. This chapter has contributed such an error analysis that allows one to choose the parameters of each method based on certain error targets. Modified numerical dispersion expressions, including both the conventional dispersion error and the PE approximation error, have been provided for the four numerical schemes that were studied. The procedure outlined in this chapter is applicable to any alternative finite-difference scheme as well.

These expressions reveal that the recently introduced PR-ADI (LOD) [15], as well as the MF-ADI method [20] present significant advantages compared to the CN scheme. Among the three ADI methods, MF-ADI is more accurate for the same execution time, or faster for the same accuracy, than the other two ADI methods. Hence, these comparisons lead to the conclusion that MF-ADI is the method of choice for the PE-based solution of tunnel propagation problems, because it employs fourth-order accurate finite differences over the electrically large tunnel cross-section.

Chapter 6

Hybrid Modeling Technique using the PE and RT Methods

6.1 Introduction

As illustrated in Section 2.3, the ray-tracing method allows the treatment of propagation modeling in very complicated scenarios, and it can provide reasonably accurate prediction of radio-wave propagation characteristics. However, ray-tracing methods typically have an exponential computational complexity for certain practical problems, as discussed in Section 2.4. The convergence issues for the results generated by ray-tracing still remain intractable, especially at large distances of separation between the transmitter and receiver. Generally, applications of the ray-tracing method are limited to modeling wave propagation in small environments such as indoor channels within buildings [24].

Parabolic equation methods, on the other hand, are far more computationally efficient for long guiding structures such as tunnels and offer prediction with higher accuracy, even if the receiver is far away from the transmitter. However, they cannot model high-order modes (large bounce angles) and corresponding rapid fluctuations accurately [23], and thus they are limited to paraxial wave propagation predictions or very large distances are considered where the higher-order modes have little contribution.

In this chapter, an advanced hybrid channel modeling technique is presented, which combines the ray-tracing and PE methods by minimizing their inherent restrictions and integrating their advantages. The proposed hybrid model enables the modeling of actual railway scenarios including stations and long guideway tunnels within a unified simulation framework. The proposed model is validated with the measured data, collected in an canonical tunnel and an actual deployment sites of communication-based

train control systems.

6.2 Hybrid Propagation Model

The principle of this proposed hybrid propagation model is to appropriately utilize the ray-tracing and PE methods by strategically sub-dividing the simulating railway environments into two subdomains. Within this hybrid model, PE is used for the long, relatively uniform tunnel sections, whereas ray-tracing is applied to complex geometries and discontinuities. The two methods are coupled by means of Huygens principle and the placement of the connecting surface is determined by convergence studies.

A practical scenario for this hybrid model is a station platform followed by a long railway tunnel. Here, for the station areas, due to the presence of the elevated platform and other objects such as escalators, stairs, and pillars, the geometry is much more complex than the connected section which is usually a long, relatively uniform tunnel.

6.2.1 Implementation

To implement this hybrid propagation technique, the primary step is to sub-divide the simulation scenario into the ray-tracing subdomain and the PE subdomain. The location of the interface is determined by taking both the accuracy and efficiency of the whole hybrid model into account. At the interface of the two subdomains, a Huygens surface is defined to couple the fields from the ray-tracing solver to the PE solver, where the former is applied to compute the fields at grid points on the interface. Then, these discrete-space fields on the interface are adopted as the initial source plane for the following PE solver, as illustrated in Fig. 6.1.

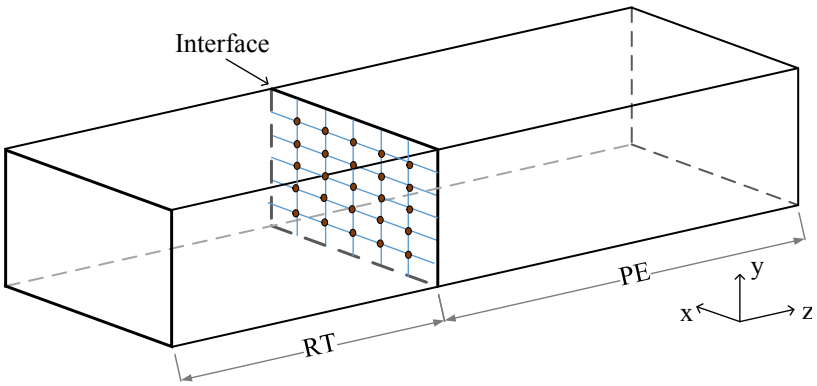


Figure 6.1: The hybrid model, where the simulation scenario is divided into two subdomains and the dots on the interface represent the discrete-space fields computed from the RT method.

Moreover, it should be guaranteed that the fields on the interface plane have converged, which means that an appropriate number of reflection levels is adopted for the simulation of the ray-tracing subdomain. Subsequently, the number of the discrete grid points projected on this surface can be chosen by making a further balance between the desired accuracy and efficiency.

6.2.2 Location of the Interface

As the PE solver is computationally more efficient than the ray-tracing solver, so to better relieve the computational burden of ray-tracing and improve the overall efficiency of the hybrid model, the interface should be placed as soon as the basic conditions for PE are satisfied. Generally, the optimal choice for the location of this interface is at the breaking point when the assumption Eqn. (3.5) is met.

Substituting the wave solution $u(x, y, z)$, Eqn. (3.8a), into Eqn. (3.5), the assumption for PE methods can be expressed as

$$k_z \approx k_0 \tag{6.1}$$

A direct perspective to understand this assumption is that the wave propagation is paraxial, which means the wave travels within a small azimuth angle to the z-axis, where z is the propagation direction. As discussed in Section 3.2.1 and according to [32], this maximum angle θ is usually chosen as 15°, within which the PE assumption is valid. Here, the angles are divided into 0°-15° and 15°-90° these two groups, as illustrated in Fig. 6.2.

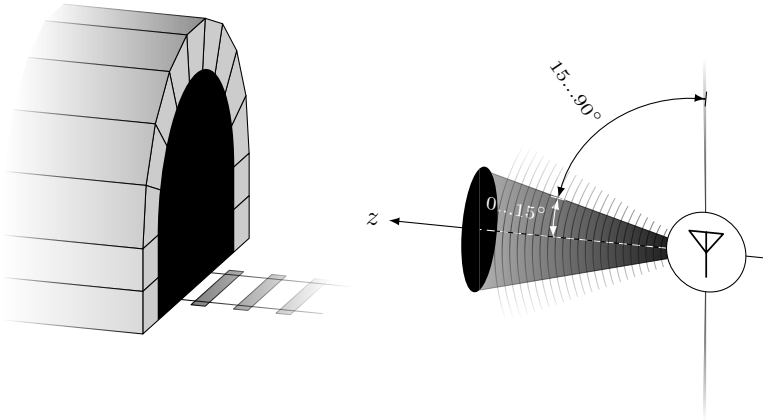


Figure 6.2: The ray diagram, where the angles are divided into 0°-15° and 15°-90° groups.

The interface should be located at the point where the relative contribution of the rays outside 15° is insignificant. Here, the total power of all the ray fields within respectively 0°-15° and 15°-90° group (P_1

and P_2), are calculated. The power weight w for the first angle group is obtained through the following expression:

$$w = \frac{P_1}{P_1 + P_2} \tag{6.2}$$

The interface is placed after the power weight w is above certain threshold. The threshold will be chosen according to the desired accuracy and computational efficiency. During the simulation, the threshold to be chosen as 95% is a reasonable choice.

Moreover, another straightforward way to determine the location of the interface is through convergence studies of the received power. Suppose the interface is placed, respectively, at d_1, d_2, and d_3; if the subsequent profiles of the received power generated by PE converge to fixed values, then it indicates that at d_1, the assumption for PE is satisfied. Hence, the minimum distance where the profile converges, will be the optimal choice for the placement of the interface.

6.2.3 Validation

To validate this hybrid model, wave propagation in a 2.5 km tunnel with cross-section 5 m × 5 m is simulated. The electrical parameters of the walls are chosen as $\varepsilon_r = 12$ and $\sigma_0 = 0.01$ S/m. A dipole antenna is placed vertically at the center of the tunnel. The operating frequency is 2.4 GHz. The results of the received power simulated by ray-tracing alone with a series number of reflection levels are depicted in Fig. 6.3.

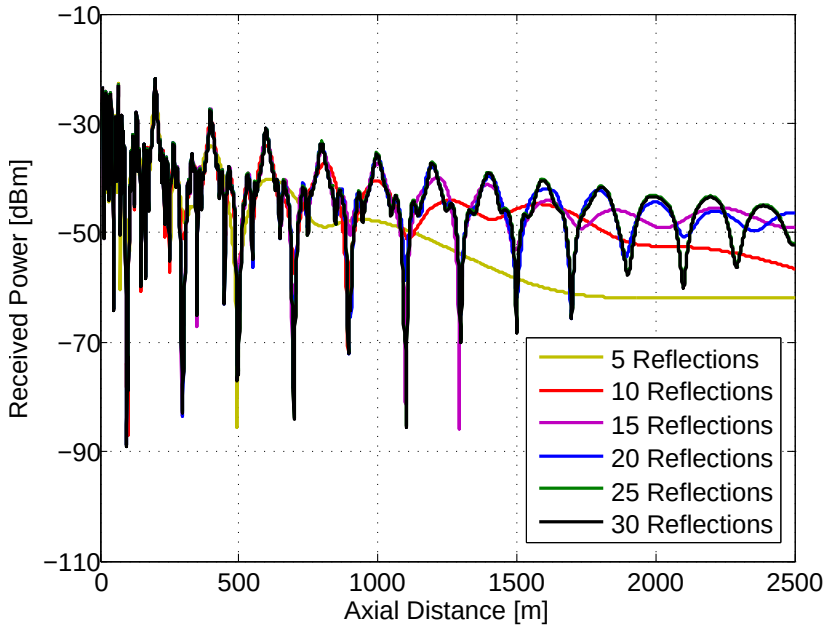


Figure 6.3: Received power generated by ray-tracing alone with a series number of reflection levels. The simulated distance is 2.5 km.

As can be seen, the near zone of the tunnel is characterized by rapid fluctuations caused by the interaction of many modes, and as higher-order modes attenuate more quickly, the fields in the far zone consist primarily of the first few lower-order modes. Comparing the received power profiles generated with different reflection levels, it can be observed that within the first 500 m region, a reflection level of 10 is already enough for the results to achieve convergence, where as at greater distance, a larger number of reflections is required. Hence, to optimize the overall efficiency of the hybrid model, PE should take over the simulation as soon as its assumptions are satisfied and ray-tracing can be reserved for the simulation of the initial regions with a minimum reflection level while still guaranteeing convergence in that region.

The trend for the power weight of the ray fields from different angle group is illustrated in Fig. 6.4. Here, ray-tracing with a reflection level of 10 is applied for the calculation of the power weight within the first 500 m region.

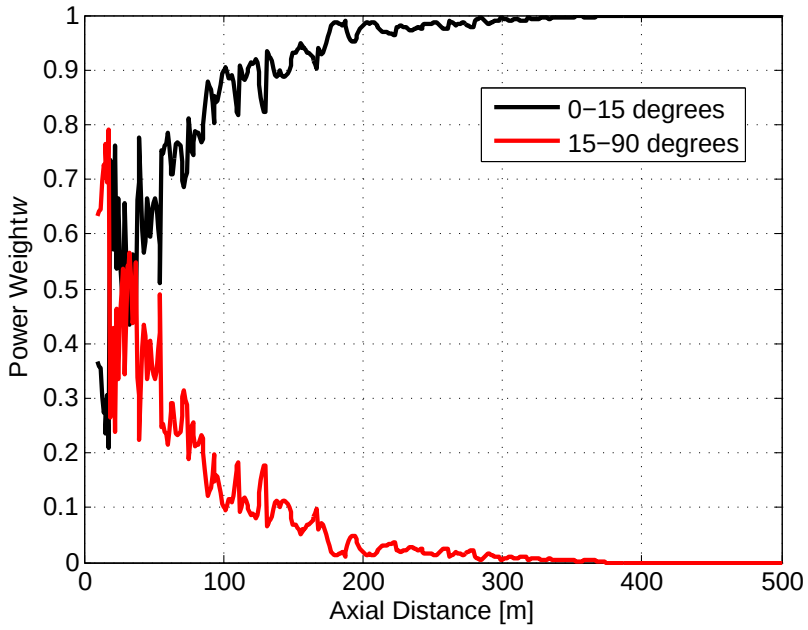
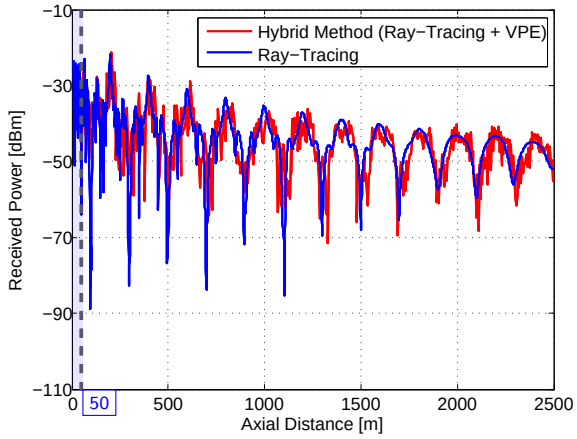


Figure 6.4: The trend for the power weight of the ray fields from different angle group within the first 500 m region, where the angles are divided into 0°-15° and 15°-90° groups.

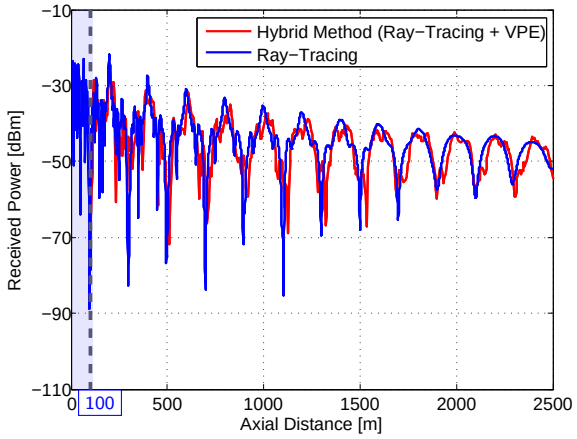
As can be seen from the graph, at the distance of about 200 m, the power weight for ray fields within 15° is above 95%, so the interface can be placed around this distance. When the interface is placed at respectively 50 m, 100 m, 200 m, and 300 m, the results of the received power are compared in Fig. 6.5.

It can be observed from Fig. 6.5 that the profiles for the received power remain almost the same after the interface is placed after 200 m, which indicates the assumption is satisfied after 200 m and the interface can be placed around this distance. This also confirms the trend of the power weight illustrated

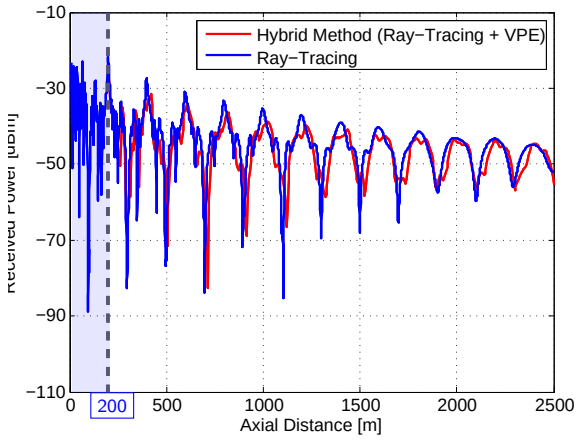
in Fig. 6.4.



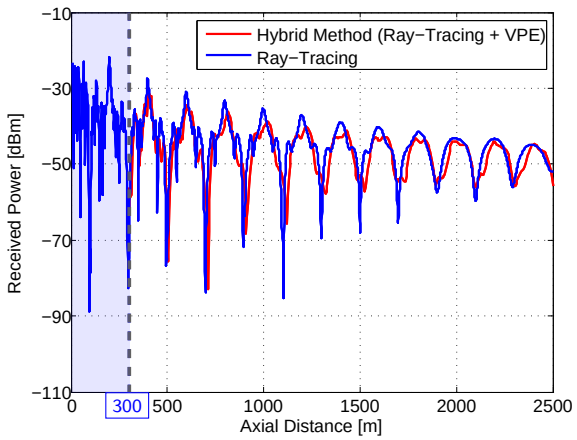
(a) Interface: 50 m



(b) Interface: 100 m



(c) Interface: 200 m



(d) Interface: 300 m

Figure 6.5: Received power at 2400 MHz, where the interface is placed at 50 m, 100 m, 200 m, and 300 m respectively. A reflection level of 10 is selected for the simulation of ray-tracing subdomain. The spatial discretizations for the PE subdomain are chosen as $\Delta x = \Delta y = 0.1$ m, and $\Delta z = 2$ m.

Moreover, the relation between the CPU time and the interface location is presented in Table 6.1. It can been seen that the CPU time increases linearly with the distance of the interface location. To reduce the overall computational cost of the hybrid model, the interface should be placed as soon as the assumption of the PE method is satisfied.

Table 6.1: CPU time vs. interface location.

Interface location [m]	50	100	200	300	400	500
CPU time [s]	56.87	124.85	261.93	395.38	534.36	673.47

6.3 Application to Realistic Railway Environments

In this section, two actual tunnels will be considered for the purpose of validation, where the measured data is available. One is the canonical Massif Central Tunnel, also shown in previous chapters, and the other one is a section selected from the Edmonton railway network, which is an actual deployment site of CBTC systems and both ends are connected with long straight tunnels.

6.3.1 The Massif Central Tunnel

Here, through similar process as described in Section 6.2.2, the total power of all the ray fields within respectively 0°-15° and 15°-90° group are calculated. The trend of the power weight is illustrated in Fig. 6.6, where the power weight of ray fields within 15° is above 95% from around 400 m.

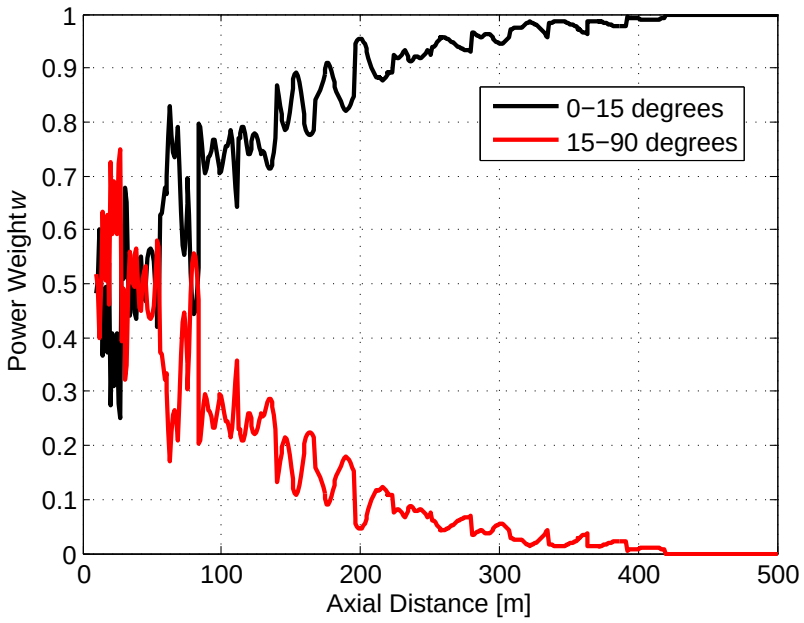
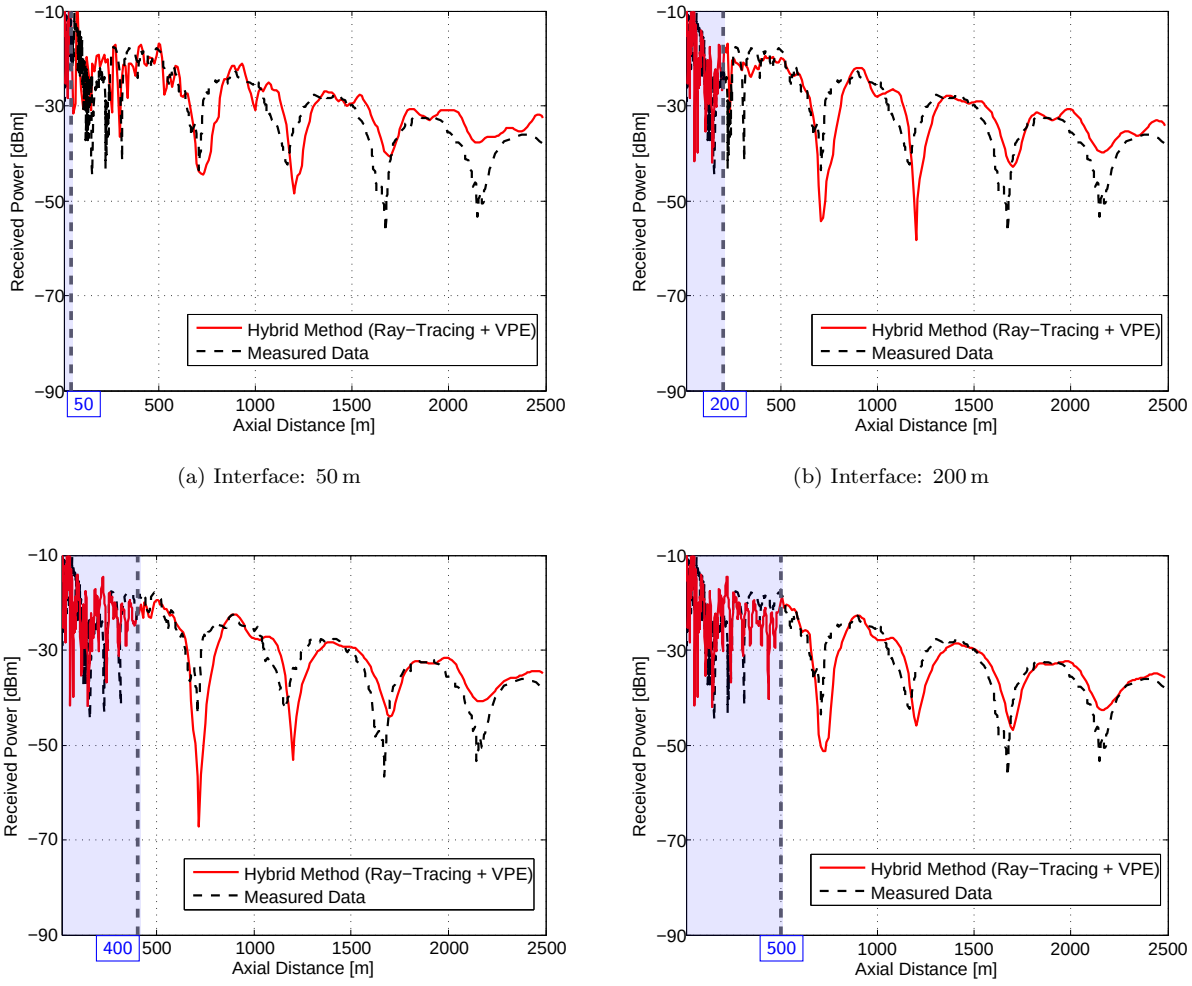


Figure 6.6: The trend for the power weight of the ray fields from different angle group within the first 500 m region of the Massif Central tunnel, where the angles are divided into 0°-15° and 15°-90° groups.

This comparison between the received power profiles generated after respectively 50 m, 200 m, 400 m, and 500 m, is depicted in Fig. 6.7. It can be observed that the trend is in agreement with the analysis of the power weight, where after 400 m, the received power profile converges.



(a) Interface: 50 m

(b) Interface: 200 m

(c) Interface: 400 m

(d) Interface: 500 m

Figure 6.7: Received power at 900 MHz in the Massif Central tunnel. The whole simulated distance is 2.5 km. A reflection level of 20 is selected for the simulation of ray-tracing subdomain. The spatial discretizations for the PE subdomain are chosen as $\Delta x = \Delta y = 0.1$ m, and $\Delta z = 2$ m.

6.3.2 The Edmonton Railway Station

The main part of this section is the railway station, both ends of which are connected with tunnels, as illustrated in Fig. 6.8. The simulated results are verified with the measured data, provided by Thales Canada Transportation Solutions, in Fig. 6.9.

Here, since the environment is very complicated and the length for the rectangular tunnel part is just 100 m, the power weight is not compared here and the interface between the two subdomains is placed right after the station area. Even if the following subdomian is not a long tunnel to let all higher-order modes attenuate significantly, it can be observed from Fig. 6.9 that the simulated results have reasonable agreement with the measured data, which further confirms the feasibility and validity of the proposed

hybrid model.

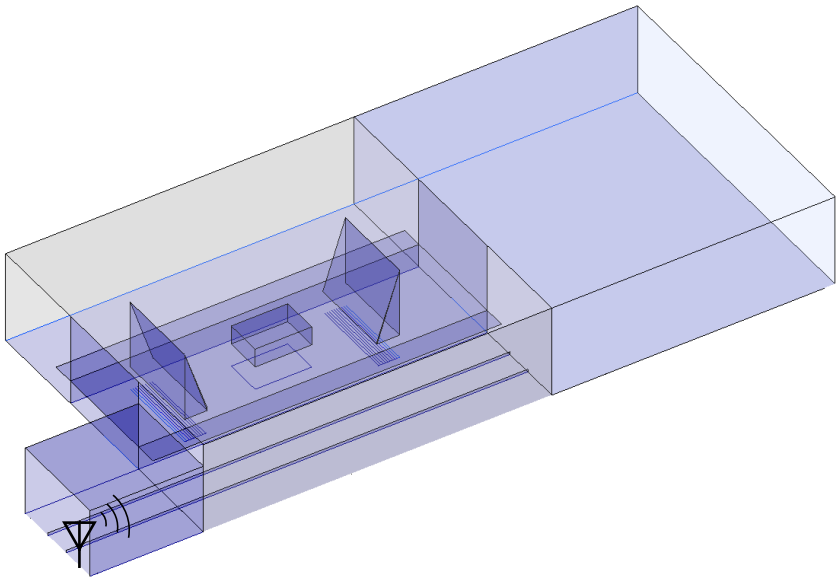


Figure 6.8: The modeling structure diagram, where the narrow one-way rectangular tunnel, the station platform, and the double-way rectangular tunnel are respectively 40 m, 123 m and 100 m in length. A transmitting antenna operating at the frequency of 2.45 GHz is placed 3.1 m in height and horizontally 0.5 m to the right wall. Two receiving antennas are fixed on the trolley with 0.9 m separation and 3.1 m to the top of the rail track.

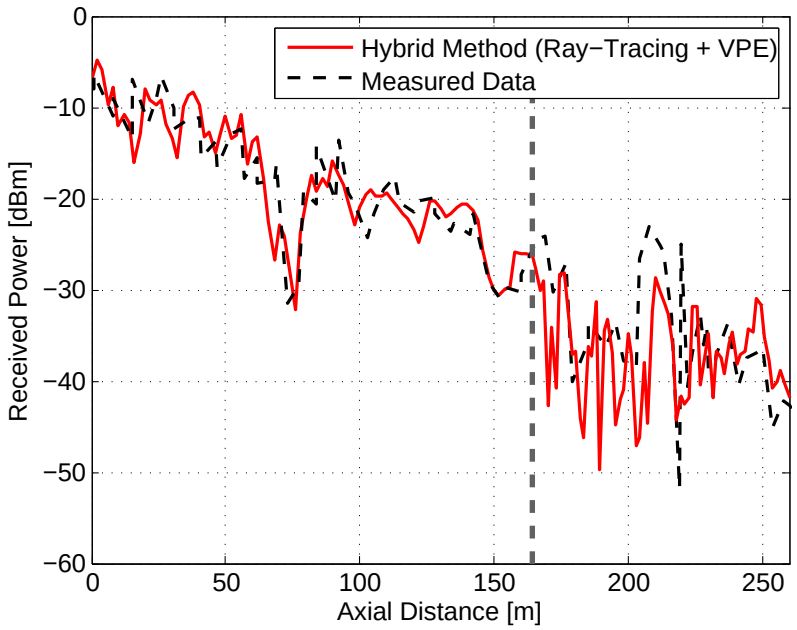


Figure 6.9: Received power for the Corona to Grandin section, where interface is placed at 164 m. Ray-tracing with a reflection level of 10 is applied for the simulation of the station area, and PE is reserved for the double-way rectangular tunnel (from $z = 163$ m to $z = 263$ m). The spatial discretizations for the PE subdomain are chosen as $\Delta x = \Delta y = 0.1$ m, and $\Delta z = 1$ m.

6.4 Conclusion

An advanced hybrid channel modeling technique, combining the versatility of ray-tracing and the computational efficiency and accuracy of PE, is proposed. Ray-tracing is reserved for the complex geometries and discontinuities, and PE is applied to the long, relatively uniform tunnels. This hybrid model offers a prospect for the modeling of wave propagation in the complete railway network.

Chapter 7

Conclusion

This thesis focused on developing an advanced modeling framework to understand and characterize the radio-wave propagation behaviors in railway scenarios, which is essential for the deployment of modern communication-based train control (CBTC) systems. As existing modeling techniques generally have their inherent limitations and trade-offs between accuracy, efficiency and generality, hence, an advanced hybrid modeling framework has been presented in this thesis by minimizing the inherent restrictions and integrating their advantages.

The ray-tracing and parabolic equation methods were combined in this hybrid model. First, the applicability of ray-tracing in solving wave propagation in complicated environments such as the railway stations has been demonstrated. Validations were carried out using the measured data, collected in an actual deployment sites of CBTC systems. Ray-tracing has proved to be flexible enough to handle arbitrary geometries with reasonable accuracy. However, it is limited by the exponentially increasing computational complexity for the long guide-way structures such as tunnels. Therefore, a computationally more efficient technique, the parabolic equation method, is required to address wave propagation in these sections.

Two three-dimensional parabolic equation solvers were developed from the ground up, the scalar PE solver and the vector PE solver, which have been validated by the case study of a rectangular tunnel with perfectly conducting walls and a well-known realistic tunnel with lossy walls. The parabolic equation method has proved to be an effective and computationally efficient modeling approach for those long, relatively uniform guideway structures.

Additionally, an improved ADI method with higher accuracy, the MF-ADI method, was proposed. The formulation itself and the numerical results indicated that this MF-ADI method has substantial

advantages over the traditional techniques that have been employed in such simulations. Compared to the standard ADI method, this scheme has proved to offer significantly better accuracy at no additional computational cost.

Moreover, different numerical finite-difference schemes such as CN, PR-ADI, LOD, and MF-ADI were analyzed and compared from modeling wave propagation in tunnel environments. A systematic study of the numerical error and the relative performance of the PE methods, which provides guidelines for the choice of the parameters and the trade-offs involved in solving parabolic equations, was provided. Numerical investigations and assessments for the performance of these methods were accompanied by the case study of wave propagation in tunnels.

7.1 Contributions

These are four major contributions that have resulted from this thesis:

X. Zhang and C. D. Sarris, "Error Analysis and Comparative Study of Numerical Methods for the Parabolic Equation Applied to Tunnel Propagation Modeling," *IEEE Transaction on Antennas and Propagation*, 2014. (in review)

X. Zhang and C. D. Sarris, "A High-Accuracy ADI Scheme for the Vector Parabolic Equation Applied to the Modeling of Wave Propagation in Tunnels," *IEEE Antennas and Wireless Propagation Letter*, vol. 13, pp. 650–653, 2014.

X. Zhang, N. Sood, J. Siu, and C. D. Sarris, "A Hybrid Vector Parabolic Equation/Ray-Tracing Propagation Modeling Technique for Rail Transportation Systems," *IEEE Antennas and Propagation Society International Symposium (APS/URSI)*, 6-11 July 2014.

X. Zhang, N. Sood, J. Siu, and C. D. Sarris, "A Hybrid Propagation Modeling Technique for Complex Railway Environments," *IEEE Antennas and Propagation Society International Symposium (APS/URSI)*, 7-13 July 2013.

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